สยามชำนาญกิจ
SIAM CHAMNAN KIT

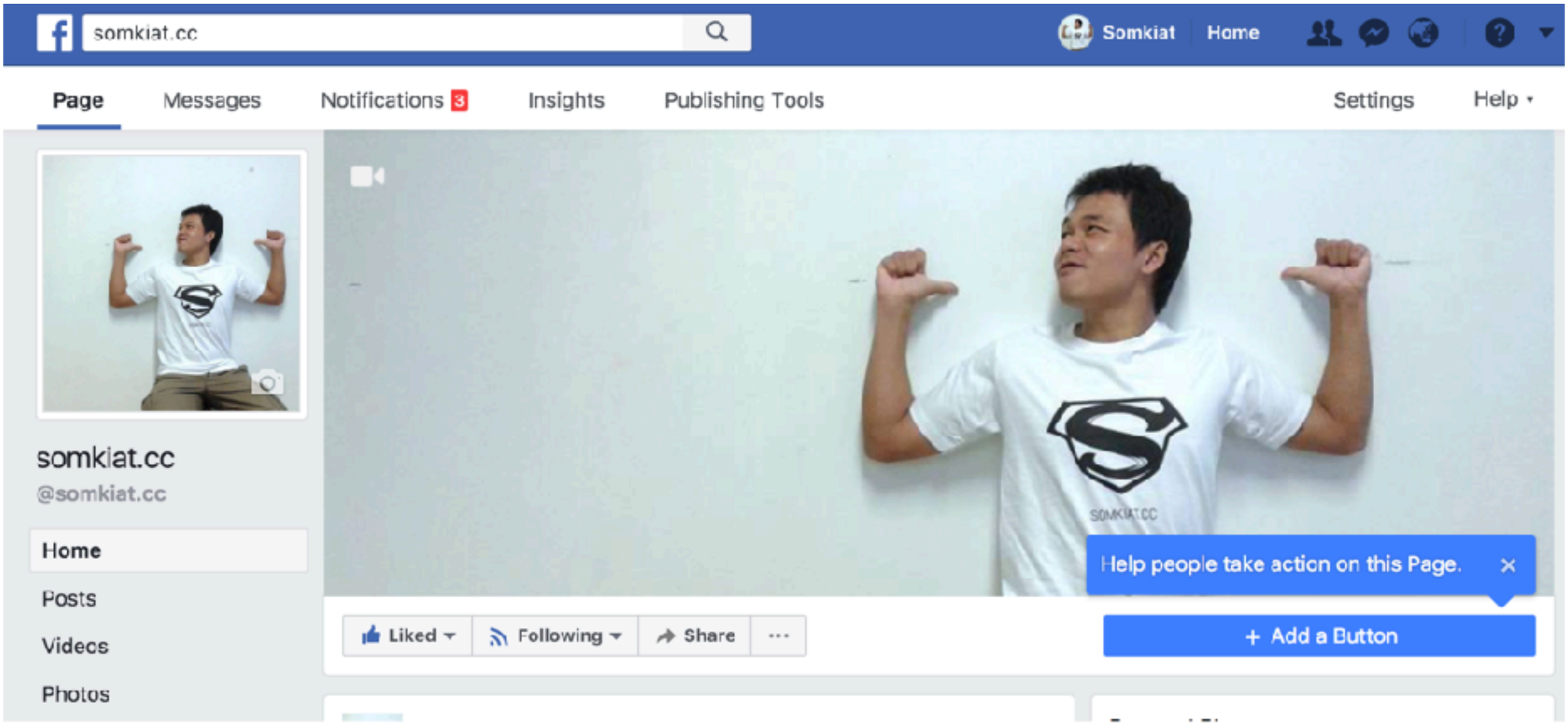
Generative AI for Software Development 2026

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https://github.com/up1/workshop-ai-with-technical-team

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Goals

Integrate Generative AI in Software Development Requirement, Coding and Testing Workshop with Copilot



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Learning Path

AI/ML

AI Model

Prompt Engineer

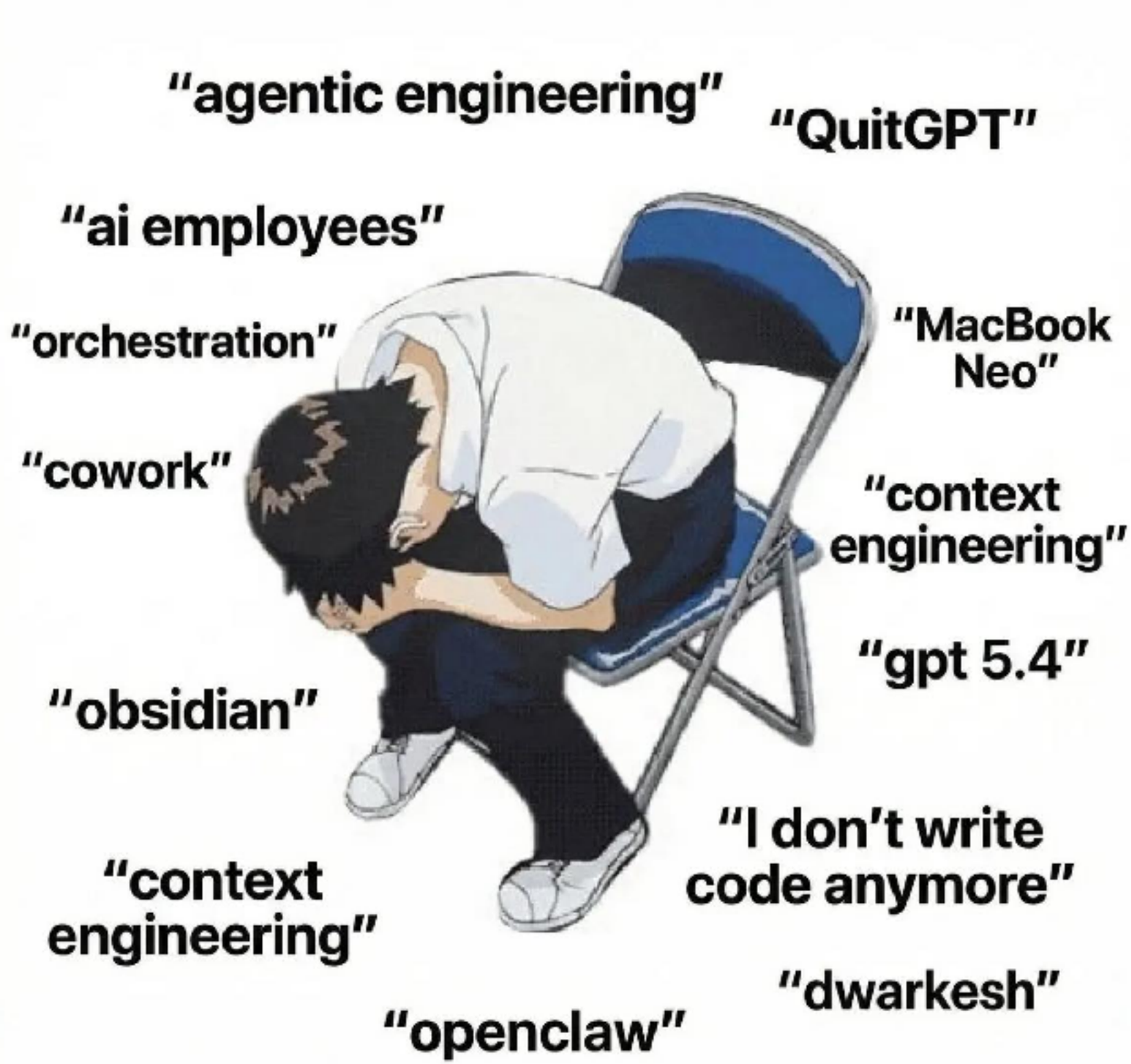
Context Engineering

AI in Software development

Develop AI/LLM app

Use cases and workshops

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How to use AI ?

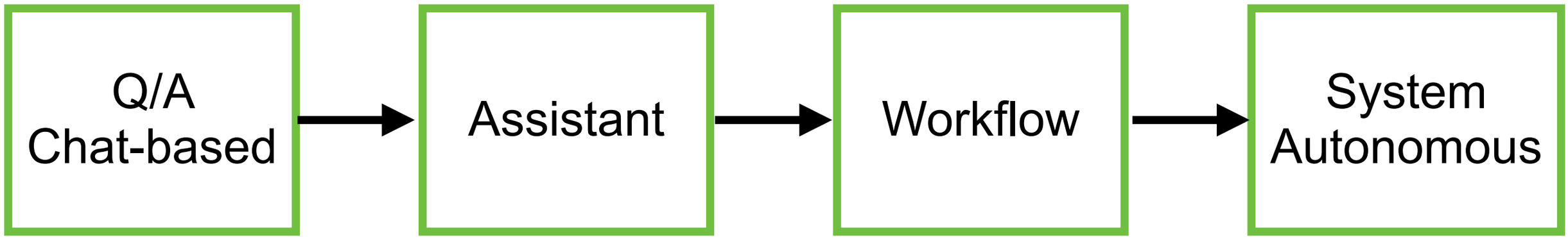
AI as
a **tool**

AI as
an **assistant**

AI as
an **agent**

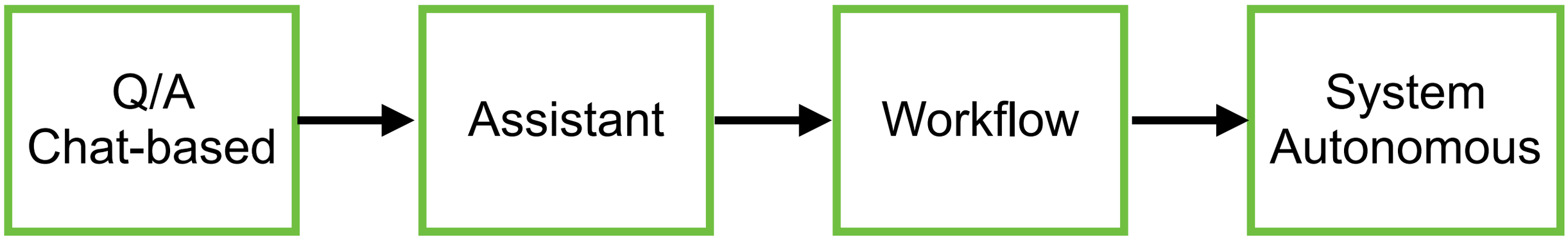
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Steps to use AI ?



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Steps to use AI ?



Prompt engineering

Manage context and knowledge for AI (Context engineering)
Human in the Loop (HITL)

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Software Development

Requirement | Design | Develop | Testing | Deploy

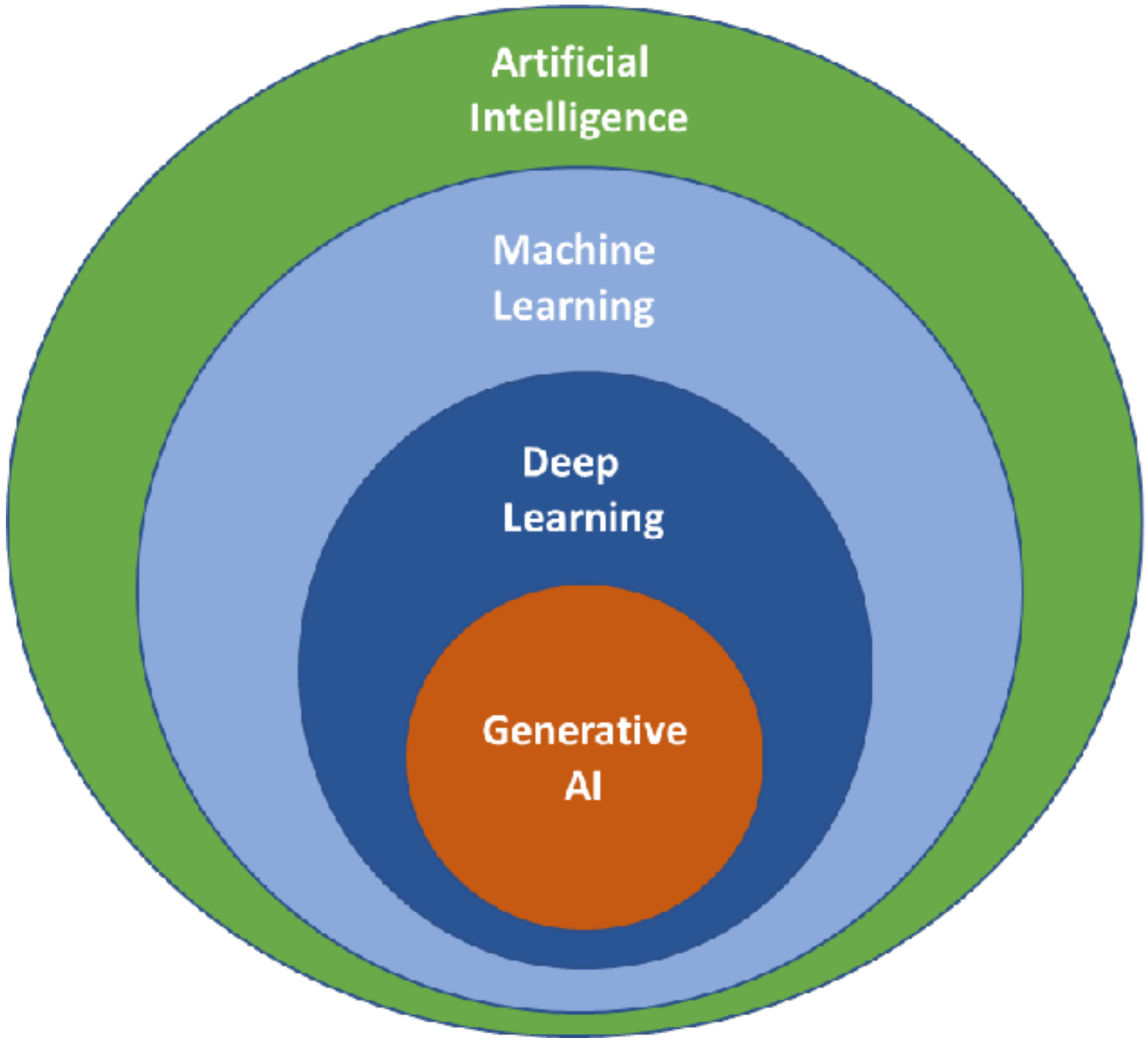
Generative AI

Improve Productivity … (Replace human !!)

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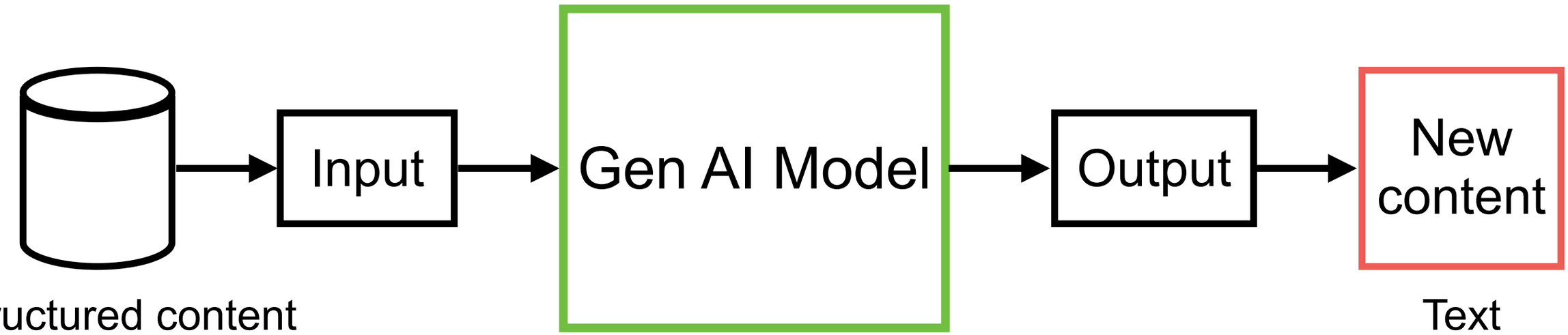
Start with AI !!

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Generative AI



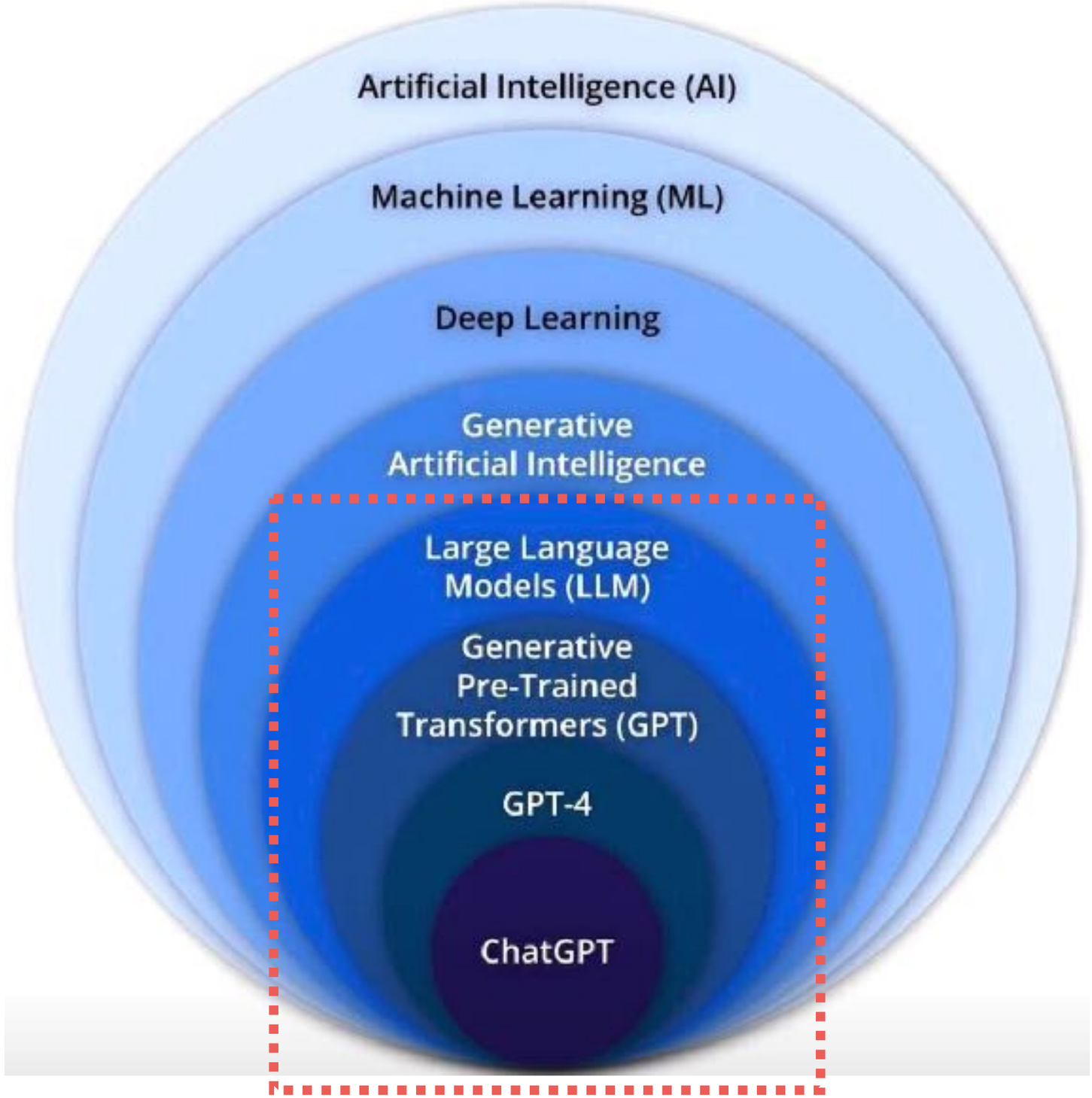
Unstructured content

Learn patterns from Unstructured content

Text
Image
Video
3D
Task

https://grow.google/ai-essentials/

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Large Language Model (LLM)

Type of AI model
Process, understand
Generate human readable data

LLM

Training data !!

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LLMs in industry

Model name	Company
Bidirectional Encoder Representation from Transformers (BERT)	Google AI
Generative Pre-trained transformer-5 (GPT-5)	OpenAI
Pathways Language Model-E (PaLM-E)	Google AI
BLOOM	NVIDIA AI
Llama 4	Facebook
Claude’s Sonnet/Opus 5.5	Anthropic



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LLM Development timeline

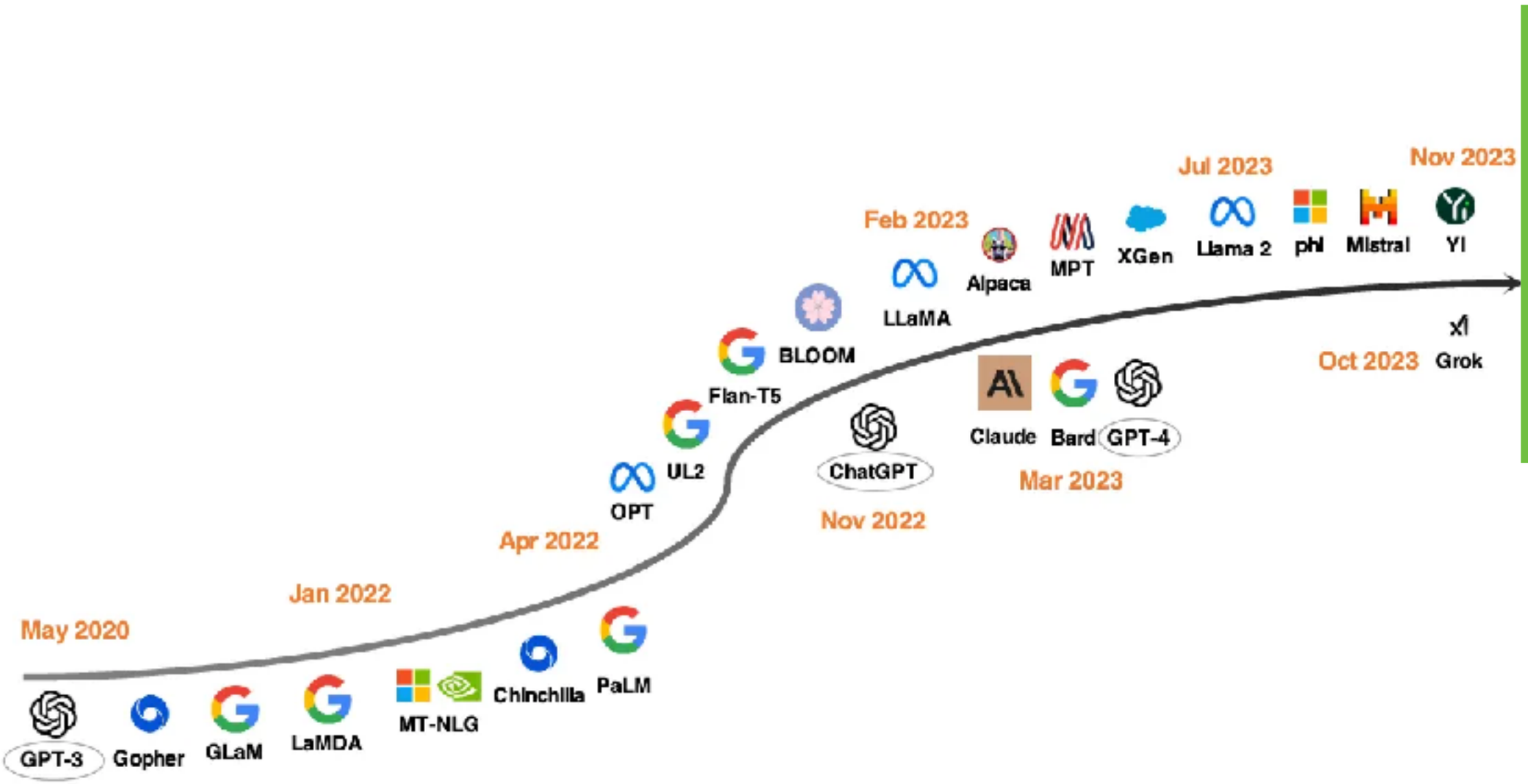


Figure 3: LLM development timeline. The models below the arrow are closed-source while those above the arrow are open-source.

https://arxiv.org/abs/2311.16989



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Let’s go !!



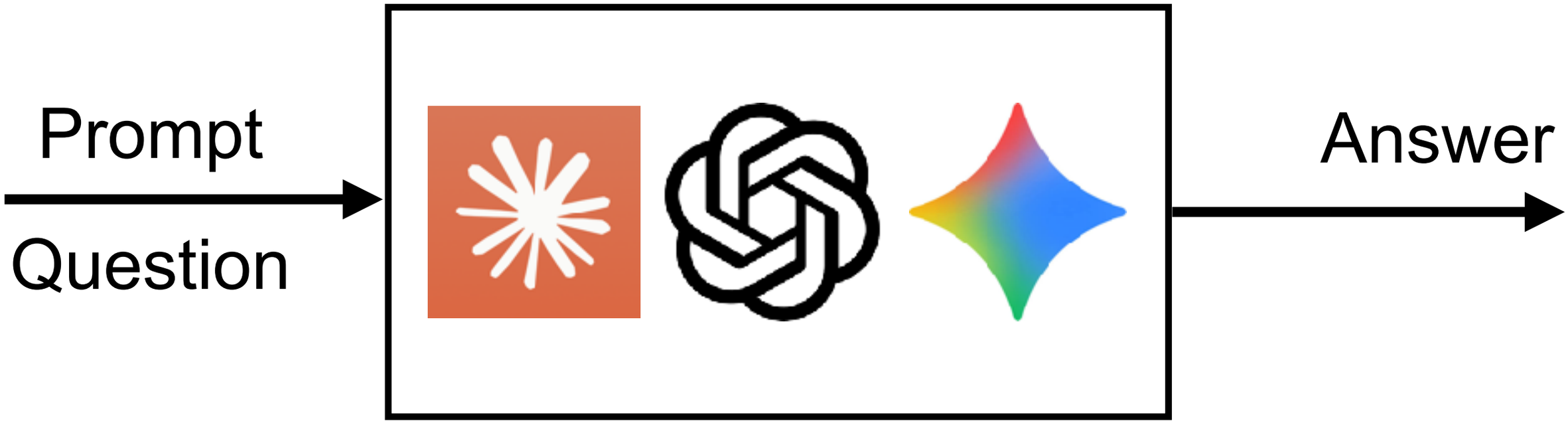
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Hello with Generative AI



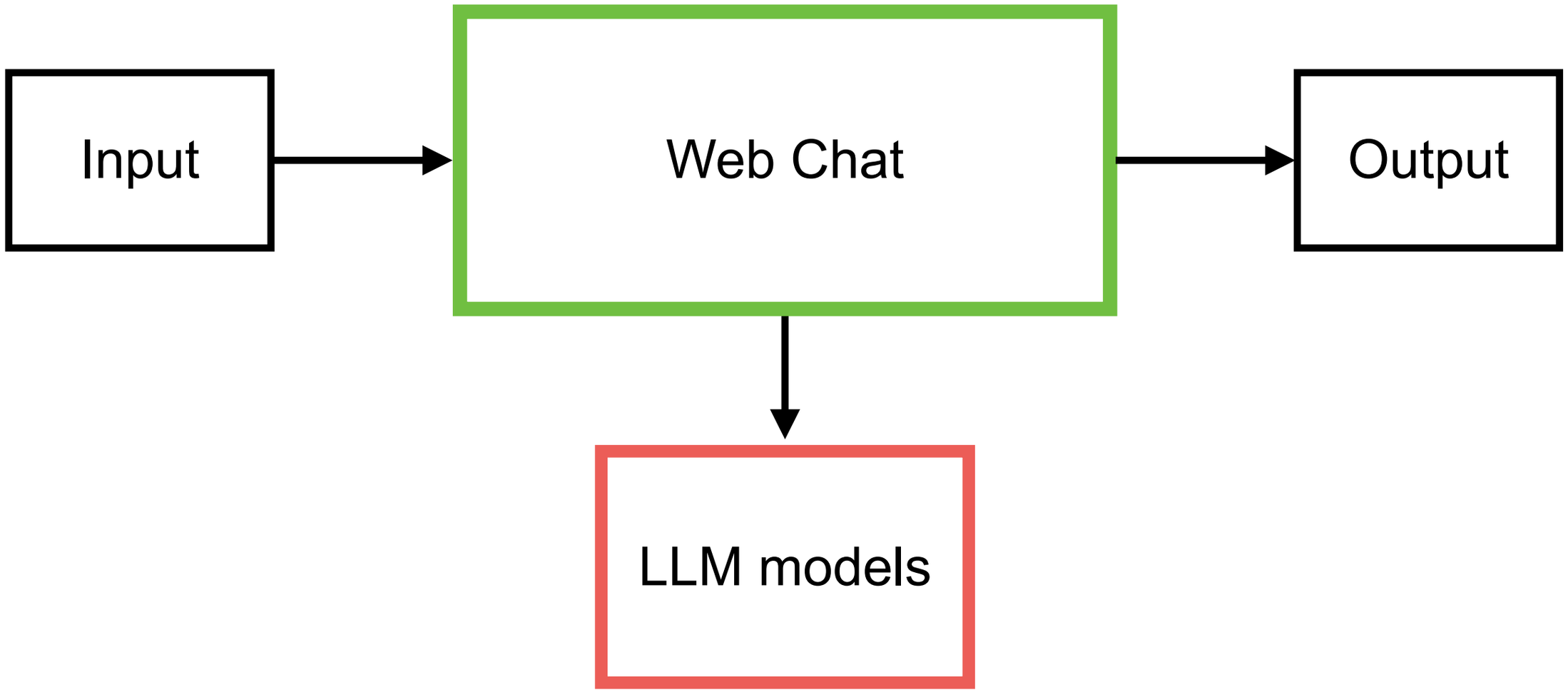
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Chat-based



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Chat-based



https://github.com/copilot

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LLM model card

GPT-6 Astra `gpt-6-astra`

Our most capable model for the most demanding work.

Compare | Try in Playground

REASONING	SPEED	PRICE	INPUT	OUTPUT
Highest	Fast	$10 · $50 Input · Output	Text, Image	Text

GPT-6 Astra is our most capable model for the most demanding work. Use it for complex reasoning, coding, computer use, research, and document creation. `reasoning.effort` supports `low`, `medium`, `high`, `xhigh`, and `max`.

Get started with GPT-6 Astra using the model guide.

- 1,050,000 context window
- 128,000 max output tokens
- Apr 30, 2026 knowledge cutoff
- Reasoning token support

Pricing

Pricing is based on the number of tokens used, or other metrics based on the model type. For tool-specific models, like search and computer use, there's a fee per tool call. See details in the pricing page.

Text tokens — Per 1M tokens

Input	Cached input	Cache writes	Output
$10.00	$1.00	$12.50	$50.00

https://developers.openai.com/api/docs/models/gpt-6-astra

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LLM model card

	Claude Fable 5.1	Claude Opus 5.5	Claude Sonnet 5.5	Claude Haiku 5.5
	For demanding reasoning and long-horizon agentic work	For long-running agentic coding and knowledge work	The best combination of speed and intelligence	For high-volume, latency-sensitive tasks such as classification, extraction, and routing
Comparative latency	Slower	Moderate	Fast	Fastest
Pricing	$10 / input MTok $50 / output MTok	$4 / input MTok $20 / output MTok	$2 / input MTok $10 / output MTok	From $0.10 / input MTok From $0.50 / output MTok
Claude API ID	`claude-fable-5-1`	`claude-opus-5-5`	`claude-sonnet-5-5`	`claude-haiku-5-5`
Capabilities				
Thinking	Adaptive (always on)	Adaptive (always on)	Adaptive	Adaptive
Default effort	`high`	`medium`	`high`	`medium`
Context window	1M tokens	1M tokens	1M tokens	1M tokens
Max output	128K tokens	128K tokens	128K tokens	128K tokens
Reliable knowledge cutoff	Jun 2026	Jun 2026	Jun 2026	Jun 2026
> Additional details				

https://platform.claude.com/docs/en/models/overview



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Token Cost ?

GPT-6 Astra `gpt-6-astra`

Our most capable model for the most demanding work.

Compare | Try in Playground

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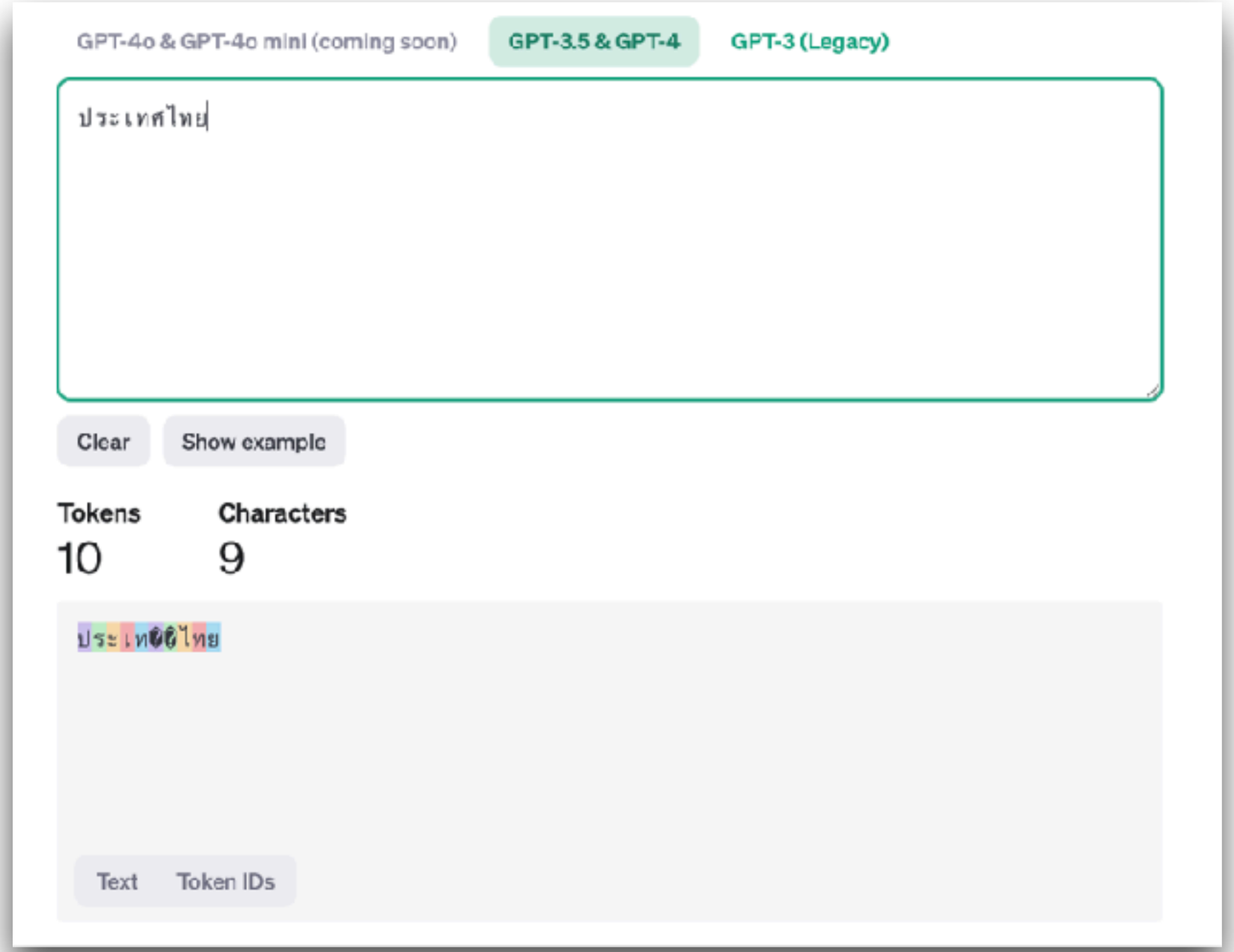
Text tokens — Per 1M tokens

Input	Cached input	Cache writes	Output
$10.00	$1.00	$12.50	$50.00

https://developers.openai.com/api/docs/models/gpt-6-astra

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OpenAI Tokenizer



https://platform.openai.com/tokenizer

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Context window ?

GPT-6 Astra `gpt-6-astra`

Our most capable model for the most demanding work.

Compare | Try in Playground

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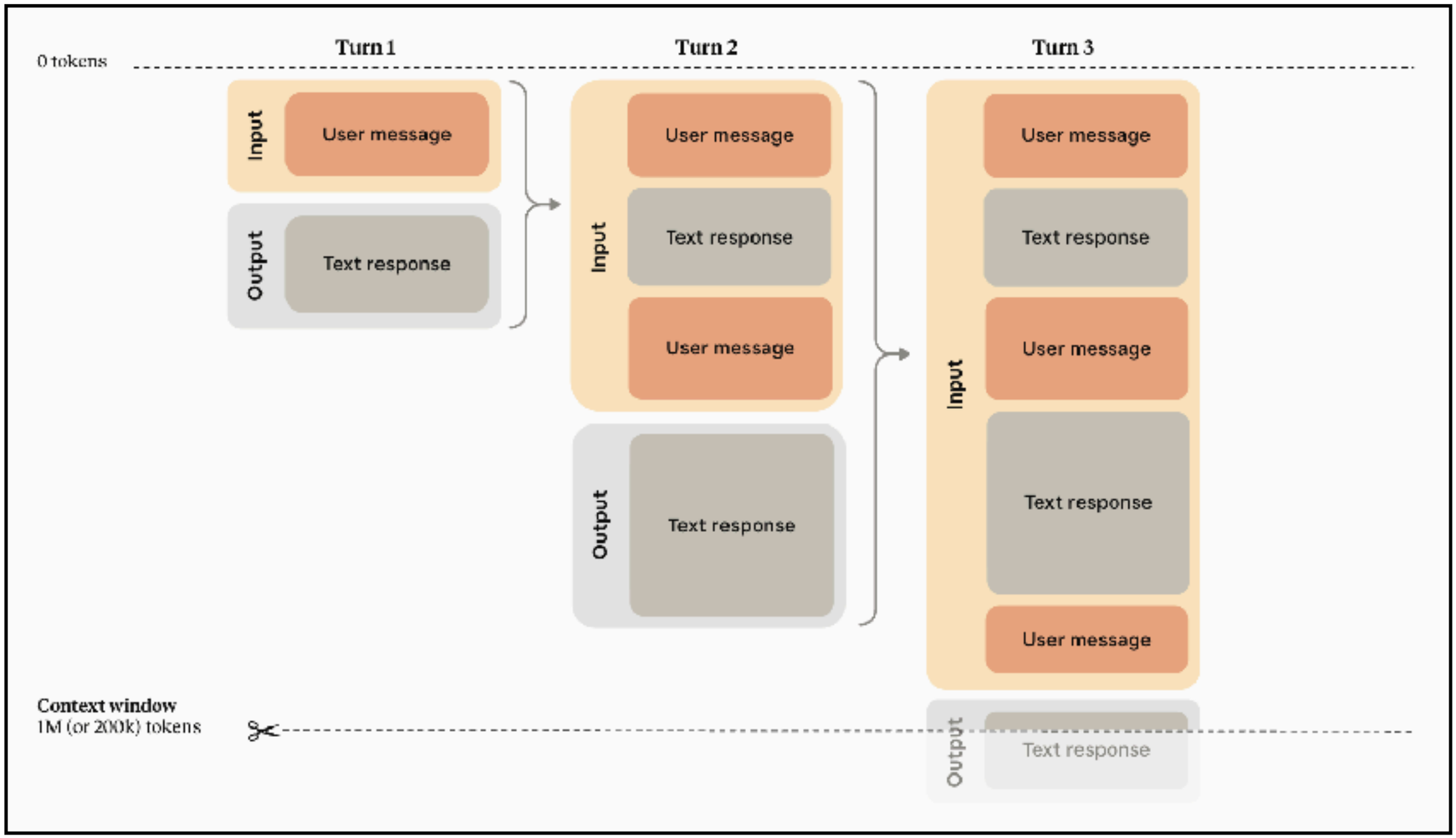
Text tokens — Per 1M tokens

Input	Cached input	Cache writes	Output
$10.00	$1.00	$12.50	$50.00

https://developers.openai.com/api/docs/models/gpt-6-astra

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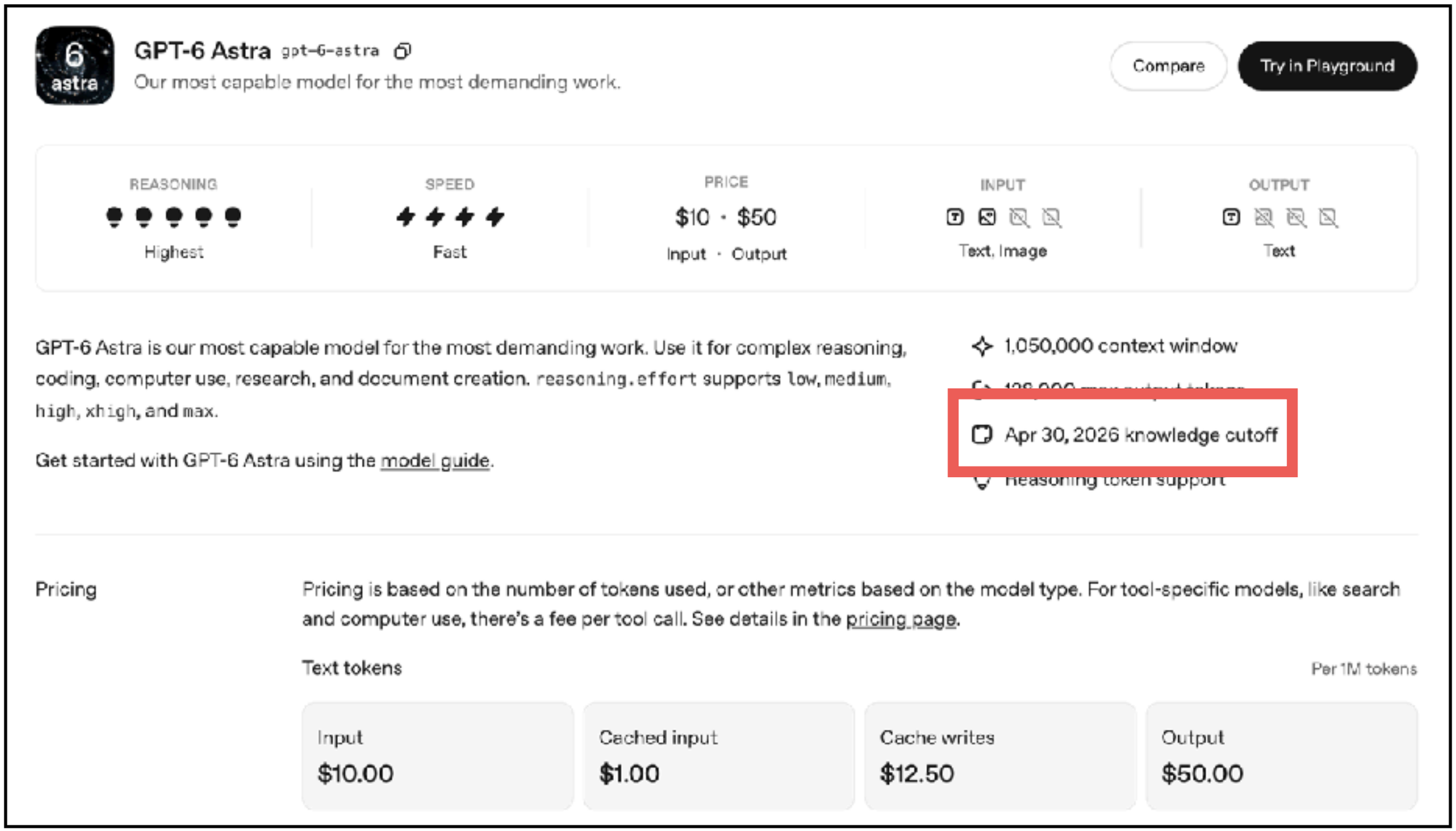
Context window ?



https://platform.claude.com/docs/en/build-with-claude/context-windows

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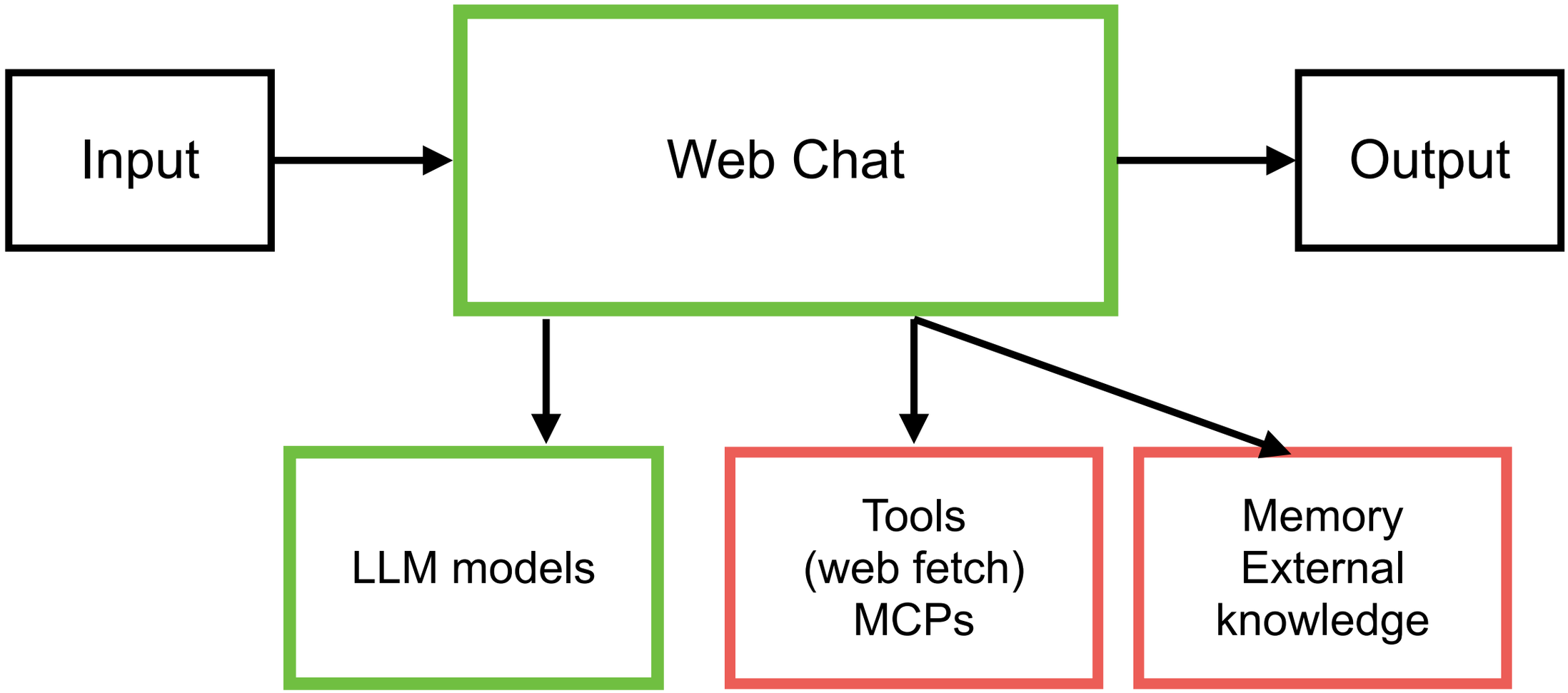
Knowledge cutoff ?



https://developers.openai.com/api/docs/models/gpt-6-astra

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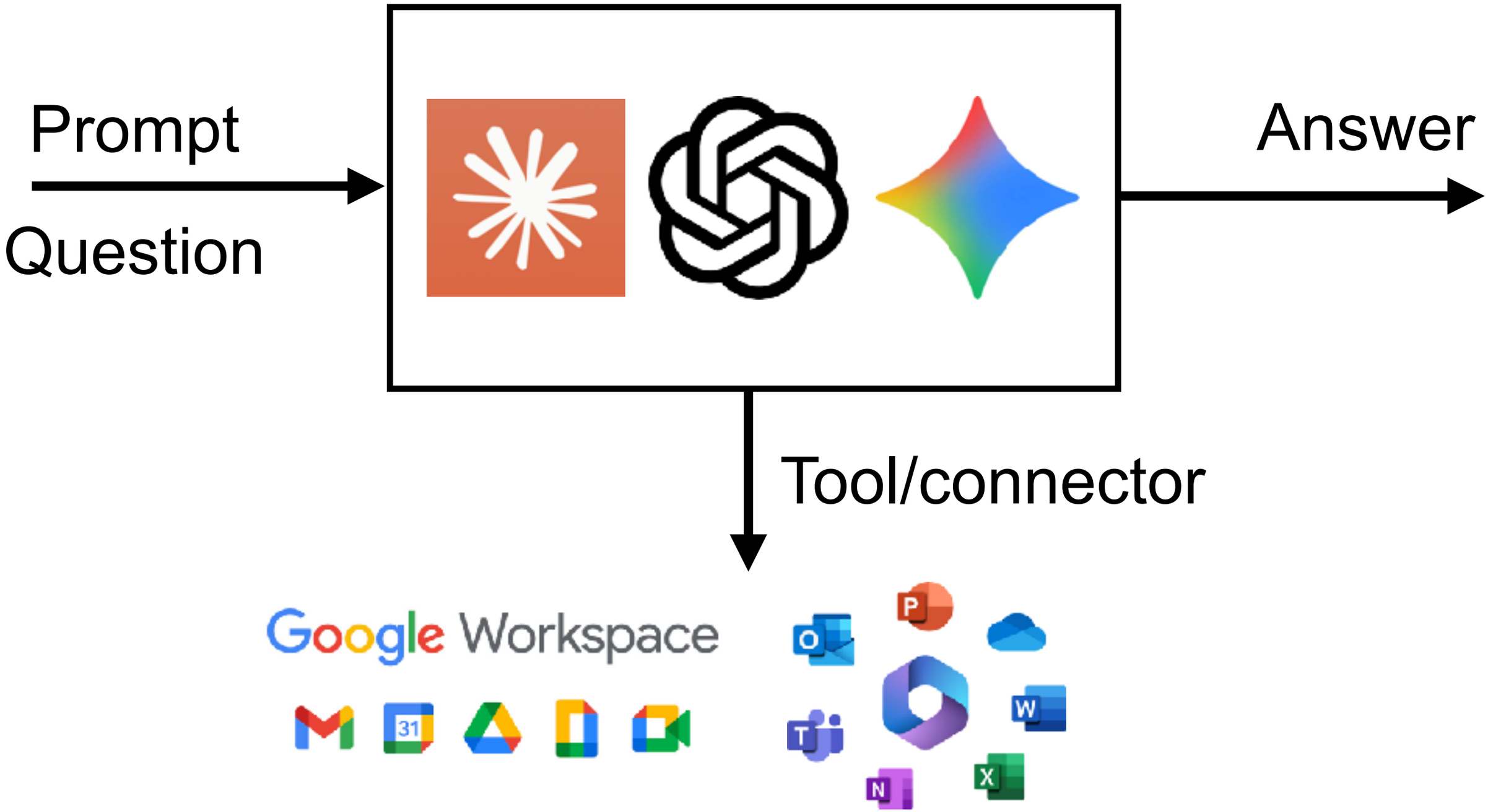
Chat-based



https://github.com/copilot

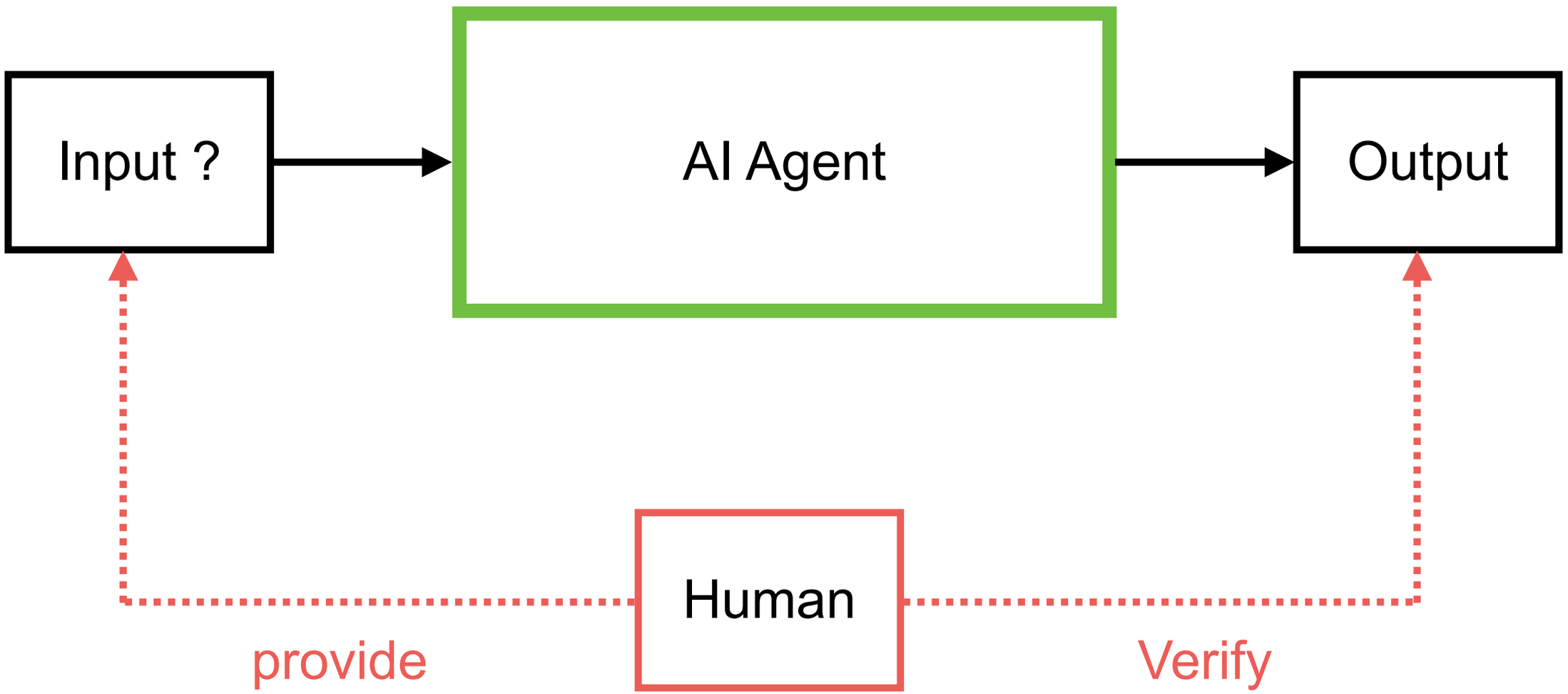
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Use tool to connect other app !!

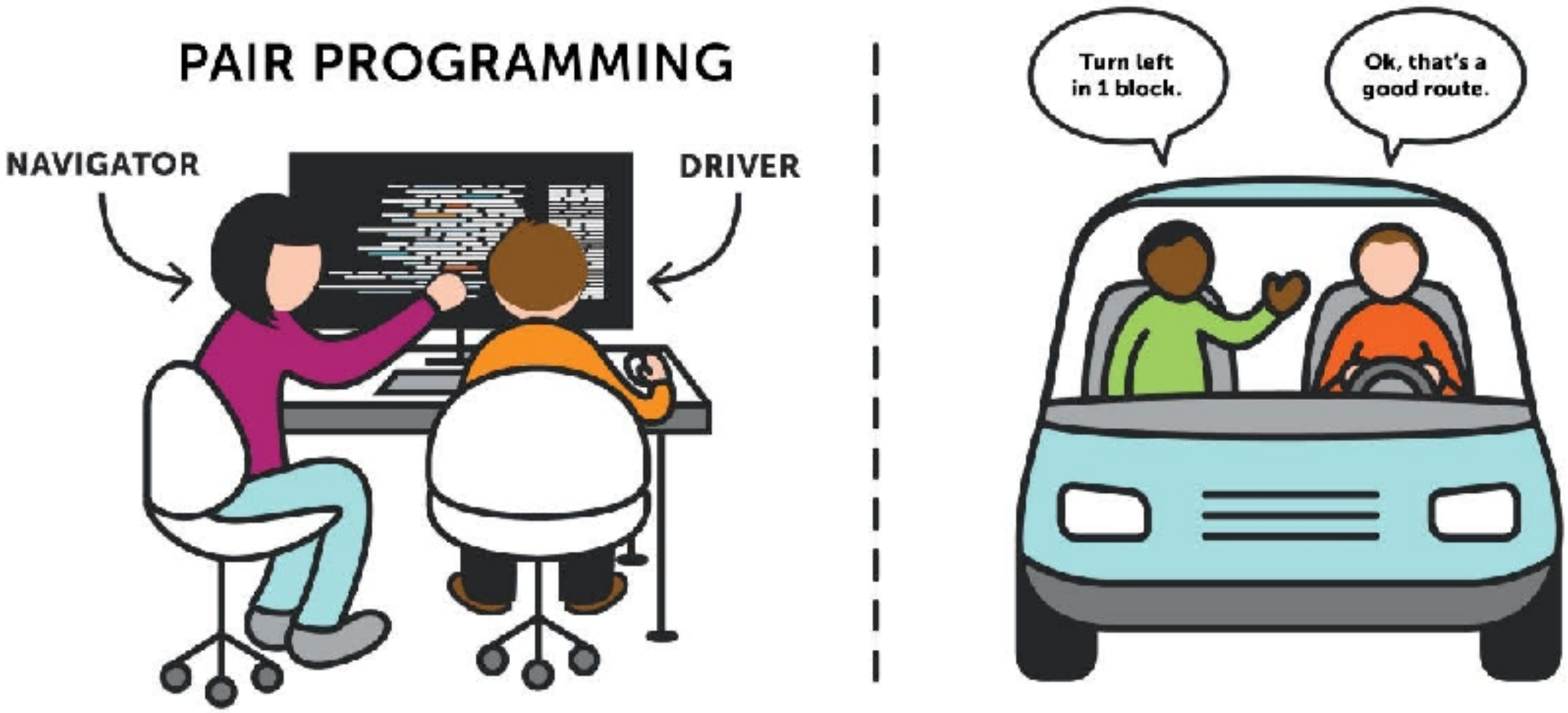


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Hello with Generative AI



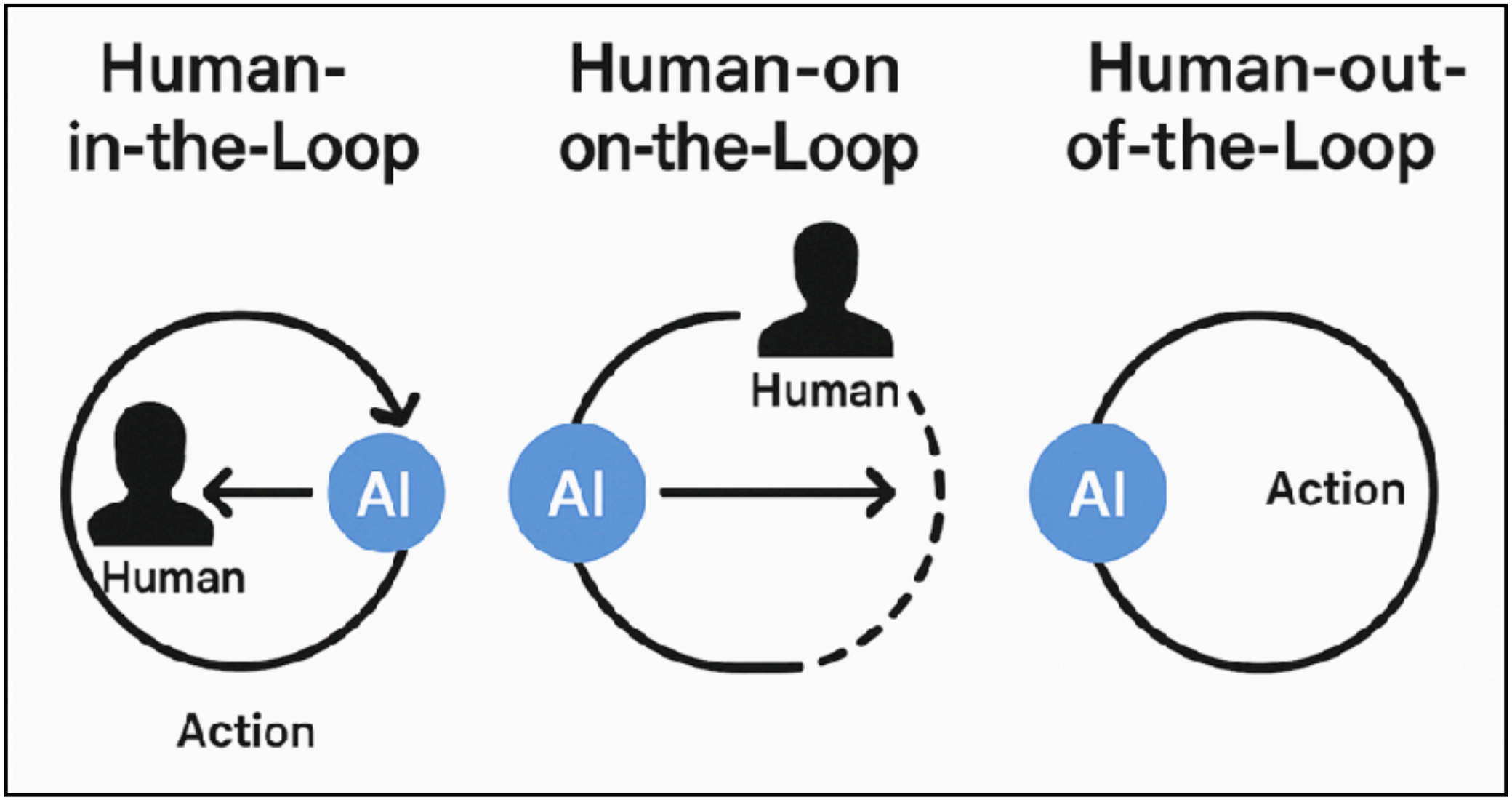
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Reviews, Tests and Understands
with experience

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AI Agent with/without Human



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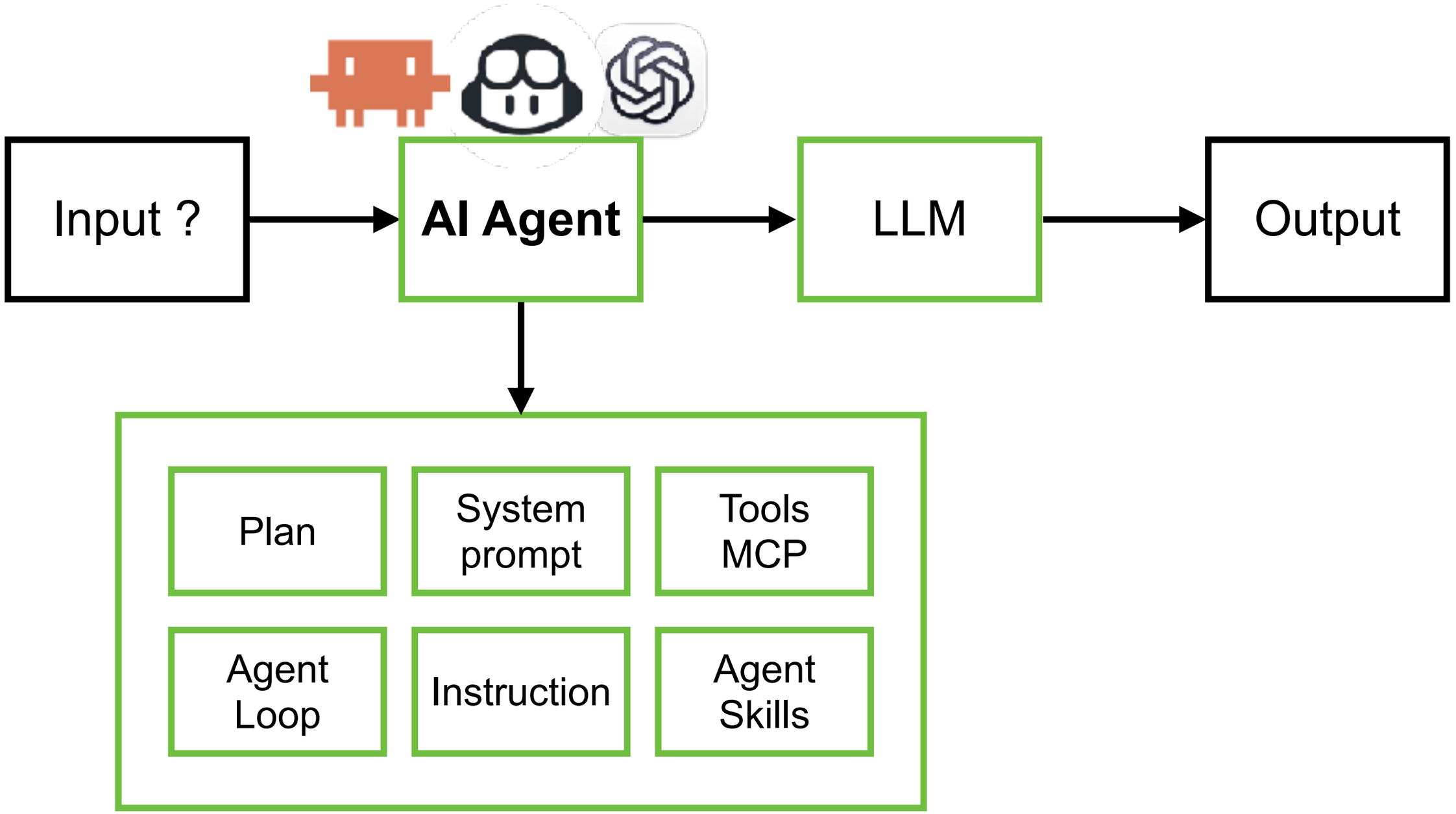
AI draft
Human craft

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Working with Copilot !!

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Working with AI Agent



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Github Copilot ?

Web

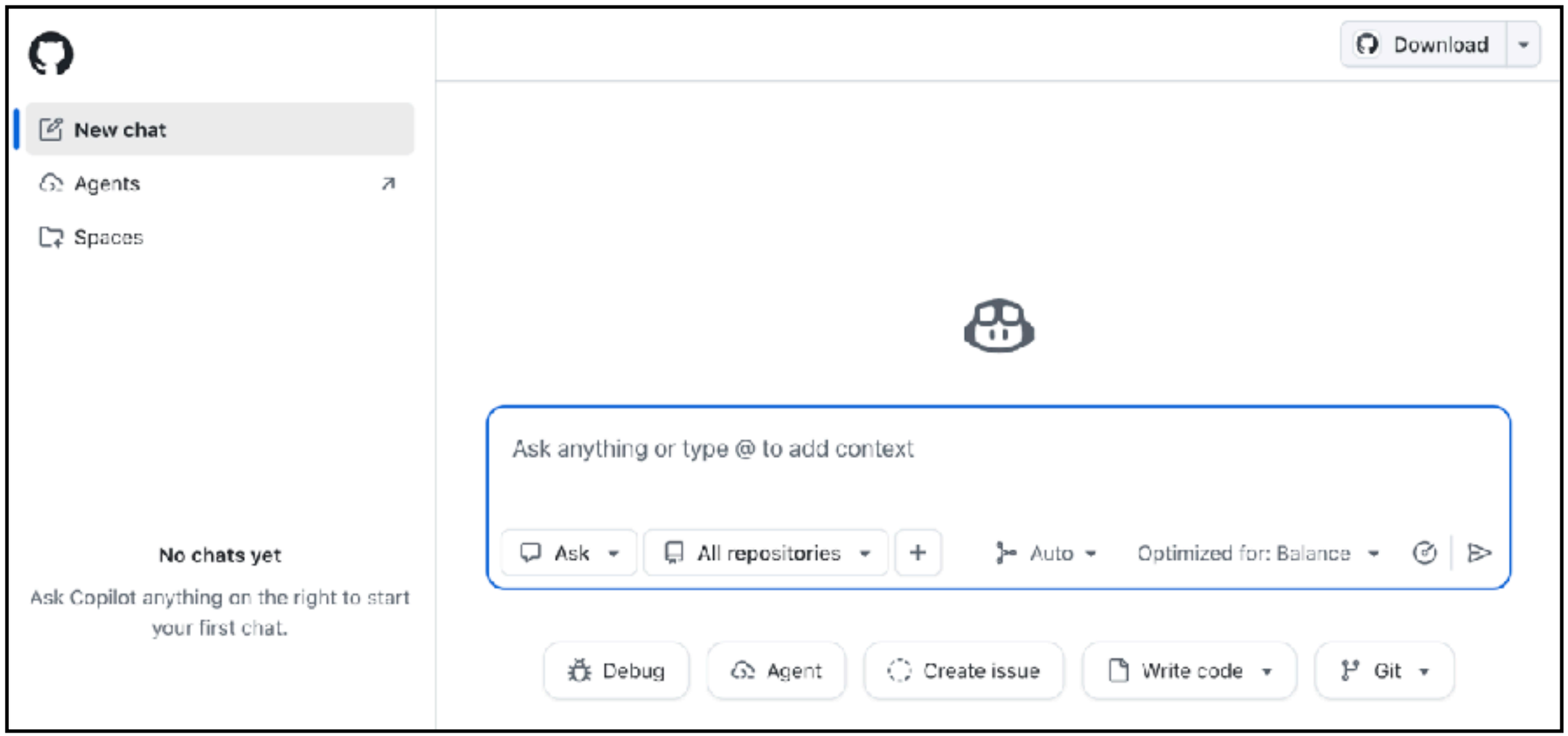
Copilot CLI

Copilot app

Copilot in VSCode

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Github Copilot



https://github.com/copilot

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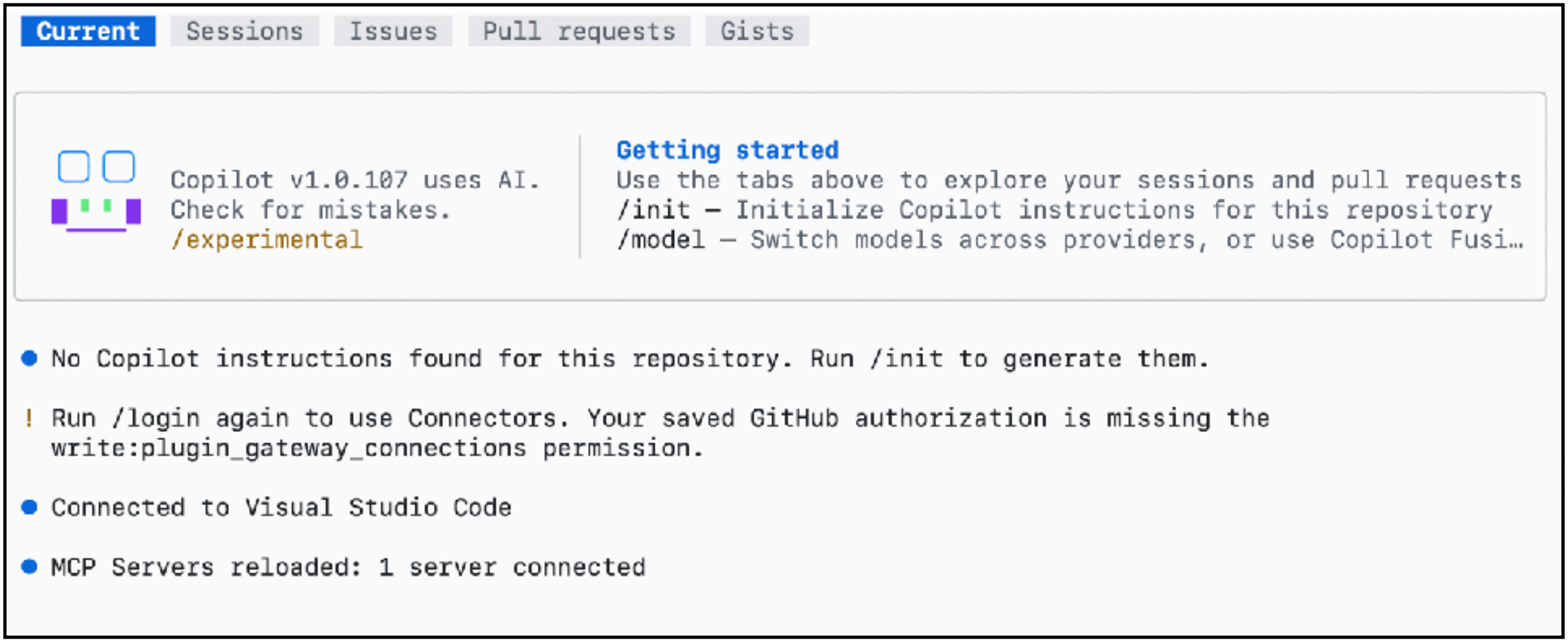
Microsoft Copilot



https://copilot.cloud.microsoft/chat

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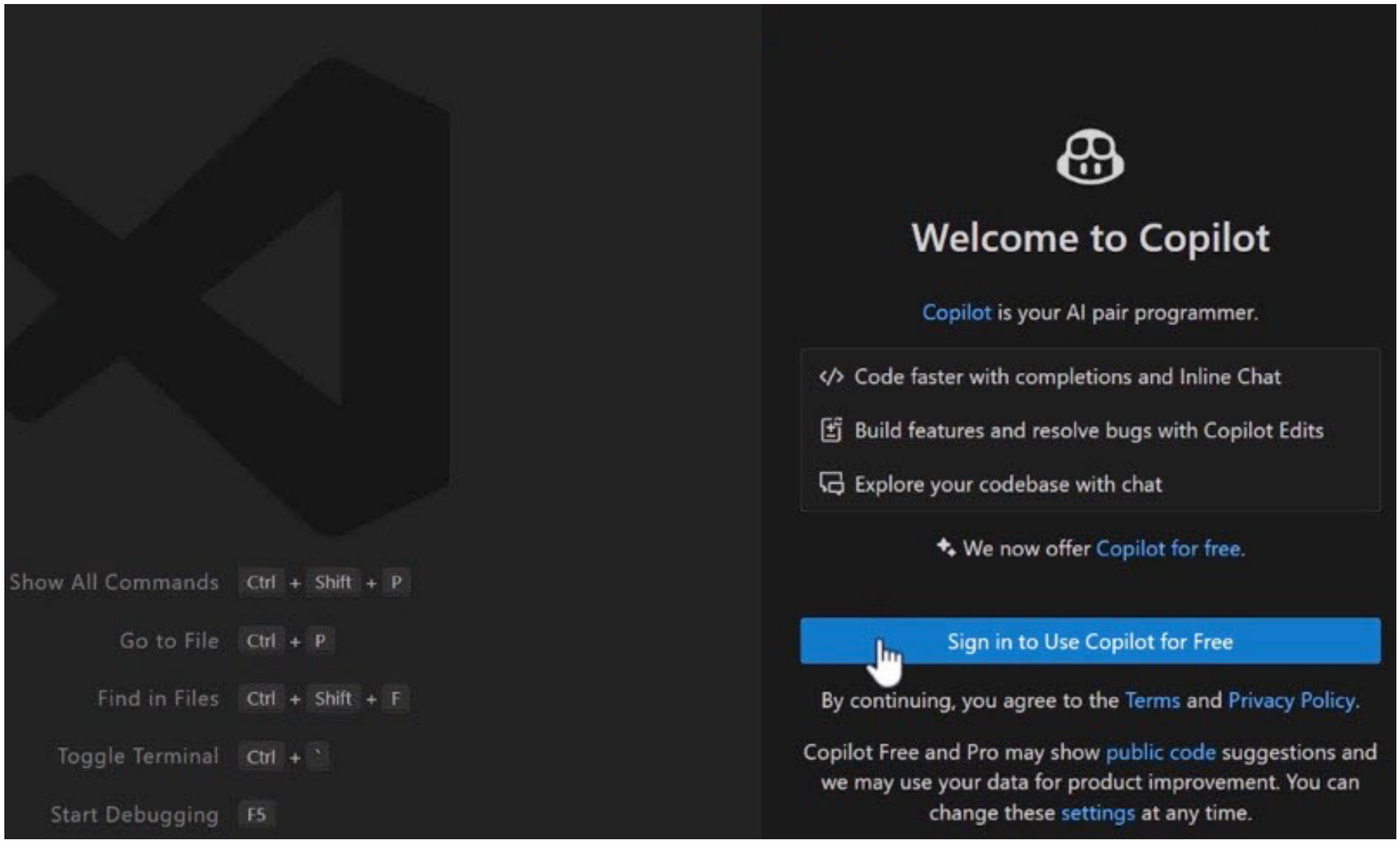
Github Copilot CLI



https://docs.github.com/en/copilot/how-tos/copilot-cli

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Github Copilot in VSCode



https://code.visualstudio.com/docs/setup/copilot

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Let’s start



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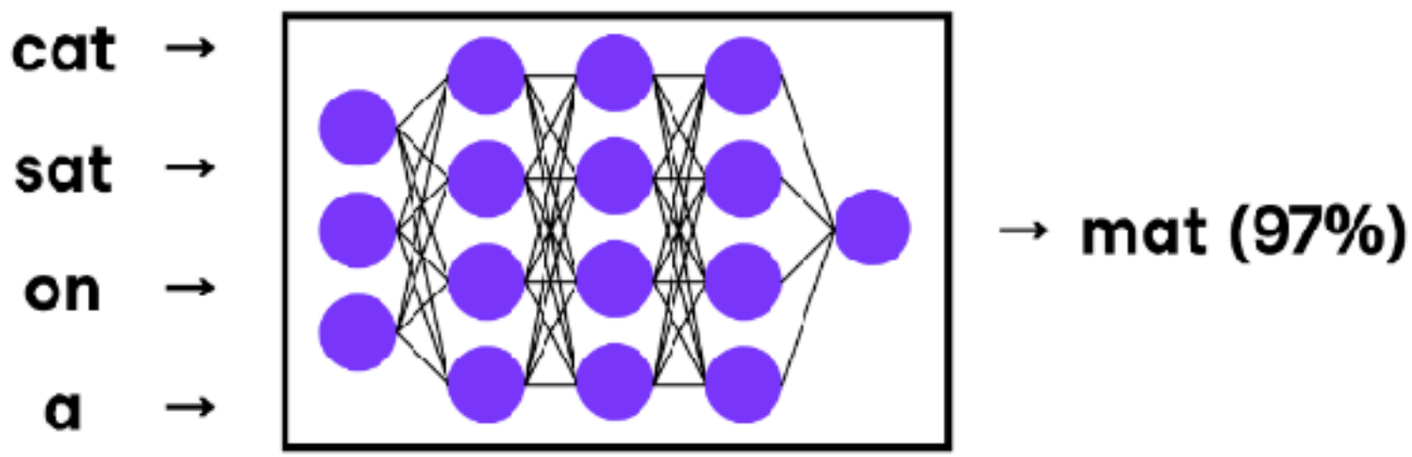
สีเหลือง

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Large Language Model (LLM)

Neural network

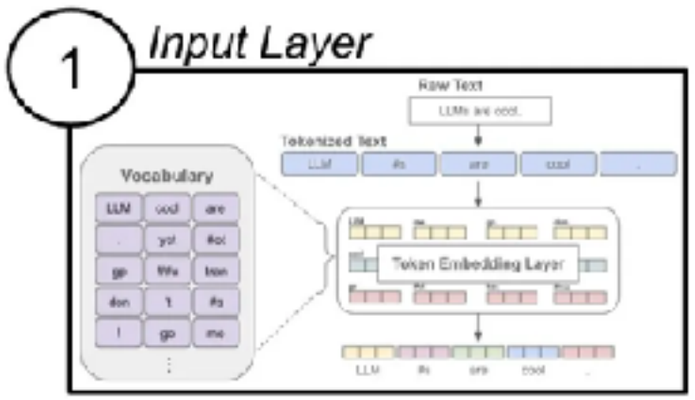
Predicts the next word in a sequence

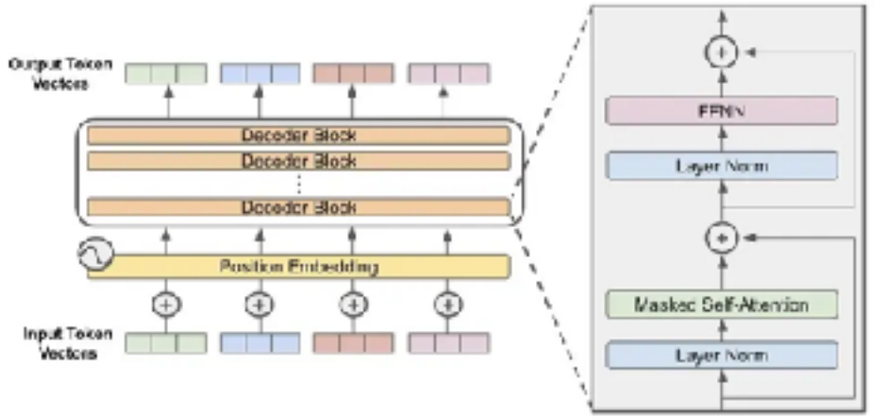


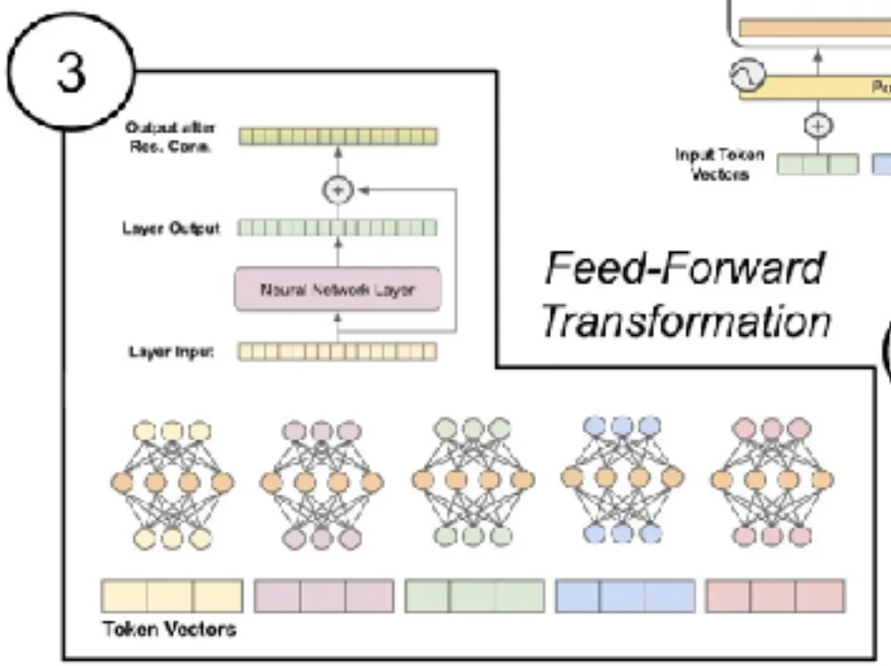
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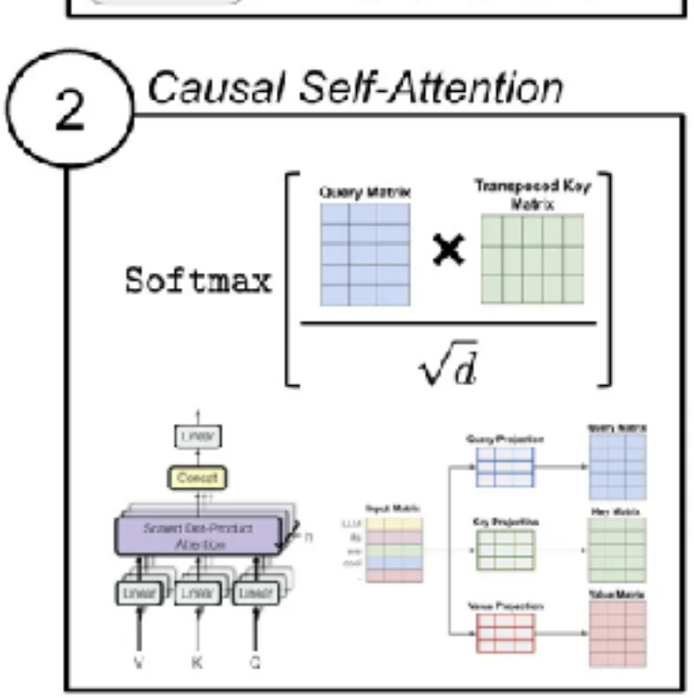
LLM components !!

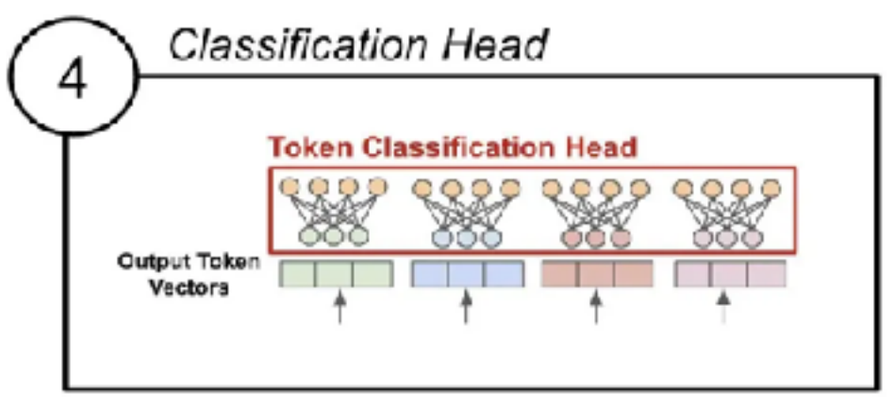
Components of the Decoder-only Transformer









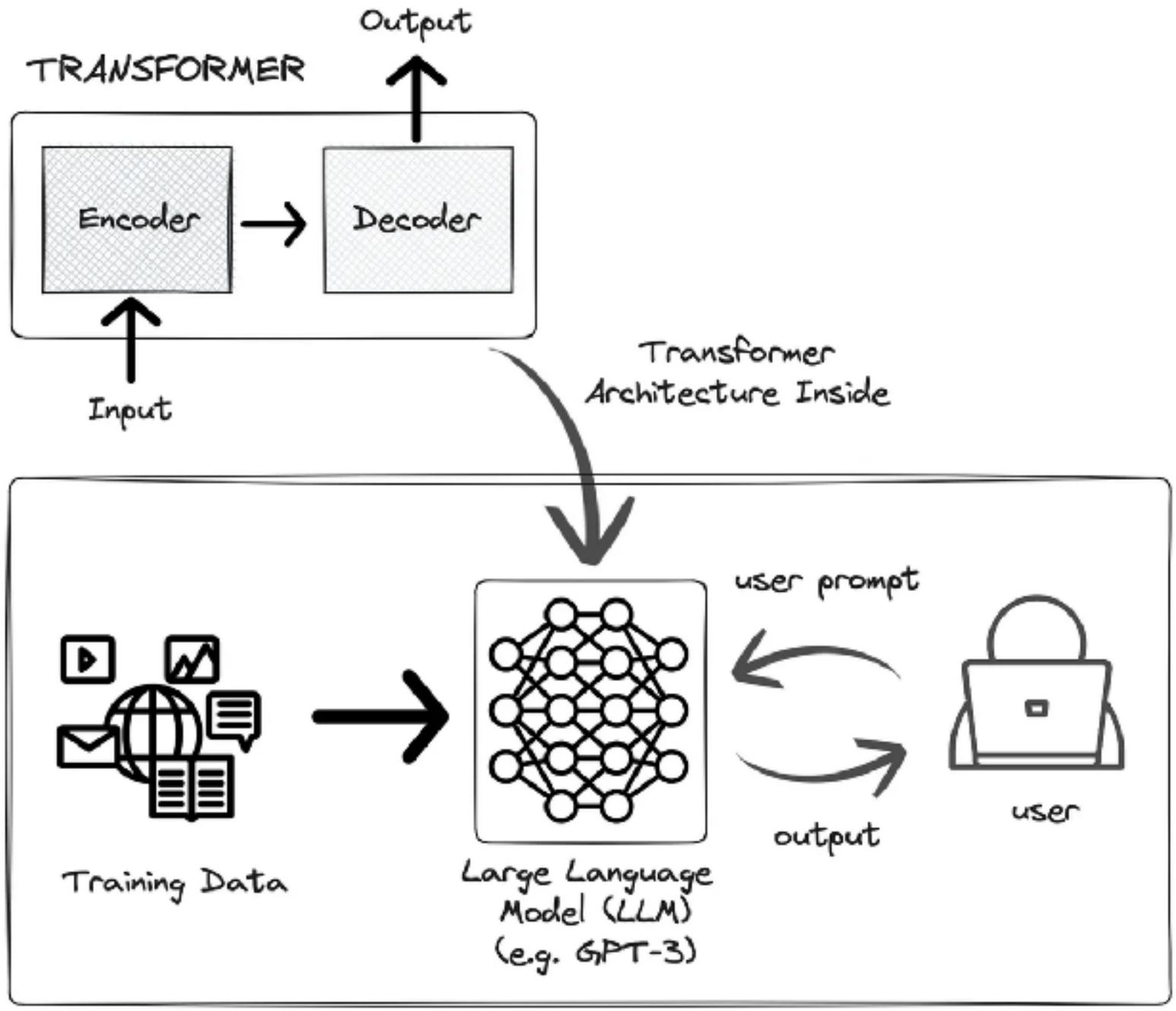




https://stackoverflow.blog/2024/08/22/llms-evolve-quickly-their-underlying-architecture-not-so-much

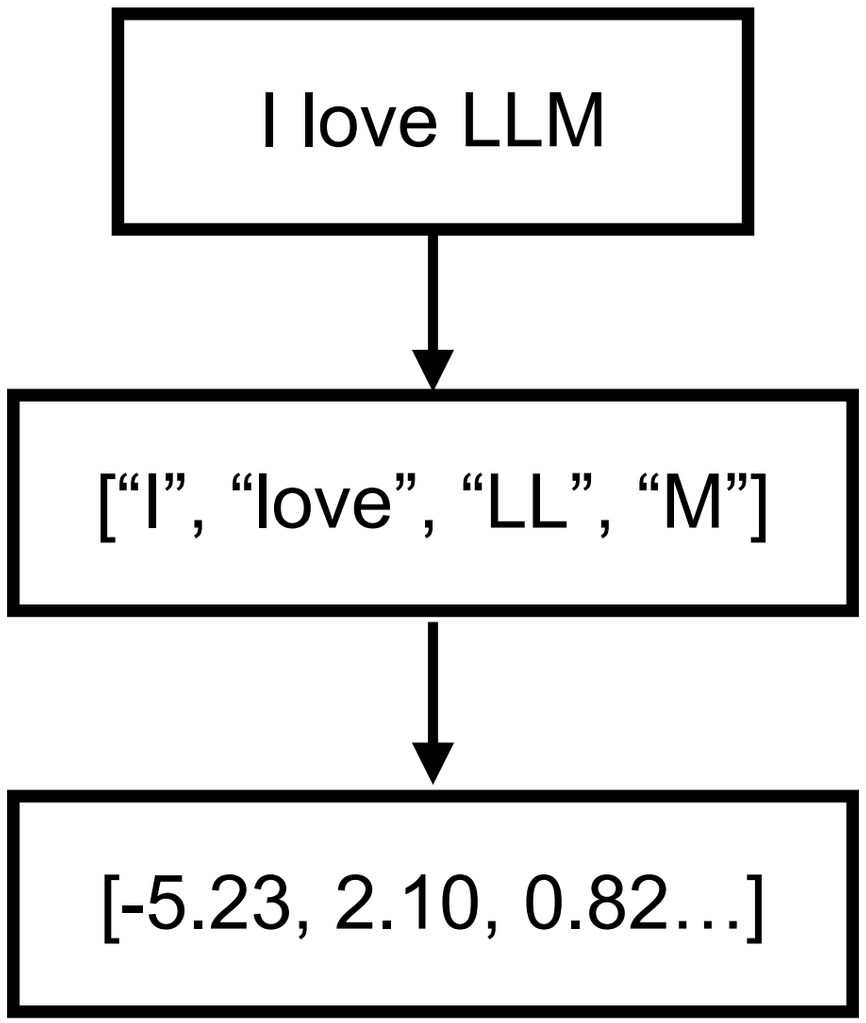
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Transformer inside



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Transformer process



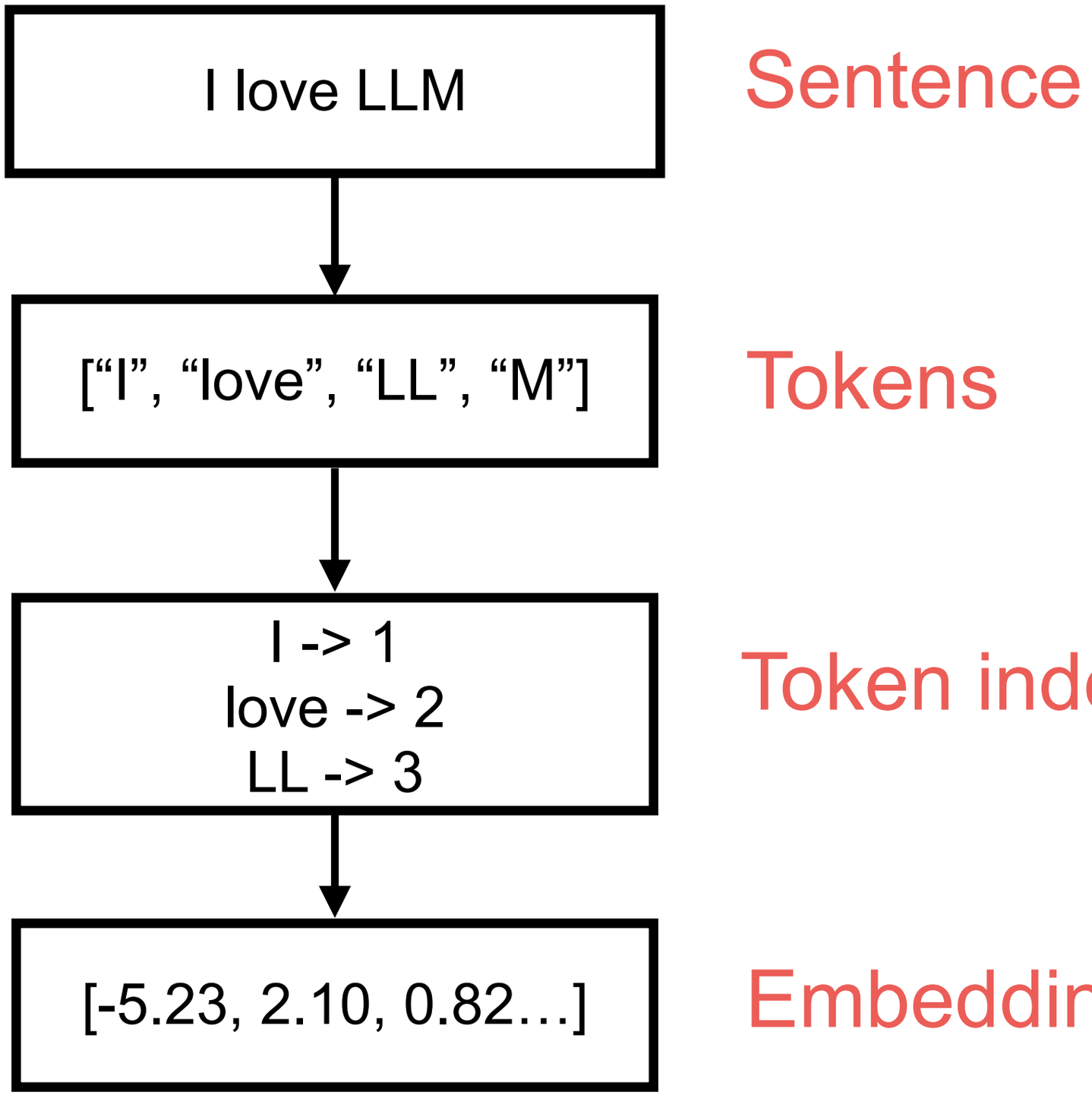
Sentence

Tokens

Embedding

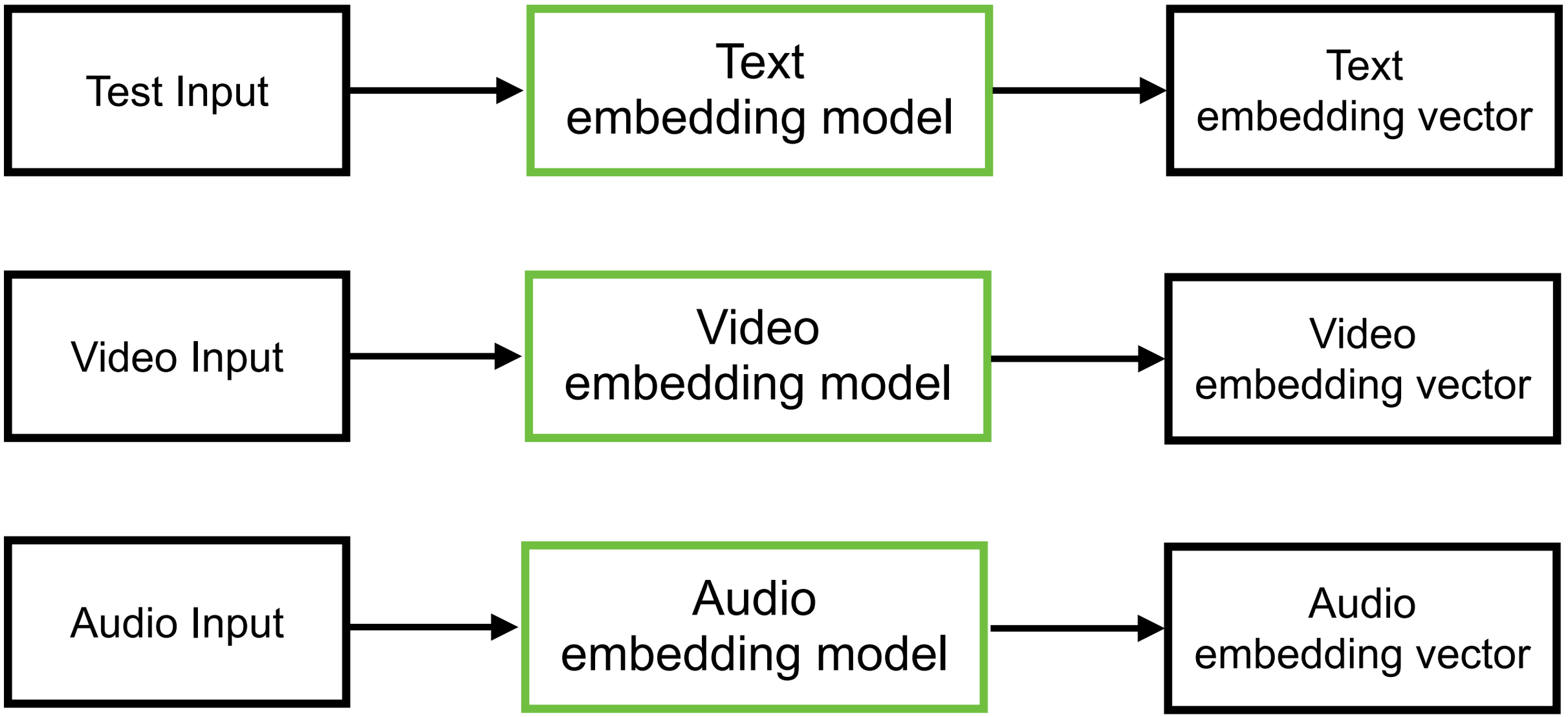
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Transformer process



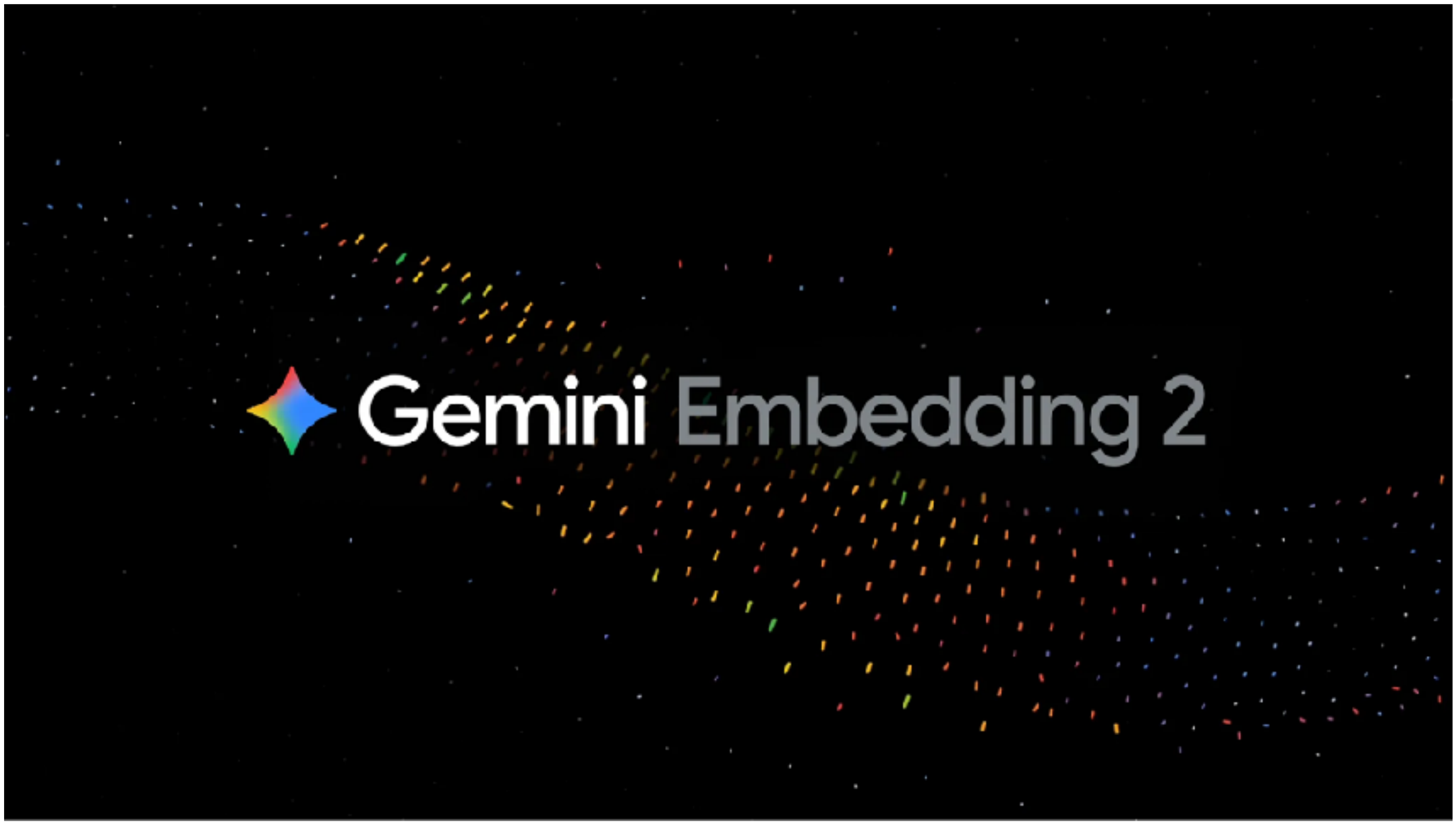
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Embedding



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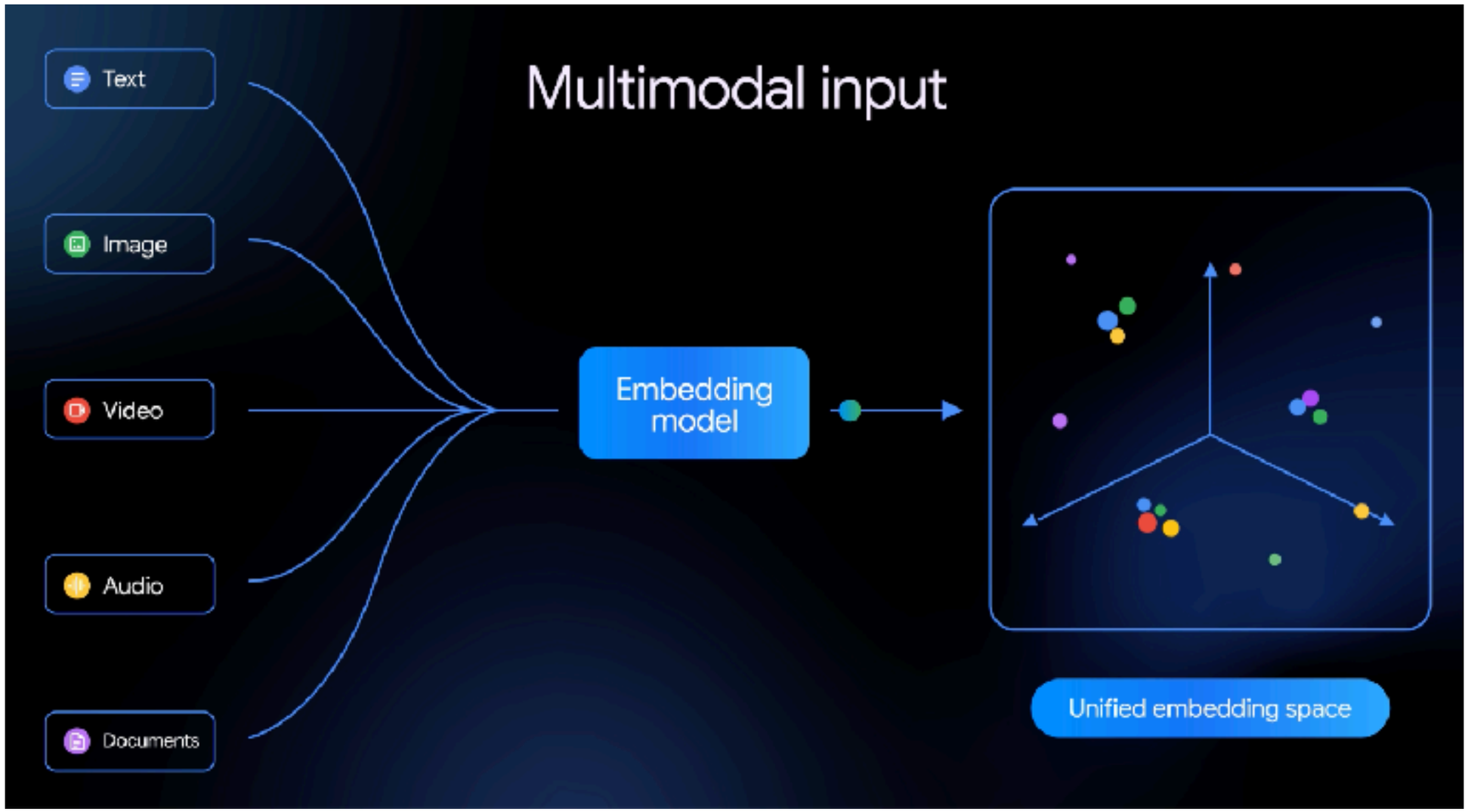
Multimodal Embedding Model



https://blog.google/innovation-and-ai/models-and-research/gemini-models/gemini-embedding-2/

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Multimodal Embedding Model

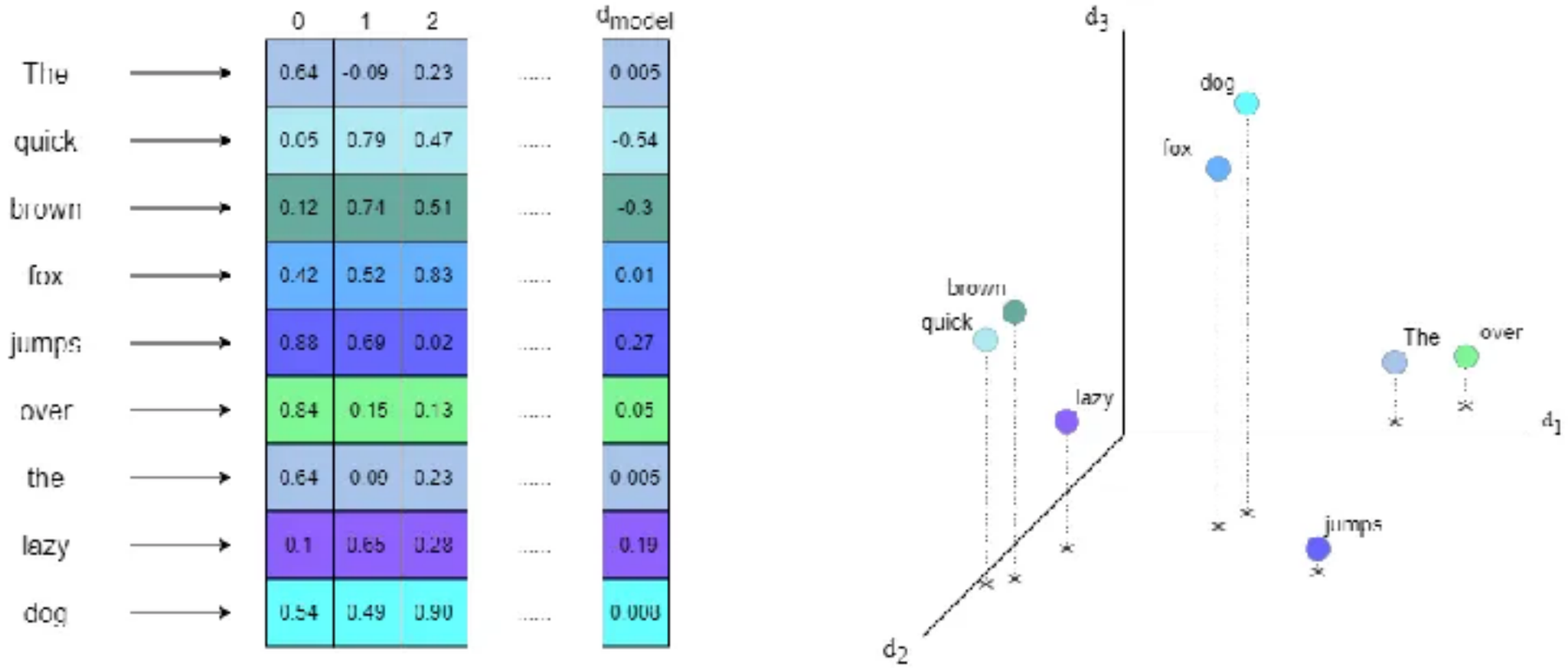


https://blog.google/innovation-and-ai/models-and-research/gemini-models/gemini-embedding-2/

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Embedding ?

Map items of **unstructured data** to high-dimensional real vectors



https://towardsdatascience.com/transformers-in-depth-part-1-introduction-to-transformer-models-in-5-minutes-ad25da6d3cca

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Embedding Leaderboard

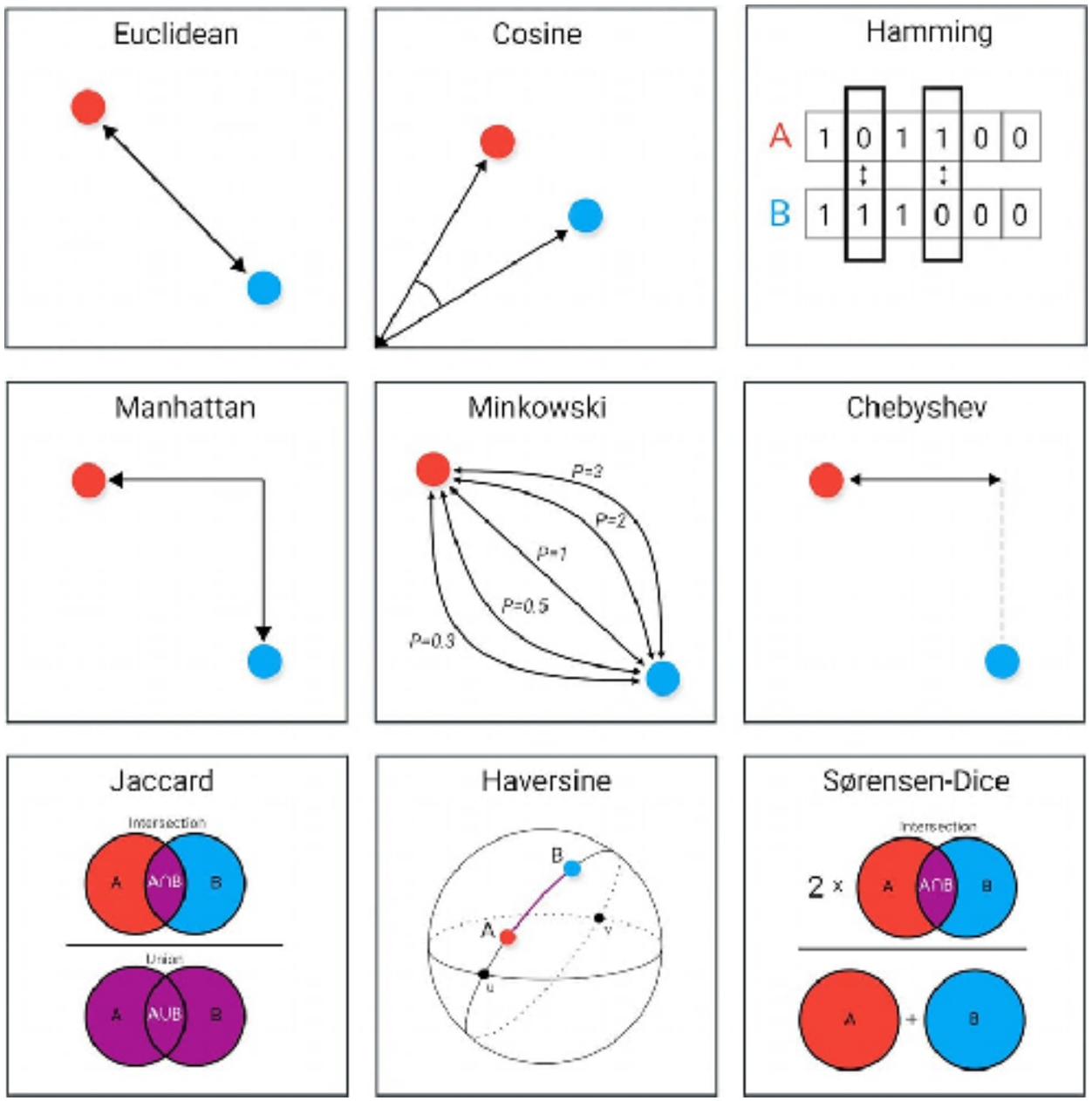
Rank (Bor…	Model	Embedding D…	Max Tokens	Mean (T…	Mean (TaskT…	Bitext …	Classification	Clustering	Instructi
1	KaLM-Embedding-Gemma3-12B-2511	3840	32768	**72.32**	**62.51**	**83.76**	**77.88**	55.77	5.49
2	llama-embed-nemotron-8b	4096	32768	69.46	61.09	81.72	73.21	54.35	10.82
3	Qwen3-Embedding-8B	4096	32768	70.58	61.69	80.89	74.00	57.65	10.06
4	gemini-embedding-001	3072	2048	68.37	59.59	79.28	71.82	54.59	5.18
5	Qwen3-Embedding-4B	2560	32768	69.45	60.86	79.36	72.33	57.15	11.56
6	Octen-Embedding-8B	4096	32768	67.84	60.28	80.35	66.68	55.68	8.98
7	F2LLM-v2-14B	5120	40960	68.74	59.45	77.15	73.60	**60.91**	0.62
8	F2LLM-v2-8B	4096	40960	68.09	58.99	75.96	71.93	60.62	0.85
9	Seed1.6-embedding-1215	2048	32768	70.26	61.34	78.68	76.75	56.78	-0.82
10	F2LLM-v2-4B	2560	40960	67.06	58.25	74.49	70.73	59.53	2.25
11	jina-embeddings-v5-text-small	1024	32768	67.00	58.90	69.71	71.32	53.41	1.35

https://huggingface.co/spaces/mteb/leaderboard



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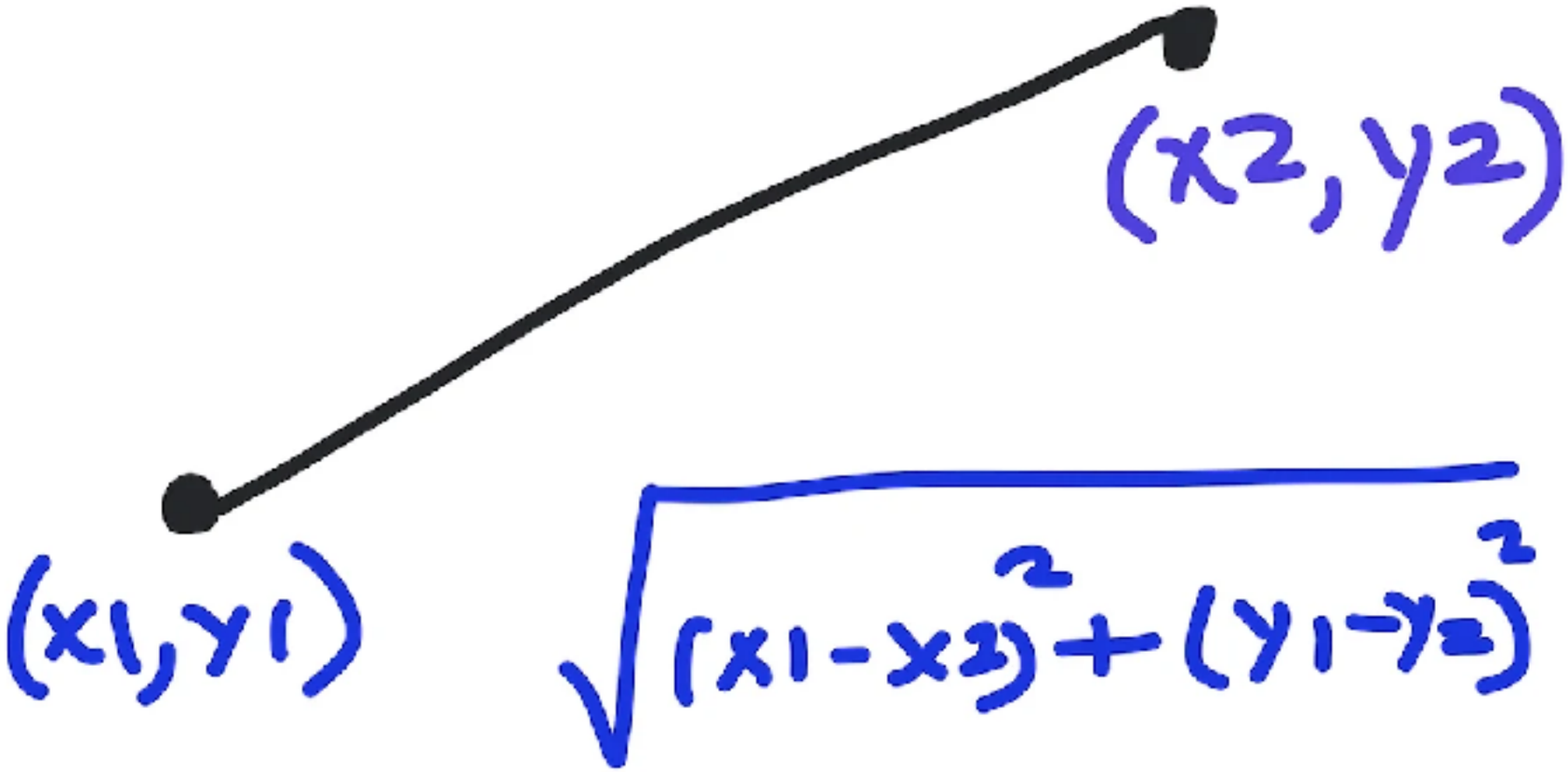
Distance measure in Data Science



https://www.maartengrootendorst.com/blog/distances/

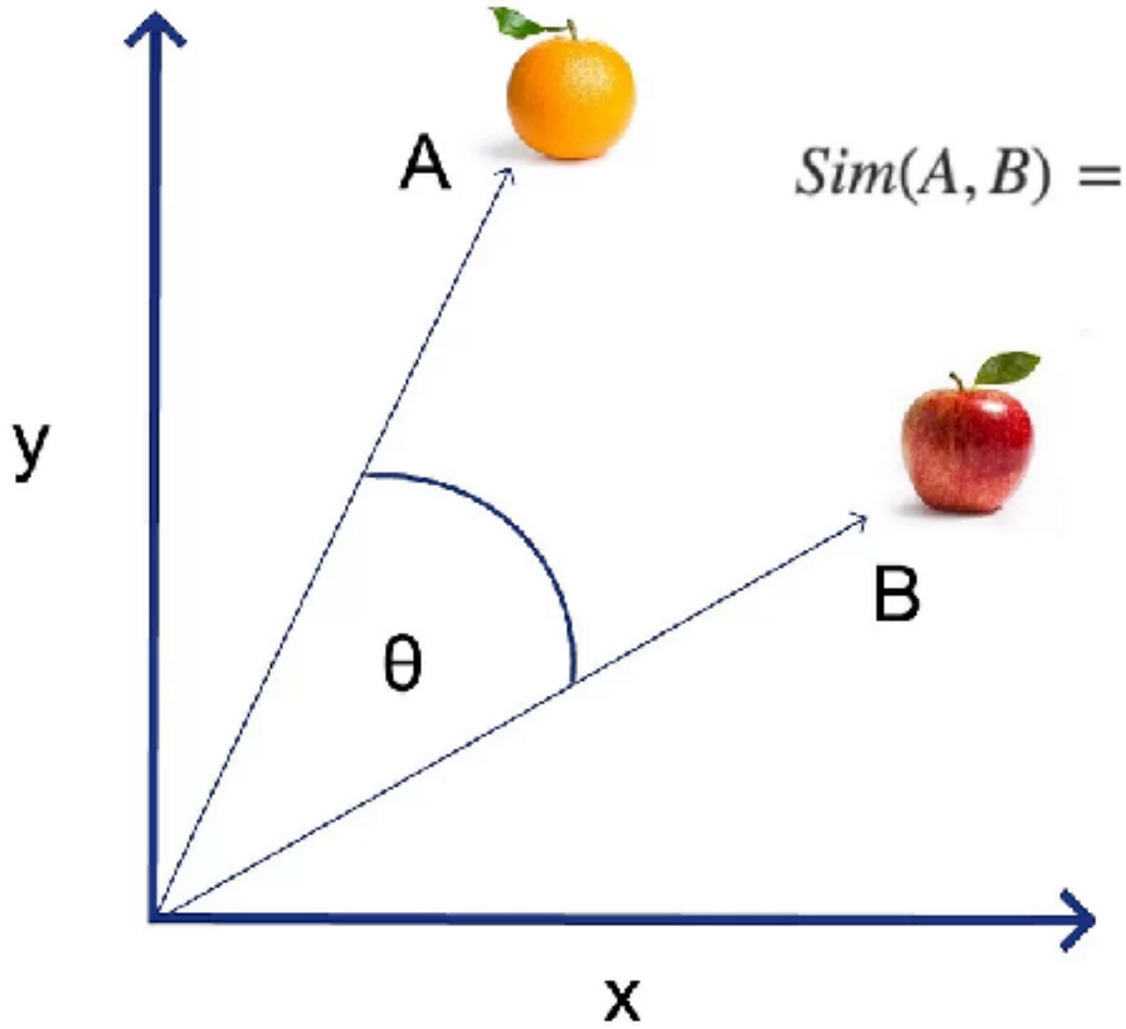
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Euclidean Distance



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Cosine Similarity

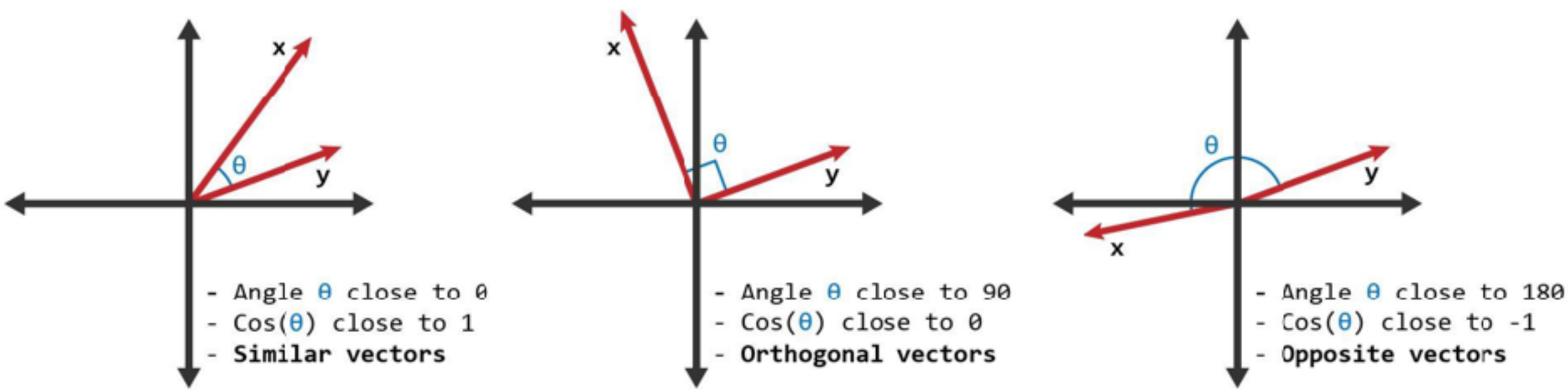


$$Sim(A, B) = \cos(\theta) = \frac{A \cdot B}{\|A\| \|B\|}$$

https://python.plainenglish.io/mastering-cosine-metrics-a-data-scientists-essential-toolkit-3ce95eda91f9

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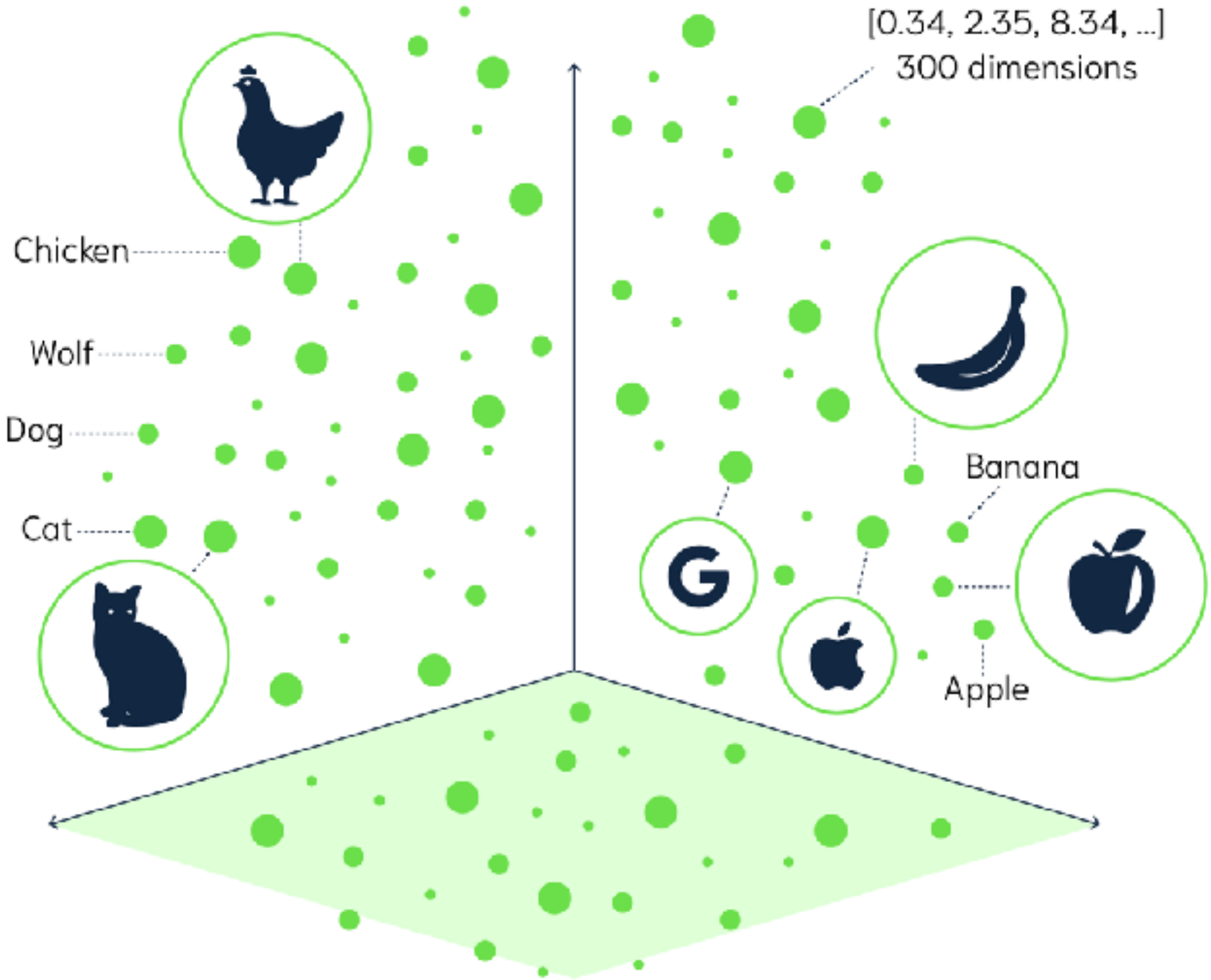
Cosign Similarity



https://www.learndatasci.com/glossary/cosine-similarity/

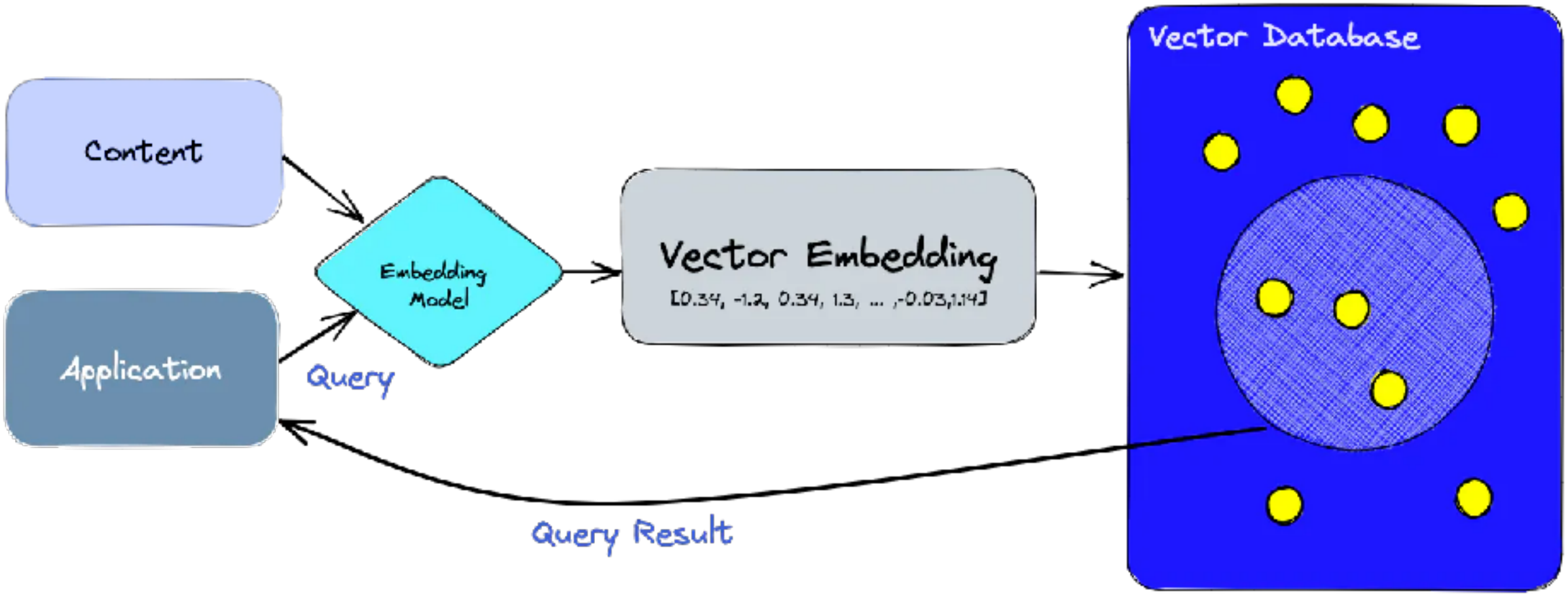
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Visual of Vector space



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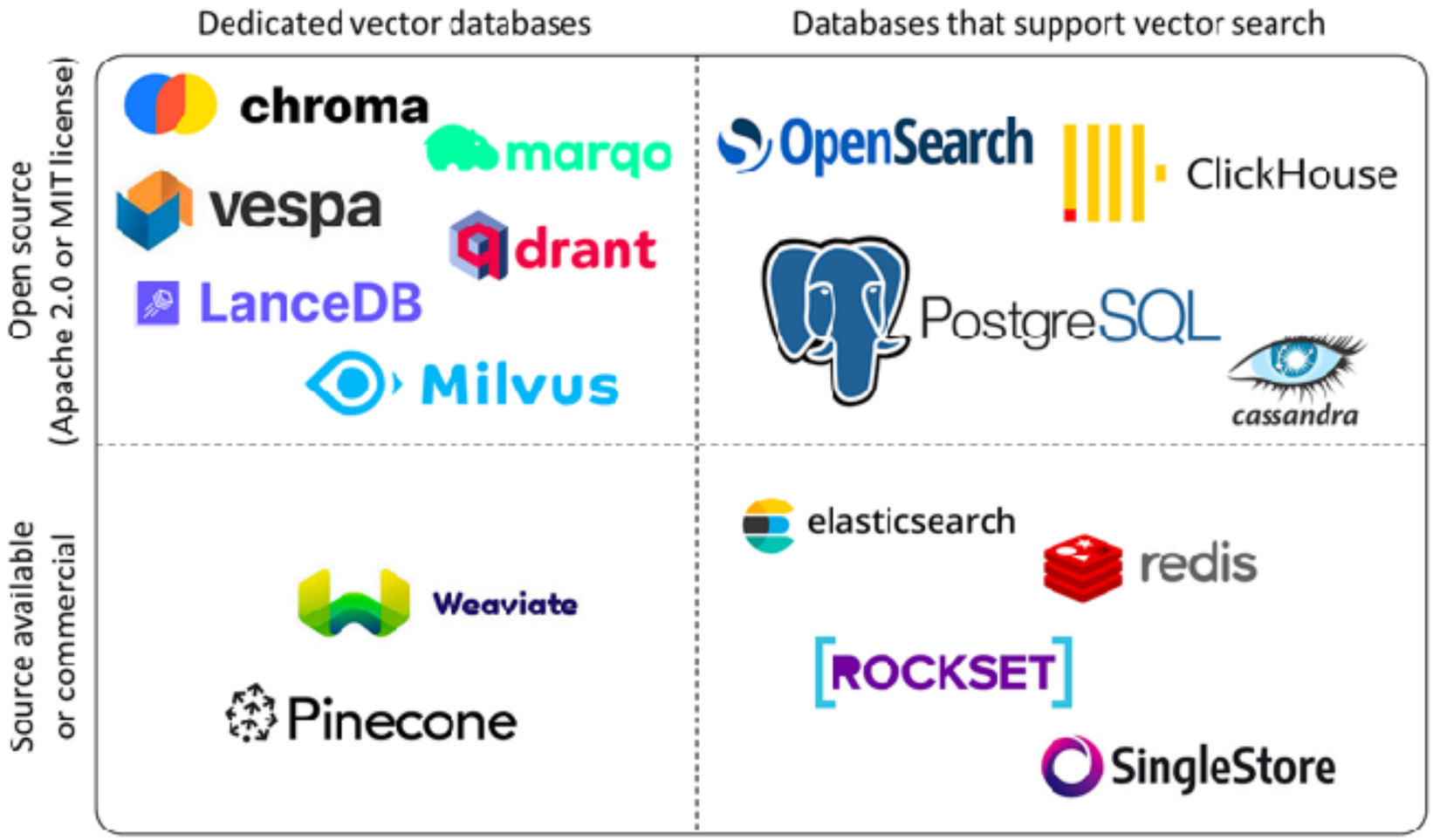
Store data in Vector Database



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Vector Database ?

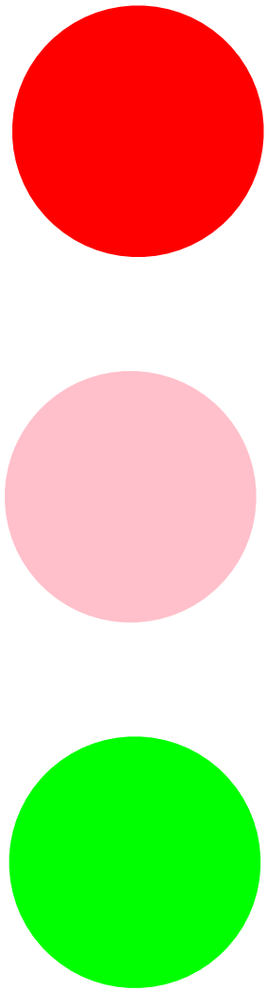
Map items of unstructured data to high-dimensional real **vectors**



https://towardsdatascience.com/transformers-in-depth-part-1-introduction-to-transformer-models-in-5-minutes-ad25da6d3cca

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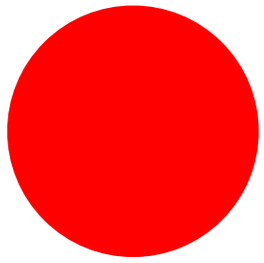
RGB ?



https://github.com/up1/workshop-ai-with-technical-team/tree/main/workshop/demo-rgb

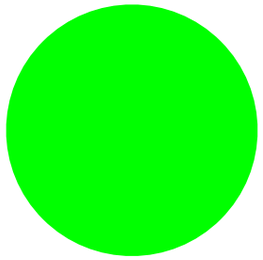
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RGB ?



[255, 0, 0]

[255, 192, 203]



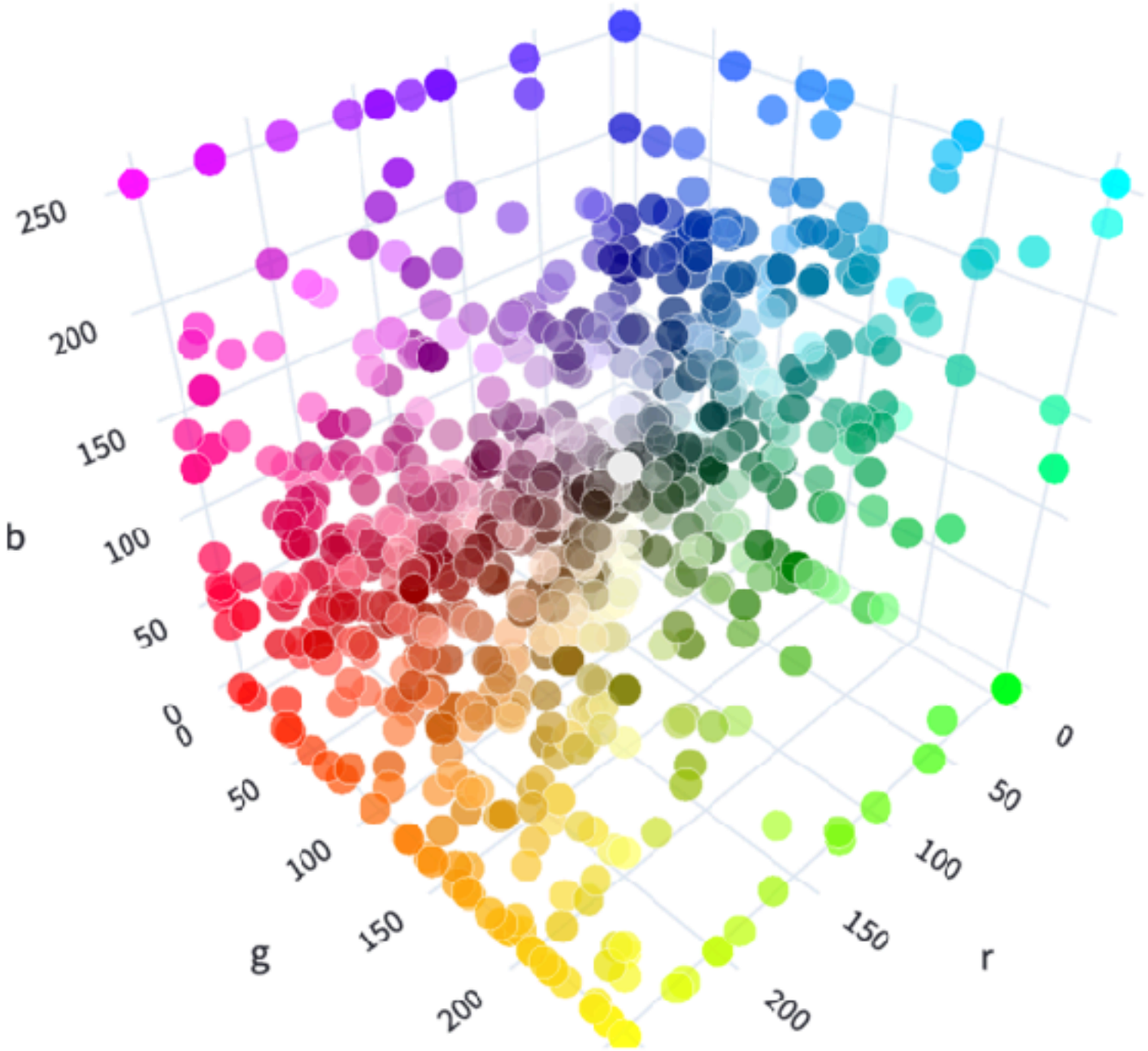
[0, 255, 0]

https://github.com/up1/workshop-ai-with-technical-team/tree/main/workshop/demo-rgb



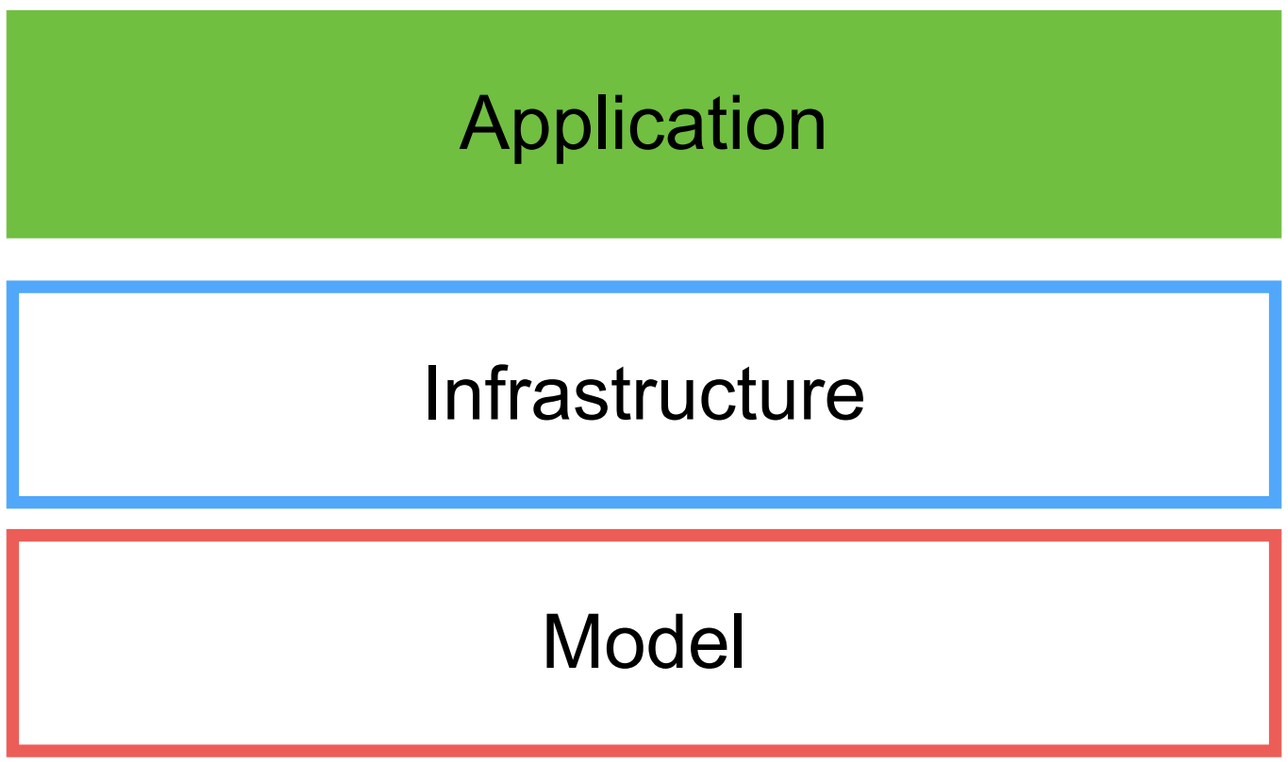
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Visualize RGB

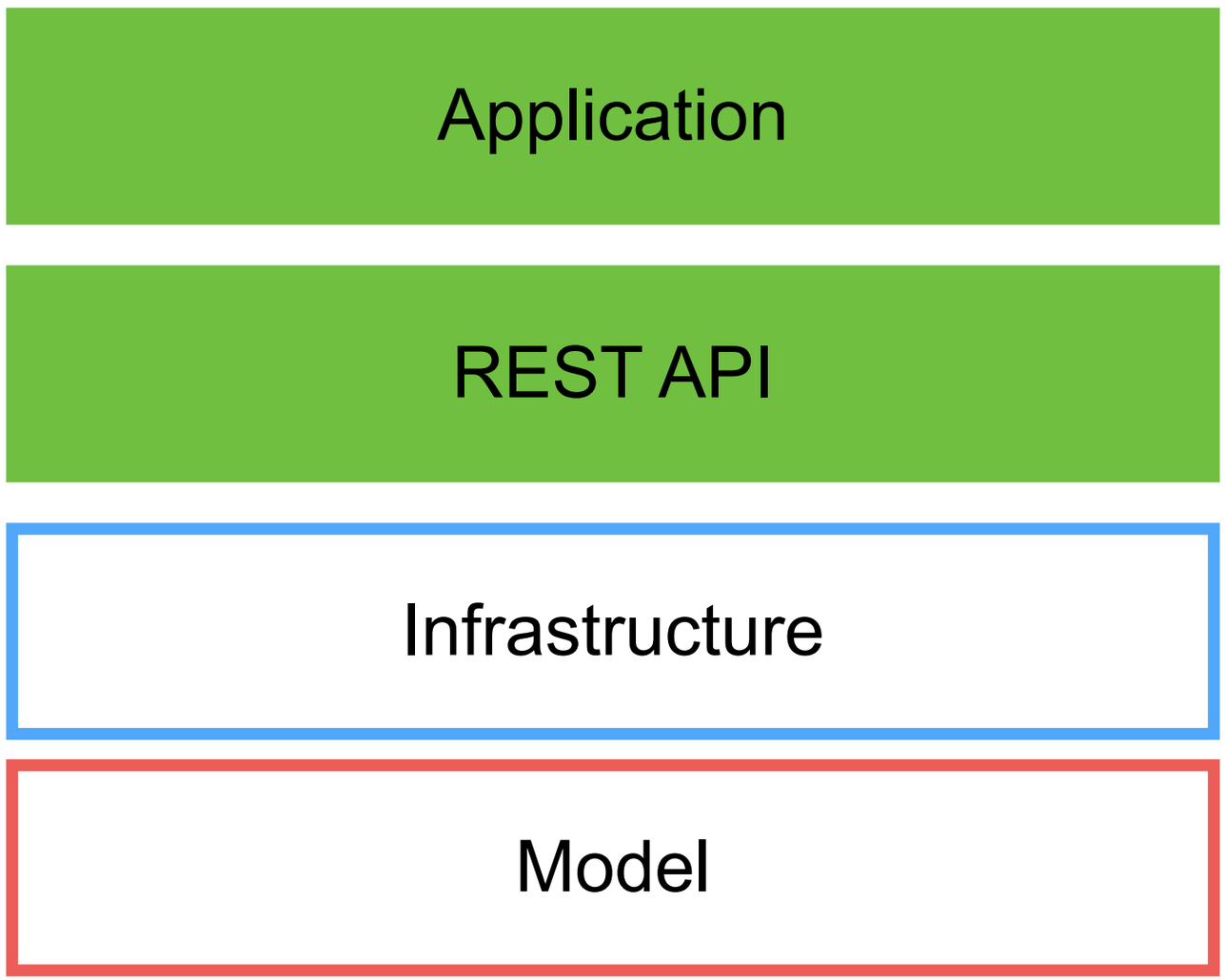


https://huggingface.co/spaces/jphwang/colorful_vectors

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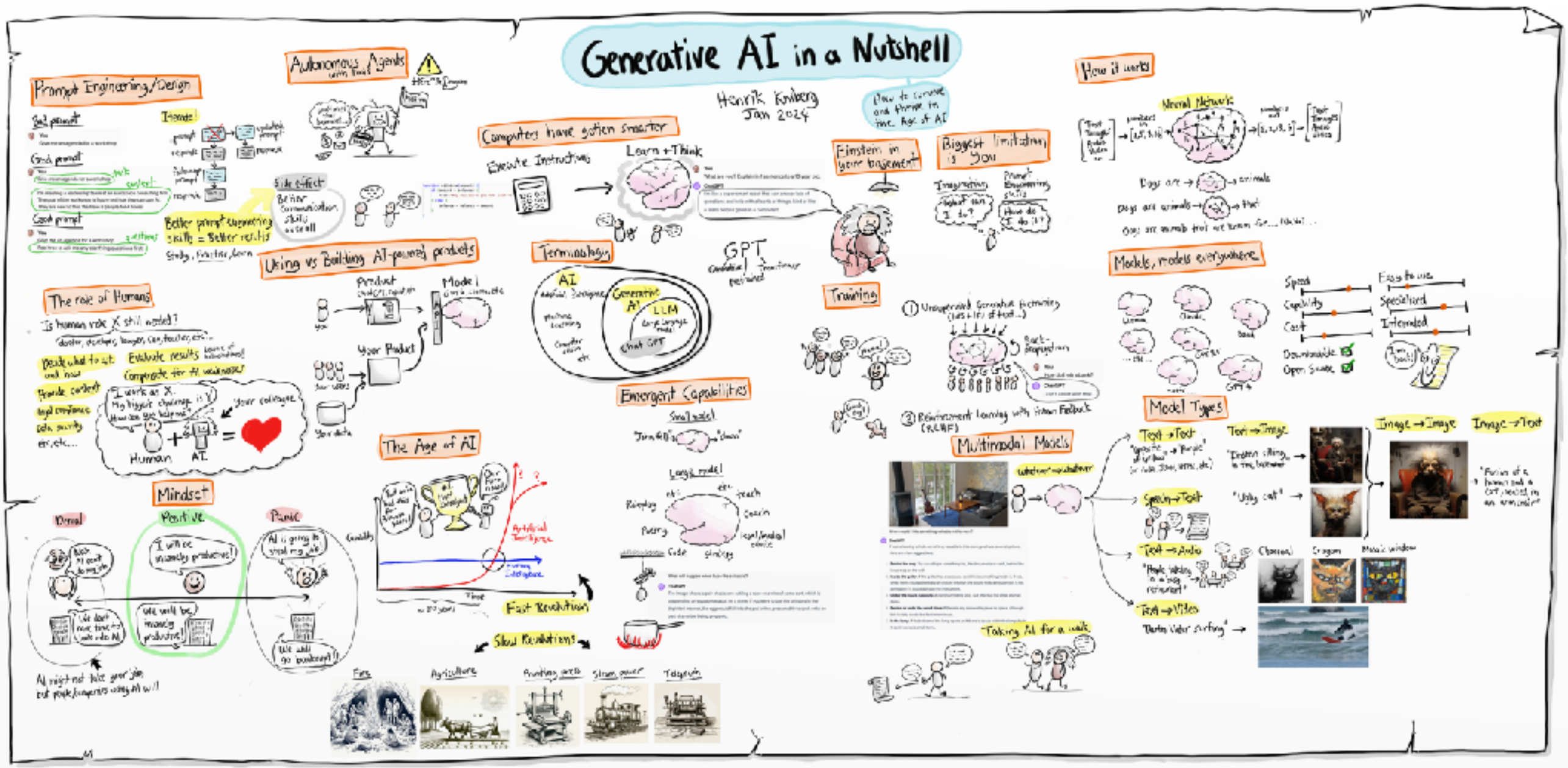


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Generative AI in Nutshell



https://www.youtube.com/watch?v=2IK3DFHRFfw

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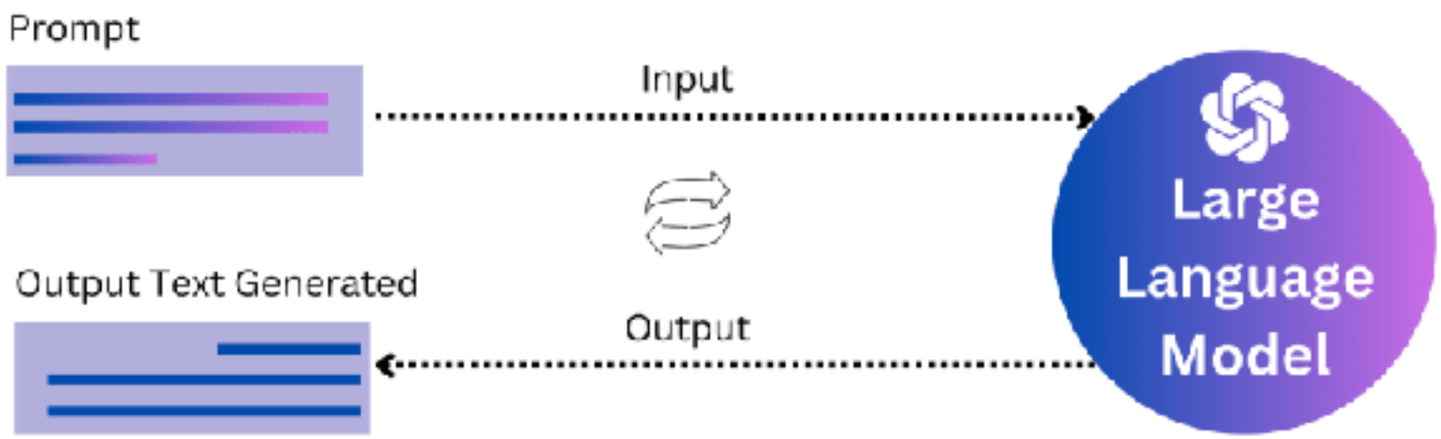
Prompt Engineering



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Prompt composition

Prompt engineering
Compose prompts from user inputs and context



https://platform.openai.com/docs/guides/prompt-engineering

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Better Prompt

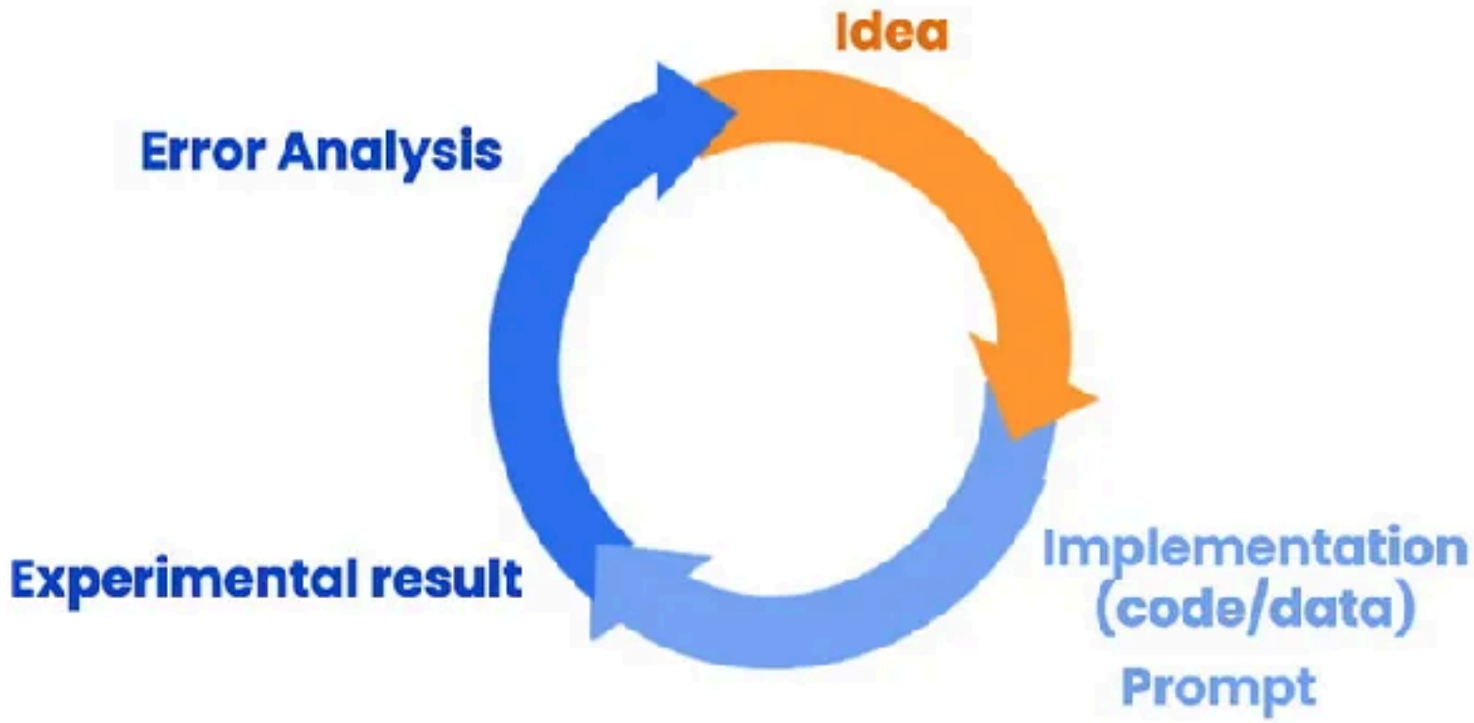
Write clear instructions
Provide reference text
Split complex tasks into simpler subtasks
Give the model time to think
Testing and improve …

https://platform.openai.com/docs/guides/prompt-engineering



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Iterative Prompt Development



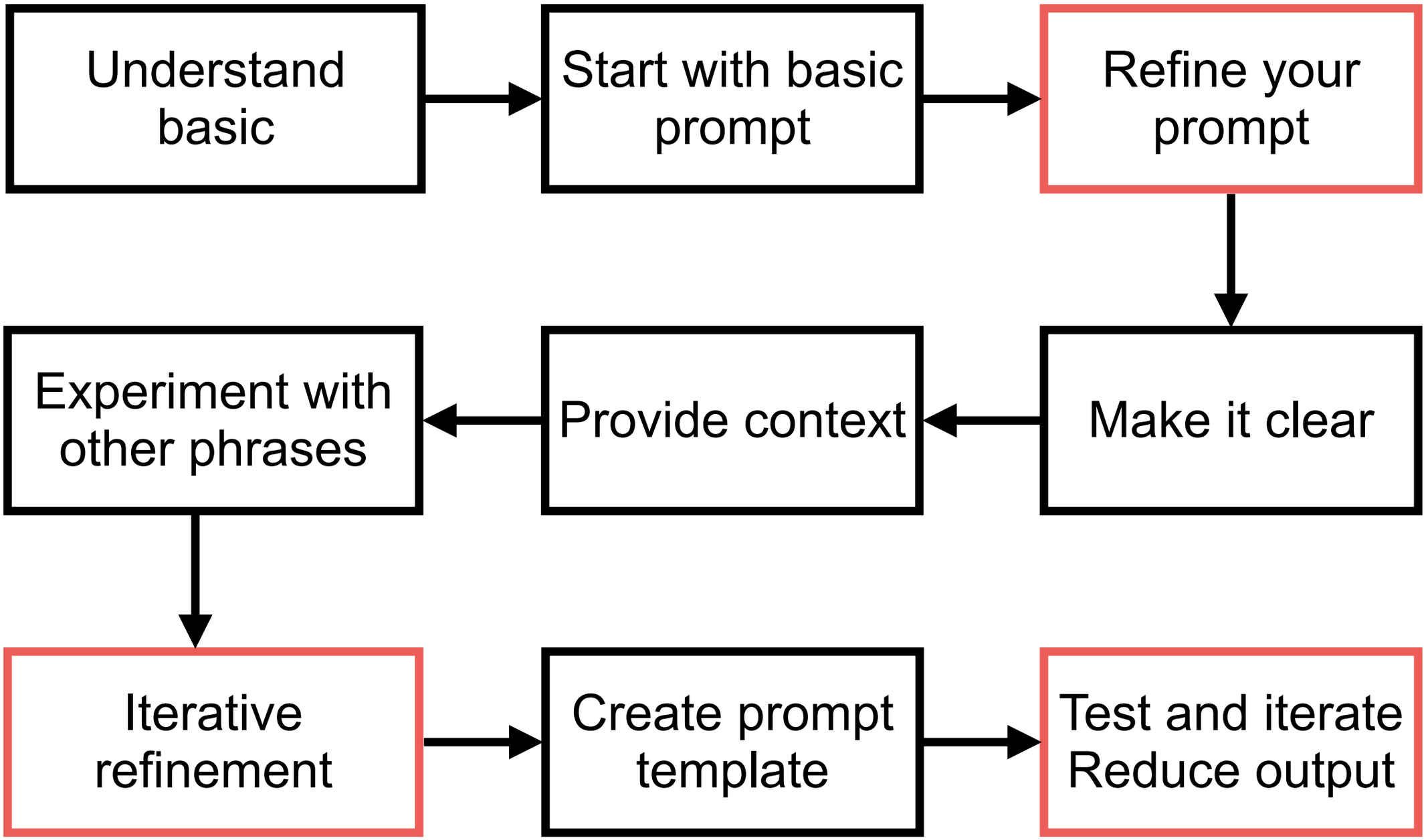
Iterative Process

- Try something
- Analyze where the result does not give what you want
- Clarify instructions, give more time to think
- Refine prompts with a batch of examples

https://www.deeplearning.ai/short-courses/chatgpt-prompt-engineering-for-developers/

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Basic of Prompt Engineer



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Better Prompt

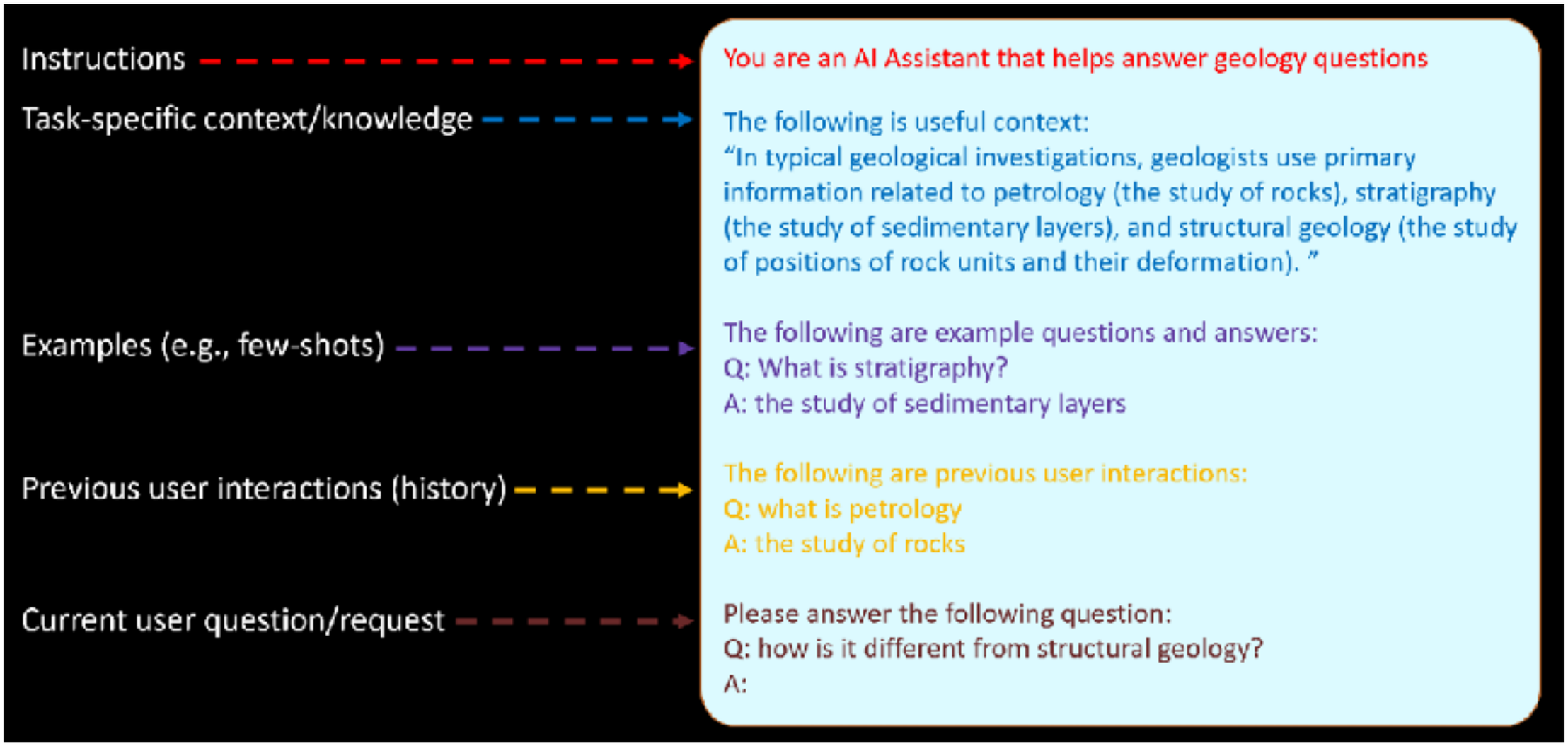
Write clear instructions
Provide reference text
Split complex tasks into simpler subtasks
Give the model time to think
Testing and improve …

https://platform.openai.com/docs/guides/prompt-engineering/six-strategies-for-getting-better-results



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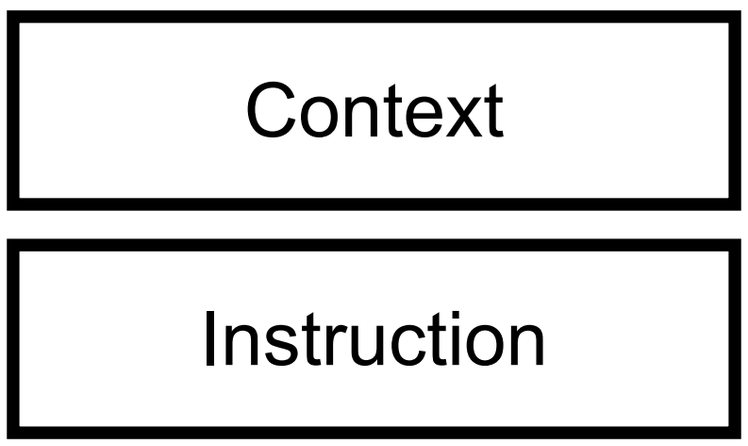
Prompt Structure



https://learn.microsoft.com/en-us/ai/playbook/technology-guidance/generative-ai/working-with-llms/prompt-engineering

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Structure of Prompt



https://platform.openai.com/docs/guides/prompt-engineering

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Prompting Guide

Prompt Engineering

Prompt Engineering Guide

Prompt engineering is a relatively new discipline for developing and optimizing prompts to efficiently use language models (LMs) for a wide variety of applications and research topics. Prompt engineering skills help to better understand the capabilities and limitations of large language models (LLMs).

Researchers use prompt engineering to improve the capacity of LLMs on a wide range of common and complex tasks such as question answering and arithmetic reasoning. Developers use prompt engineering to design robust and effective prompting techniques that interface with LLMs and other tools.

Prompt engineering is not just about designing and developing prompts. It encompasses a wide range of skills and techniques that are useful for interacting and developing with LLMs. It's an important skill to interface, build with, and understand capabilities of LLMs. You can use prompt engineering to improve safety of LLMs and build new capabilities like augmenting LLMs with domain knowledge and external tools.

https://www.promptingguide.ai/



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Structure of Prompt !!

APE
Action, Purpose, Expectation

RACE
Role, Action, Context, Expectation

TAG
Task, Action, Goal

COAST
Context, Objective, Action, Scenario, Task

RISE
Role, Input, Step, Expectation

RCCF
Role, Context, Constraint, Format

https://twitter.com/pradeepeth/status/1673271866696544257



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Prompt Techniques

Zero-shot

Chain-of Thought (CoT)

Few-shot

Meta or structure

https://www.promptingguide.ai/techniques

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Chain of Thought Prompting (CoT)

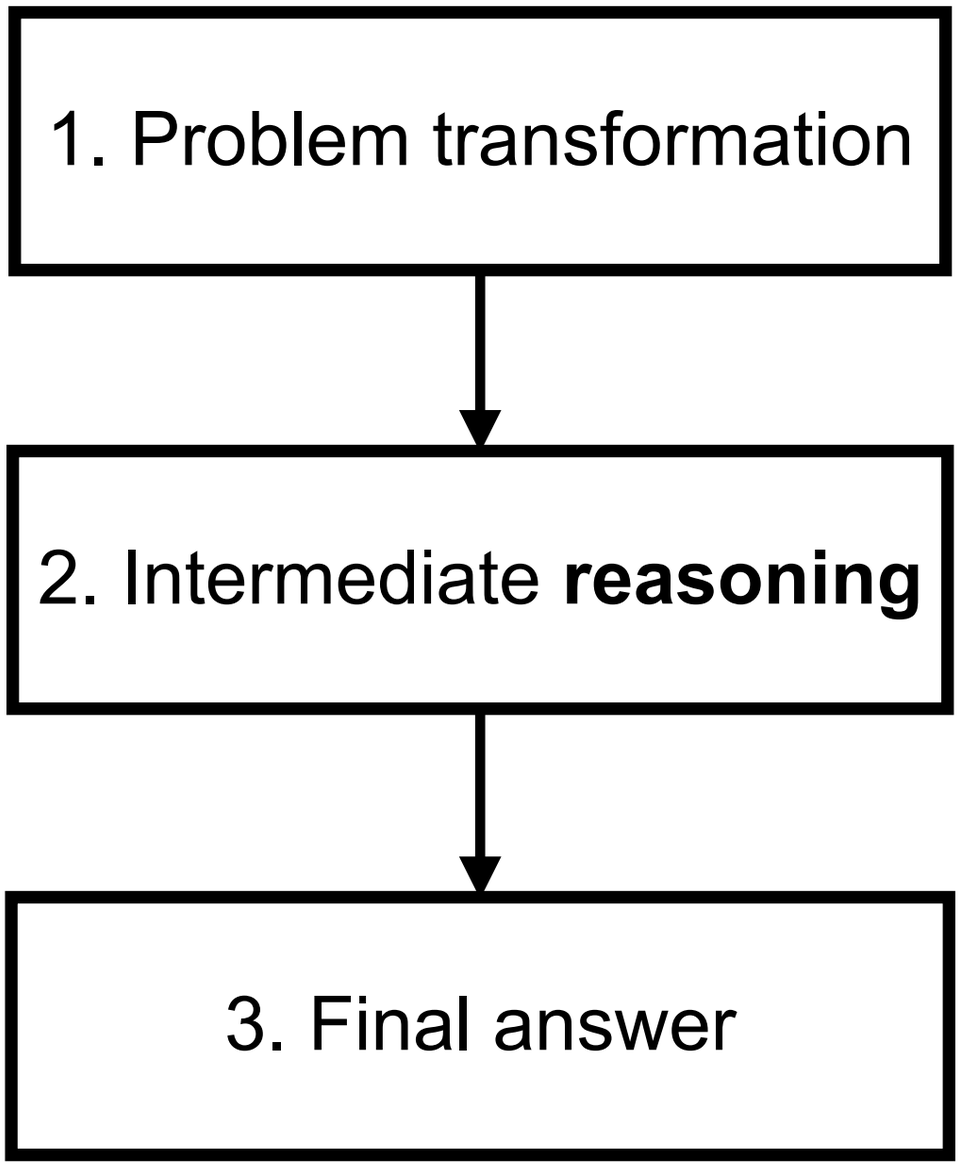
Technique used to improve the reasoning ability of LLM

Try to break down a complex problem into smaller, More manageable steps, lead to final answer

Reasoning model !!

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Chain of Thought Prompting (CoT)



eg. step-by-step

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Chain of Thought Prompting (CoT)

(a) Few-shot

Q: Roger has 5 tennis balls. He buys 2 more cans of tennis balls. Each can has 3 tennis balls. How many tennis balls does he have now?
A: The answer is 11.

Q: A juggler can juggle 16 balls. Half of the balls are golf balls, and half of the golf balls are blue. How many blue golf balls are there?
A:

(Output) The answer is 8. ✗

(b) Few-shot-CoT

Q: Roger has 5 tennis balls. He buys 2 more cans of tennis balls. Each can has 3 tennis balls. How many tennis balls does he have now?
A: Roger started with 5 balls. 2 cans of 3 tennis balls each is 6 tennis balls. 5 + 6 = 11. The answer is 11.

Q: A juggler can juggle 16 balls. Half of the balls are golf balls, and half of the golf balls are blue. How many blue golf balls are there?
A:

(Output) The juggler can juggle 16 balls. Half of the balls are golf balls. So there are 16 / 2 = 8 golf balls. Half of the golf balls are blue. So there are 8 / 2 = 4 blue golf balls. The answer is 4. ✓

(c) Zero-shot

Q: A juggler can juggle 16 balls. Half of the balls are golf balls, and half of the golf balls are blue. How many blue golf balls are there?
A: The answer (arabic numerals) is

(Output) 8 ✗

(d) Zero-shot-CoT (Ours)

Q: A juggler can juggle 16 balls. Half of the balls are golf balls, and half of the golf balls are blue. How many blue golf balls are there?
A: ***Let's think step by step.***

(Output) There are 16 balls in total. Half of the balls are golf balls. That means that there are 8 golf balls. Half of the golf balls are blue. That means that there are 4 blue golf balls. ✓

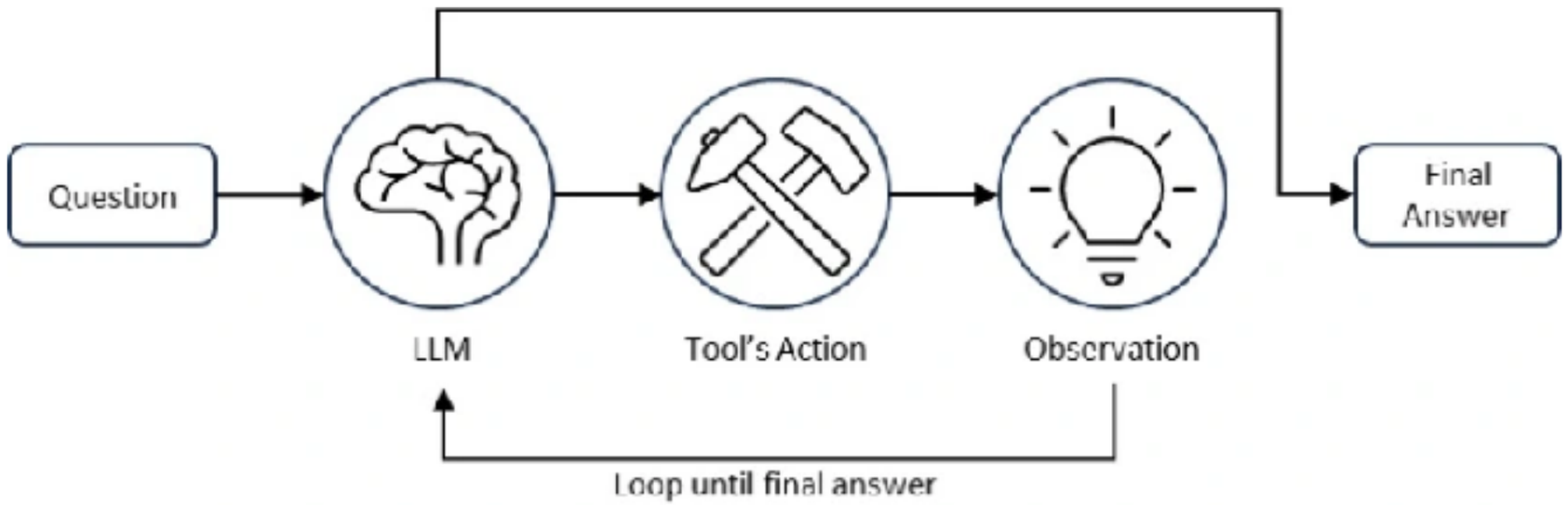
https://www.promptingguide.ai/techniques/cot



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ReAct Prompt

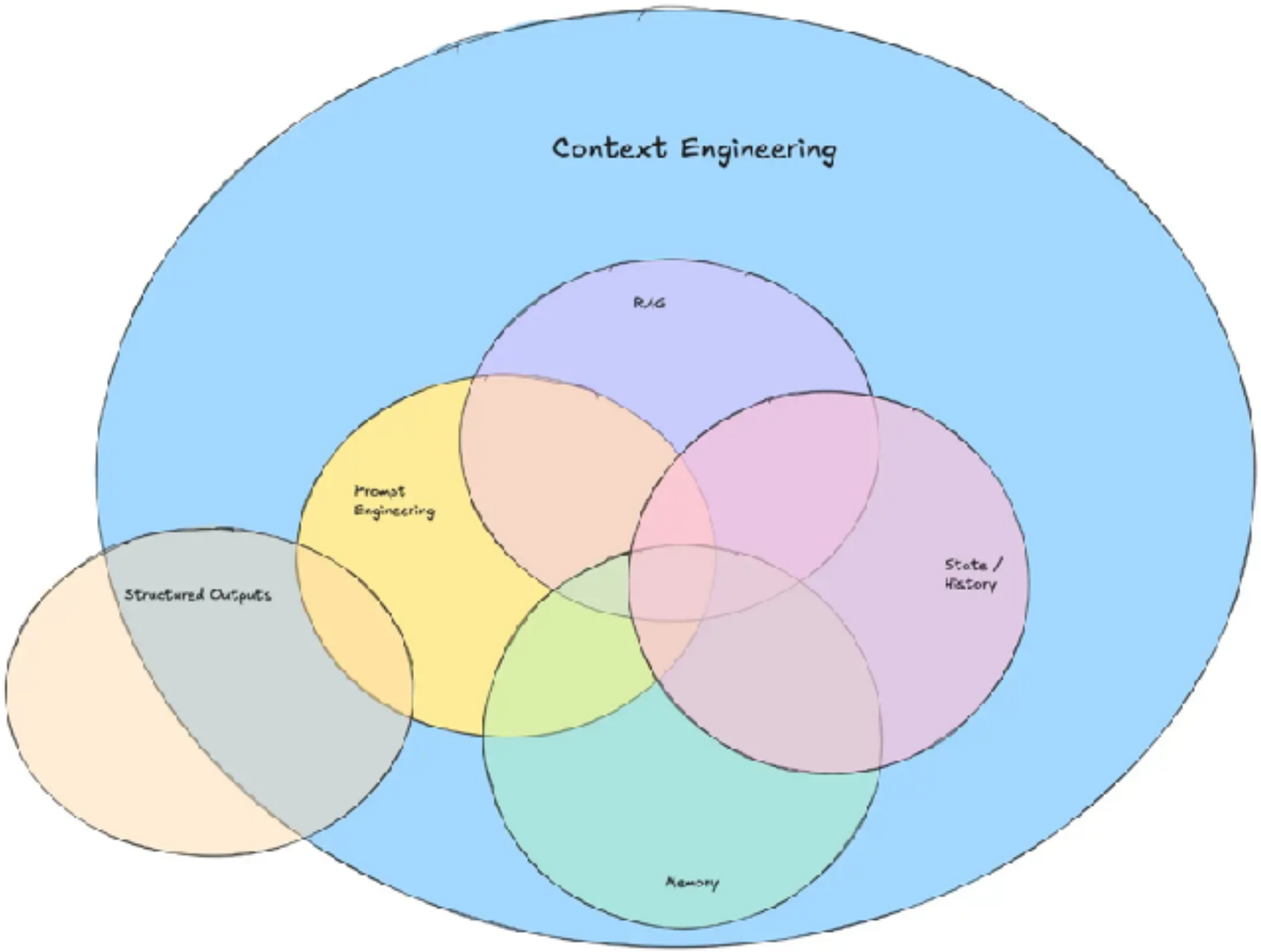
LLM reasoning and additional tools (expert)
Improve better answer



https://www.promptingguide.ai/techniques/react

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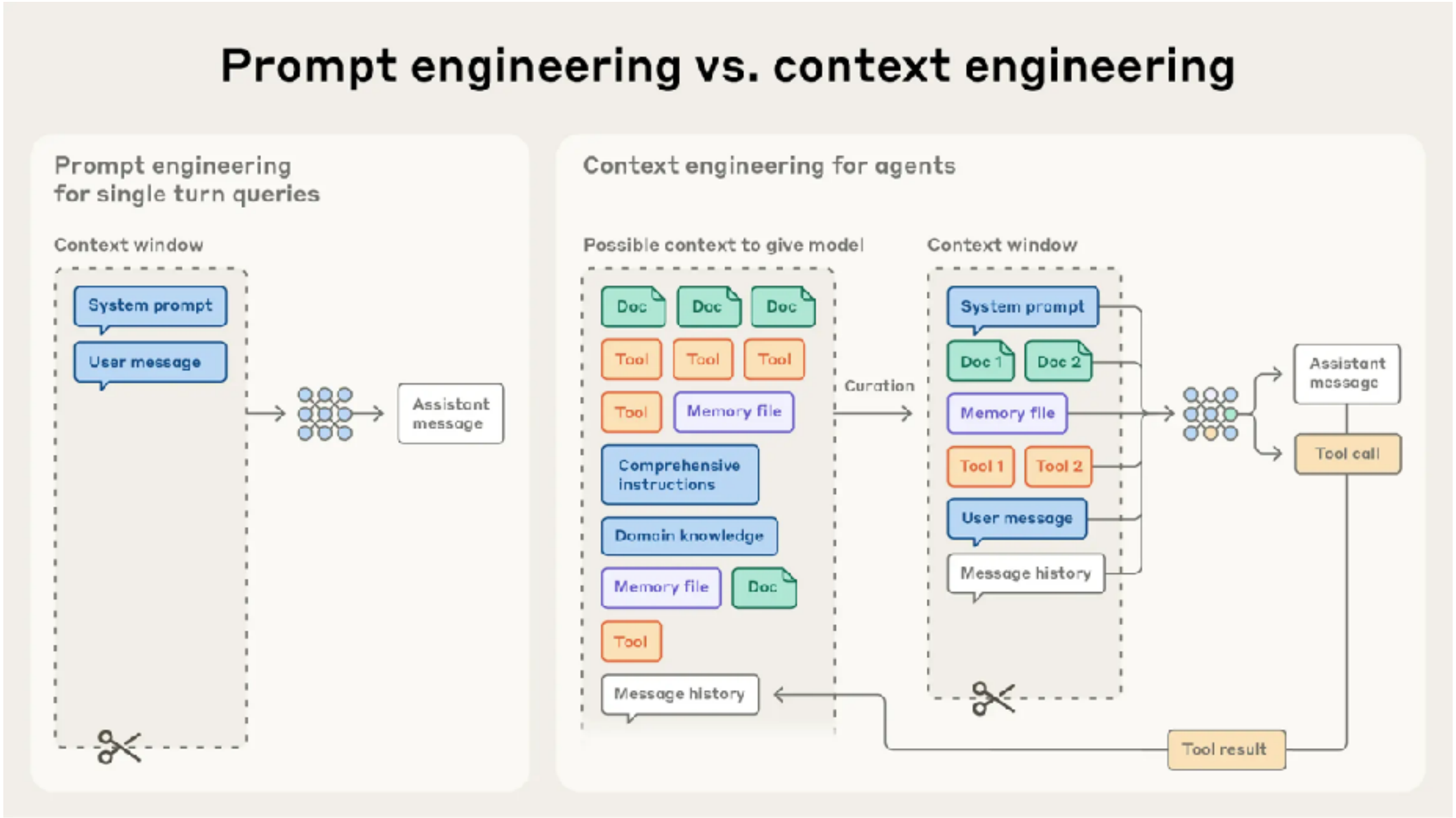
Context Engineering



https://www.promptingguide.ai/guides/context-engineering-guide

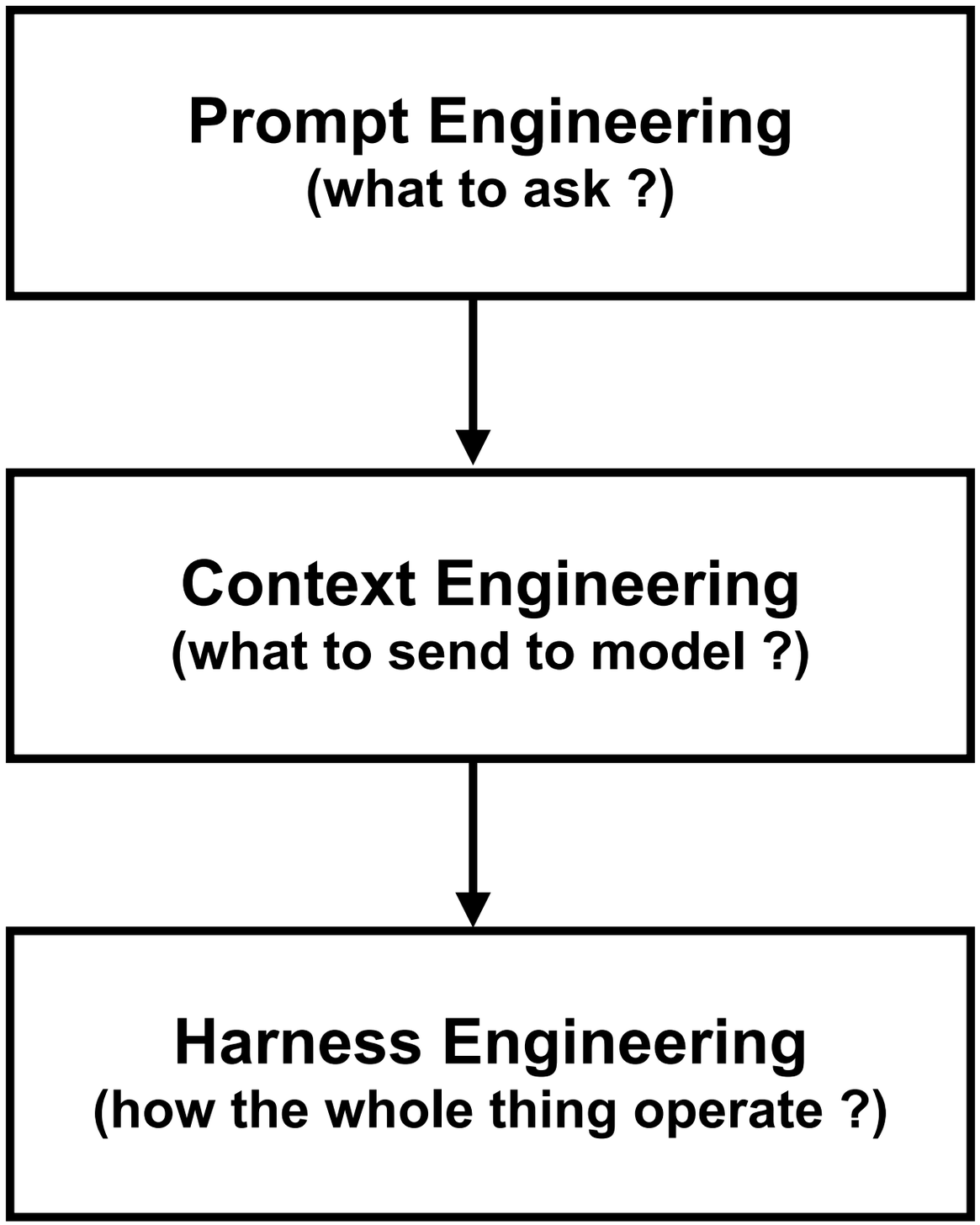
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Effective Context Engineering

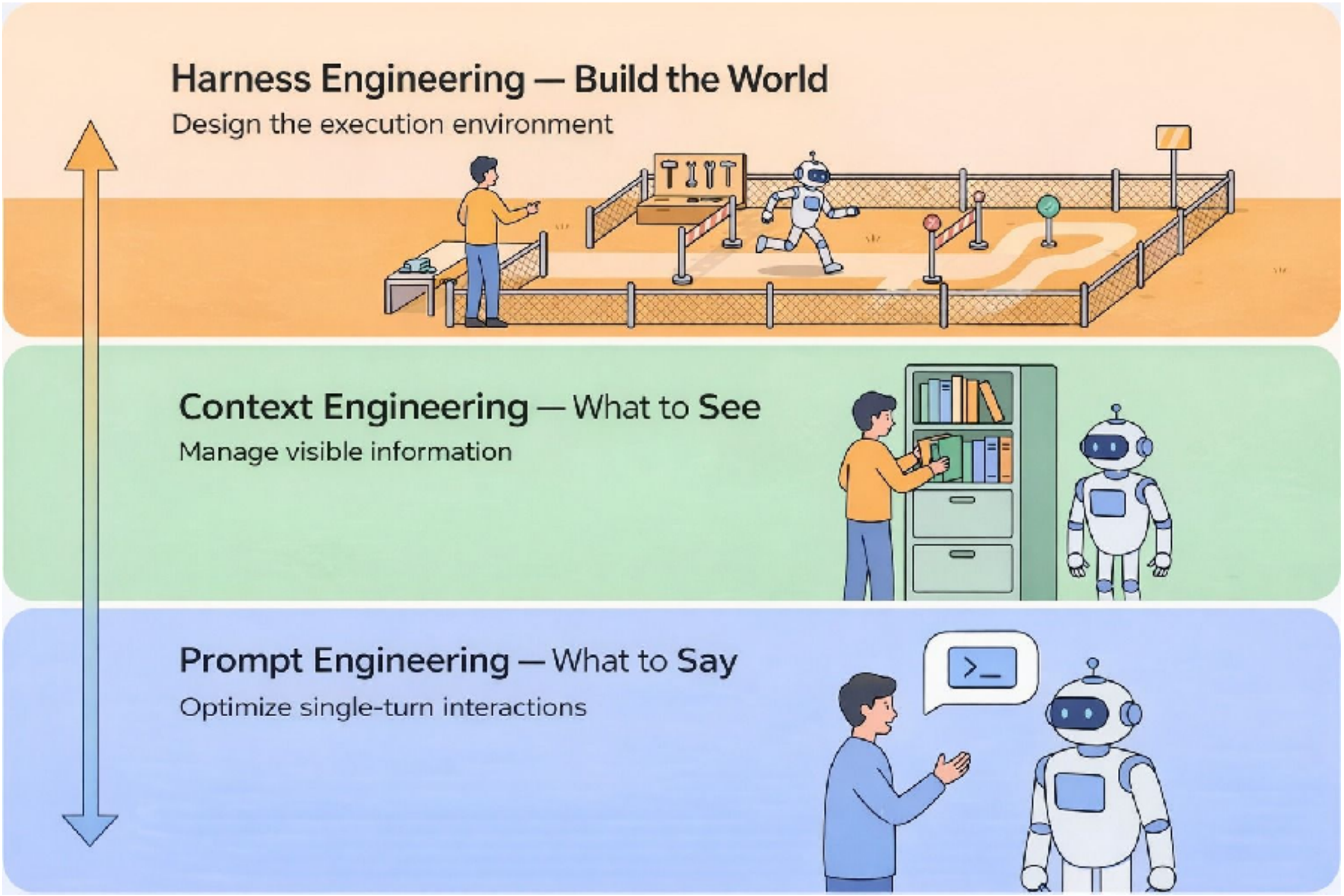


https://www.anthropic.com/engineering/effective-context-engineering-for-ai-agents

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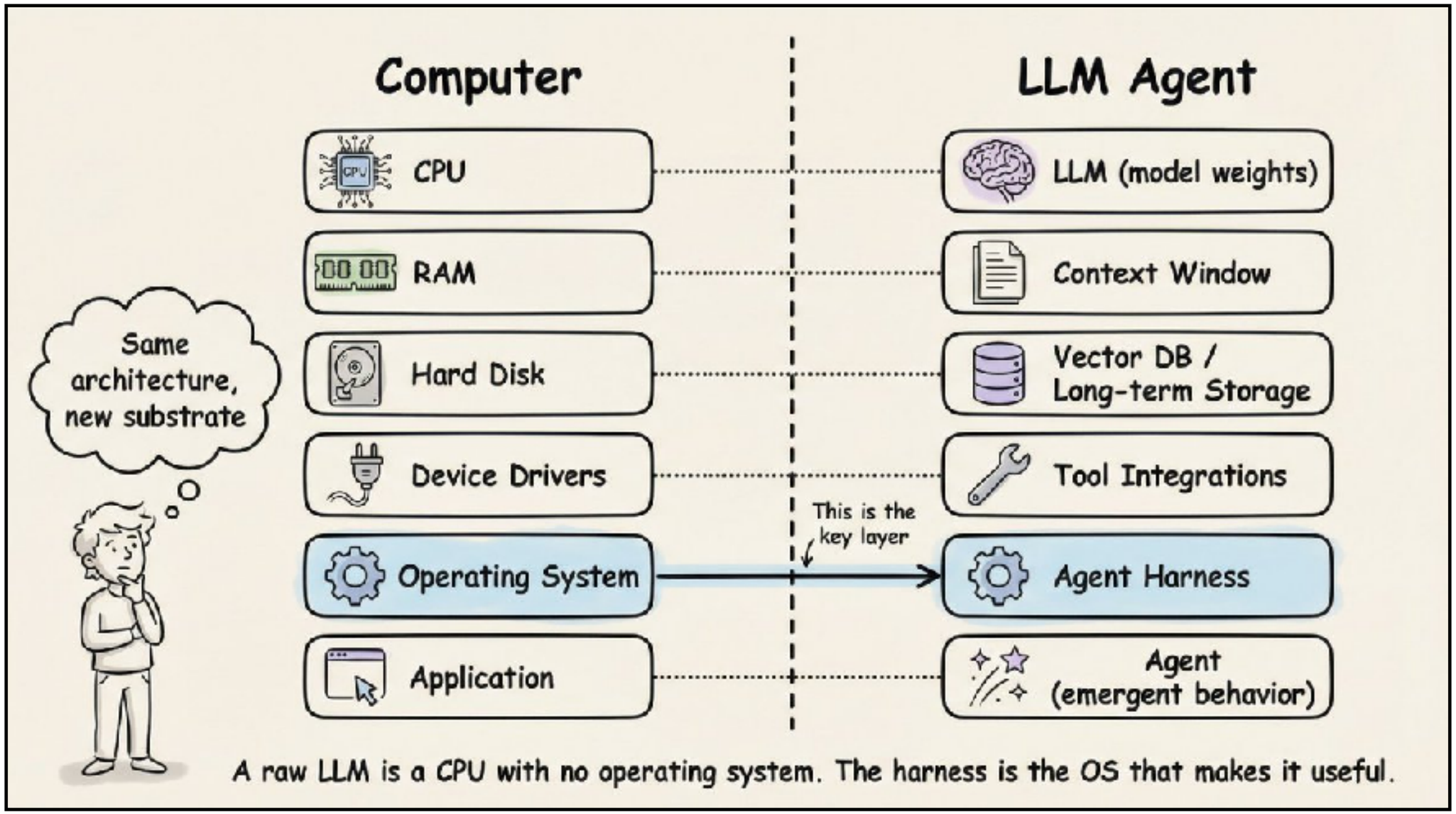
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https://milvus.io/blog/harness-engineering-ai-agents.md

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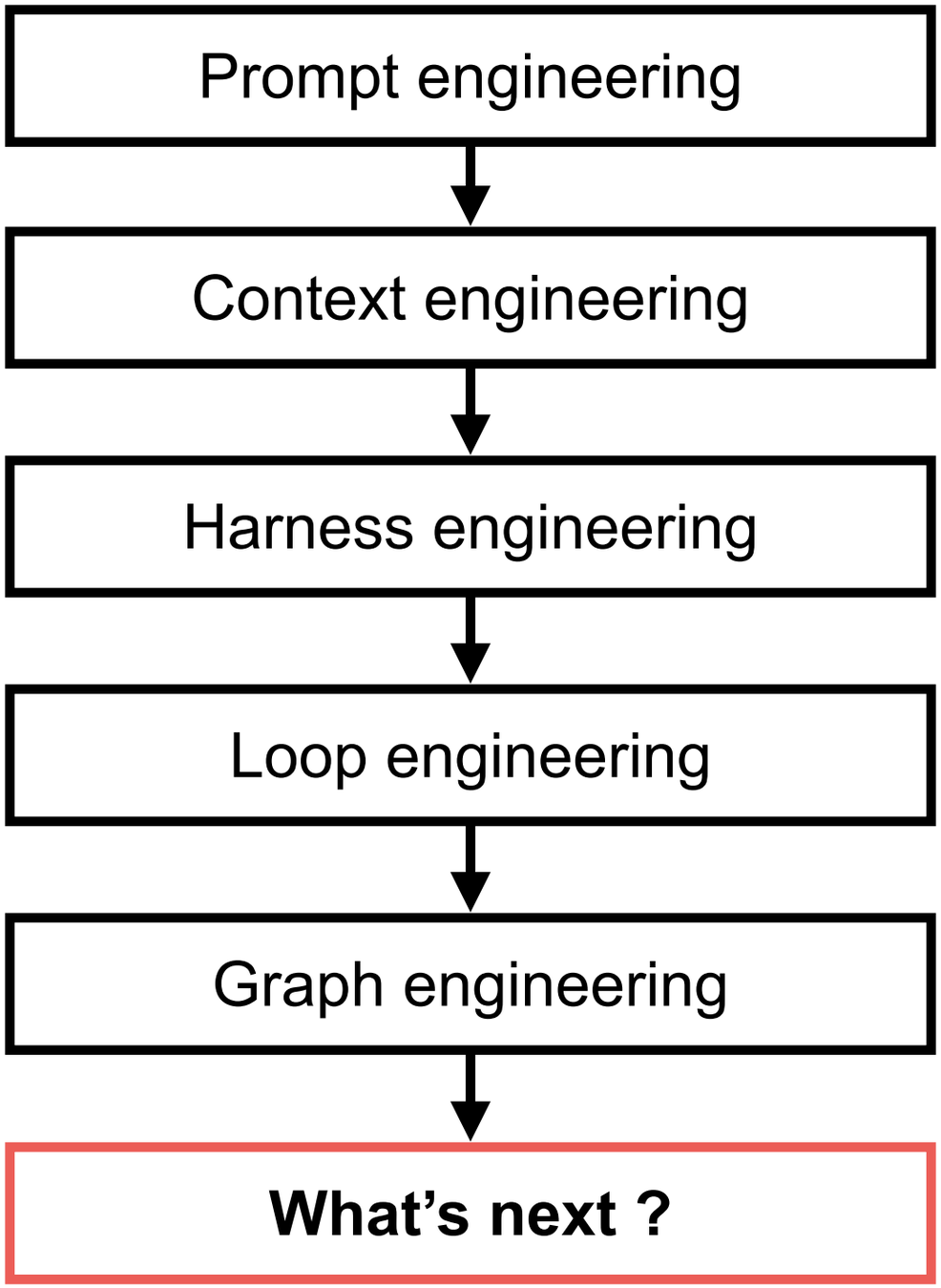
Harness Engineering



https://x.com/akshay_pachaar/status/2041146899319971922

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Current ?



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Workshop with Coding

Chat

Text Editor with AI

Pair programming with AI

SDD and Rules

BMAD-METHOD

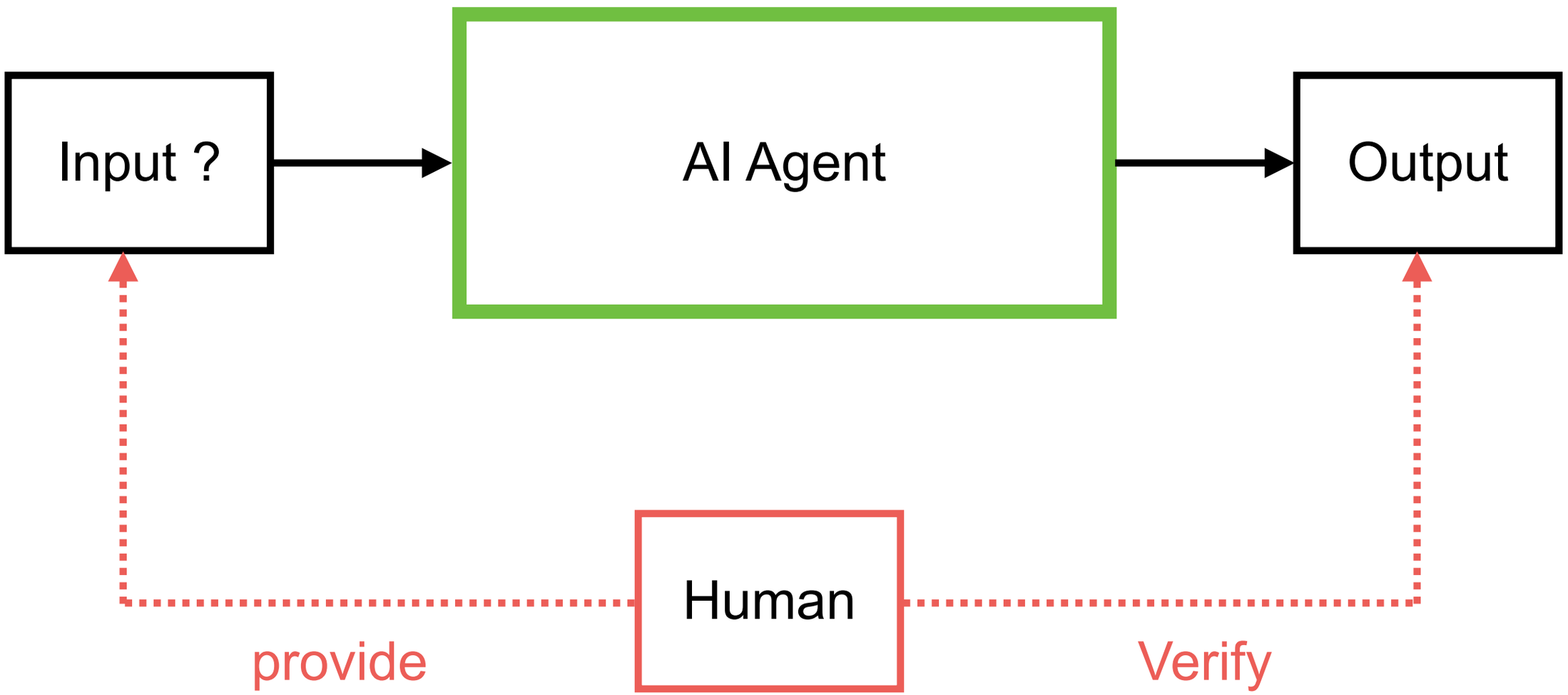
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AI Agent for Coding

Claude Code
Github Copilot Chat
Cline
Qwen code
Gemini CLI
Pi coding agent

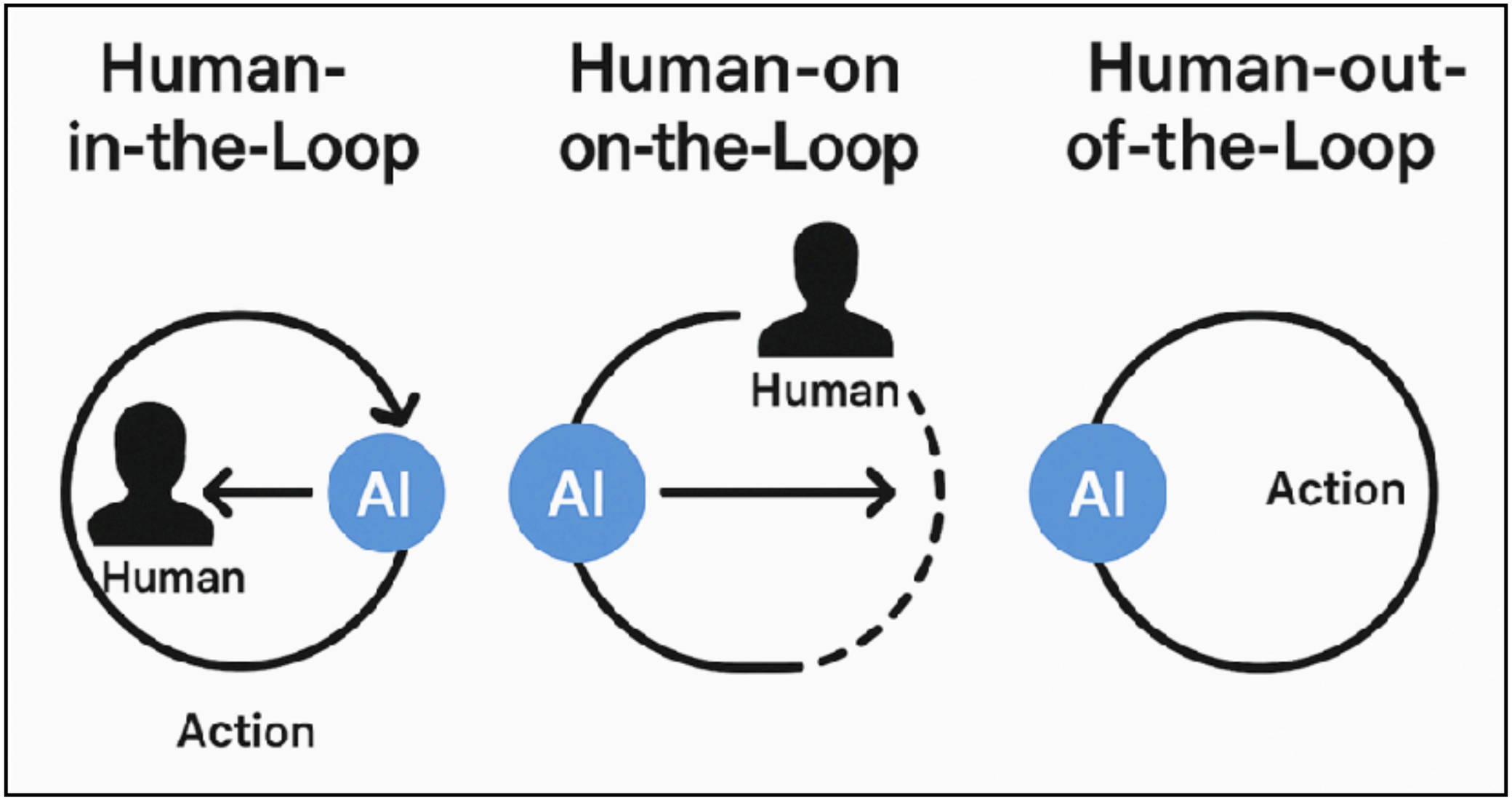
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Coding with AI Agent



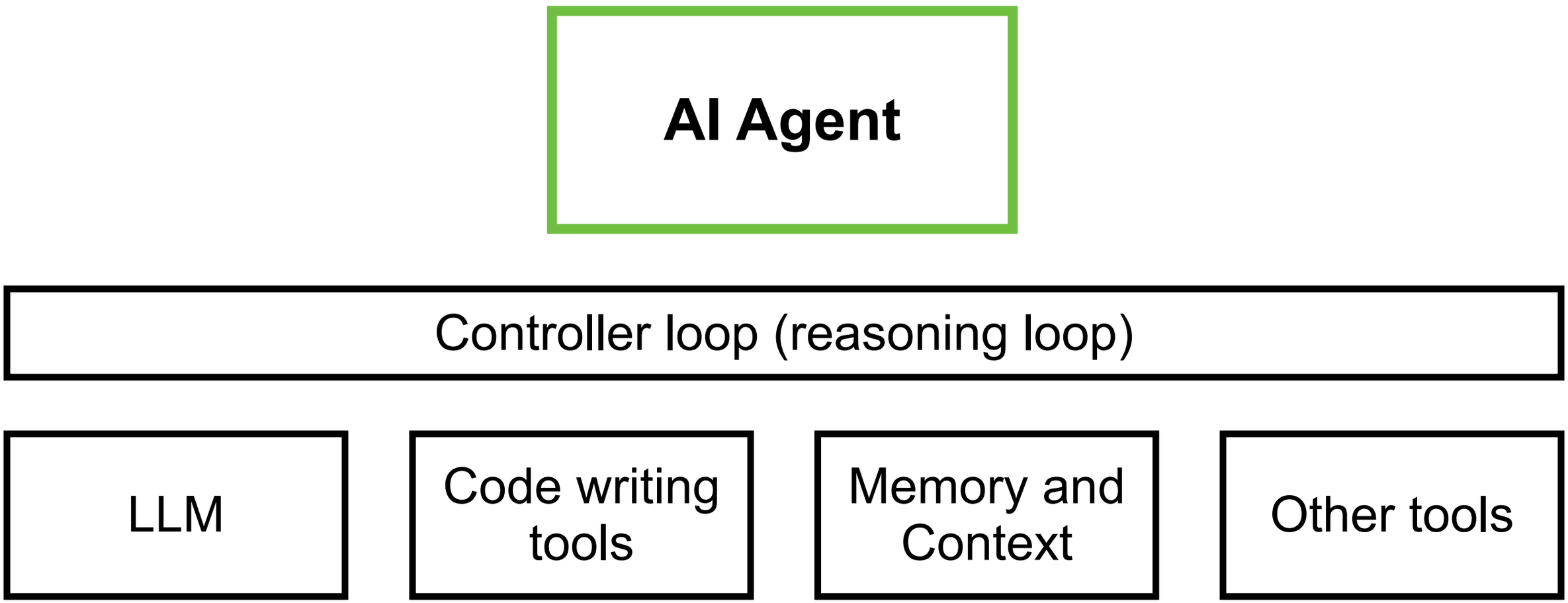
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AI Agent with/without Human



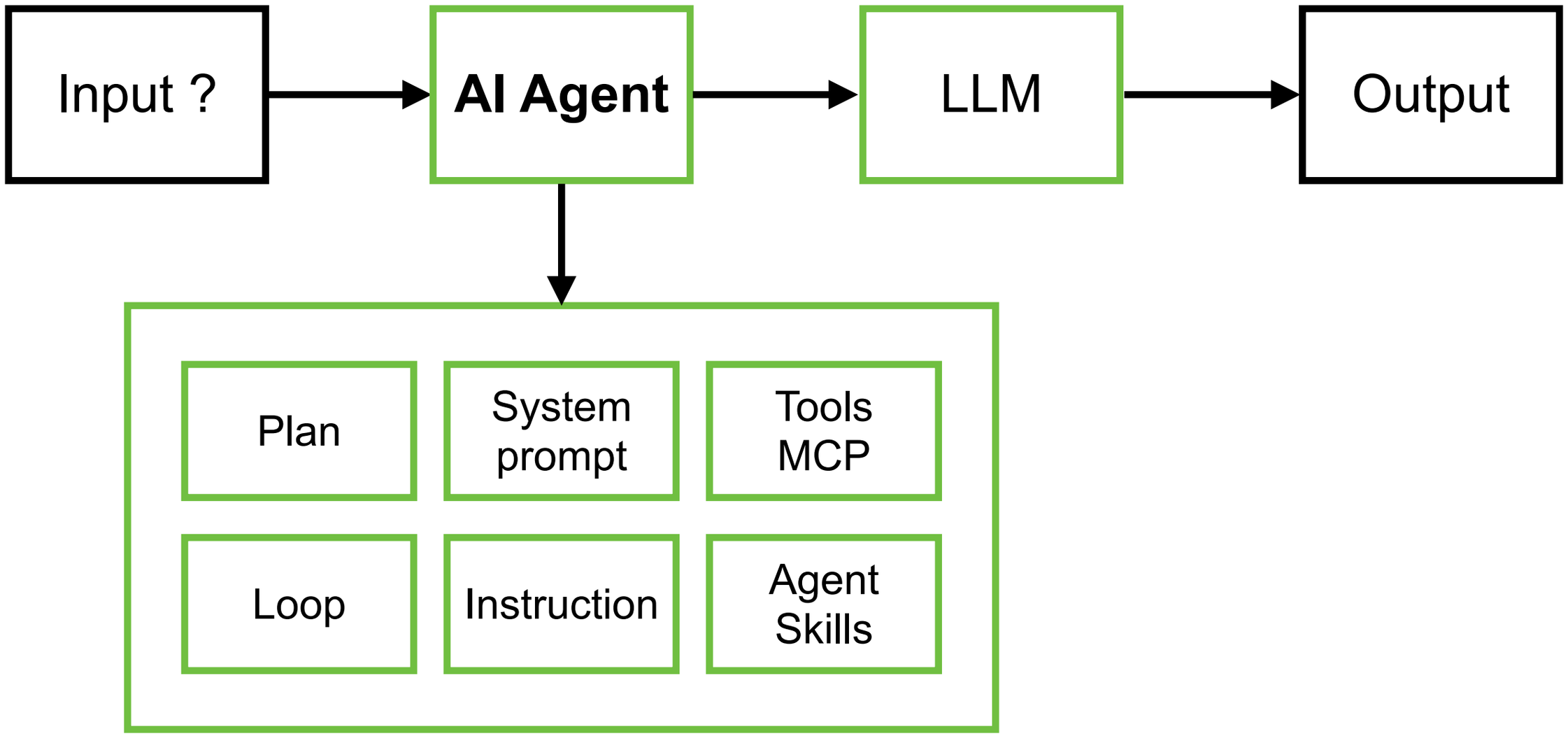
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AI Agent for Coding



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Working with AI Agent



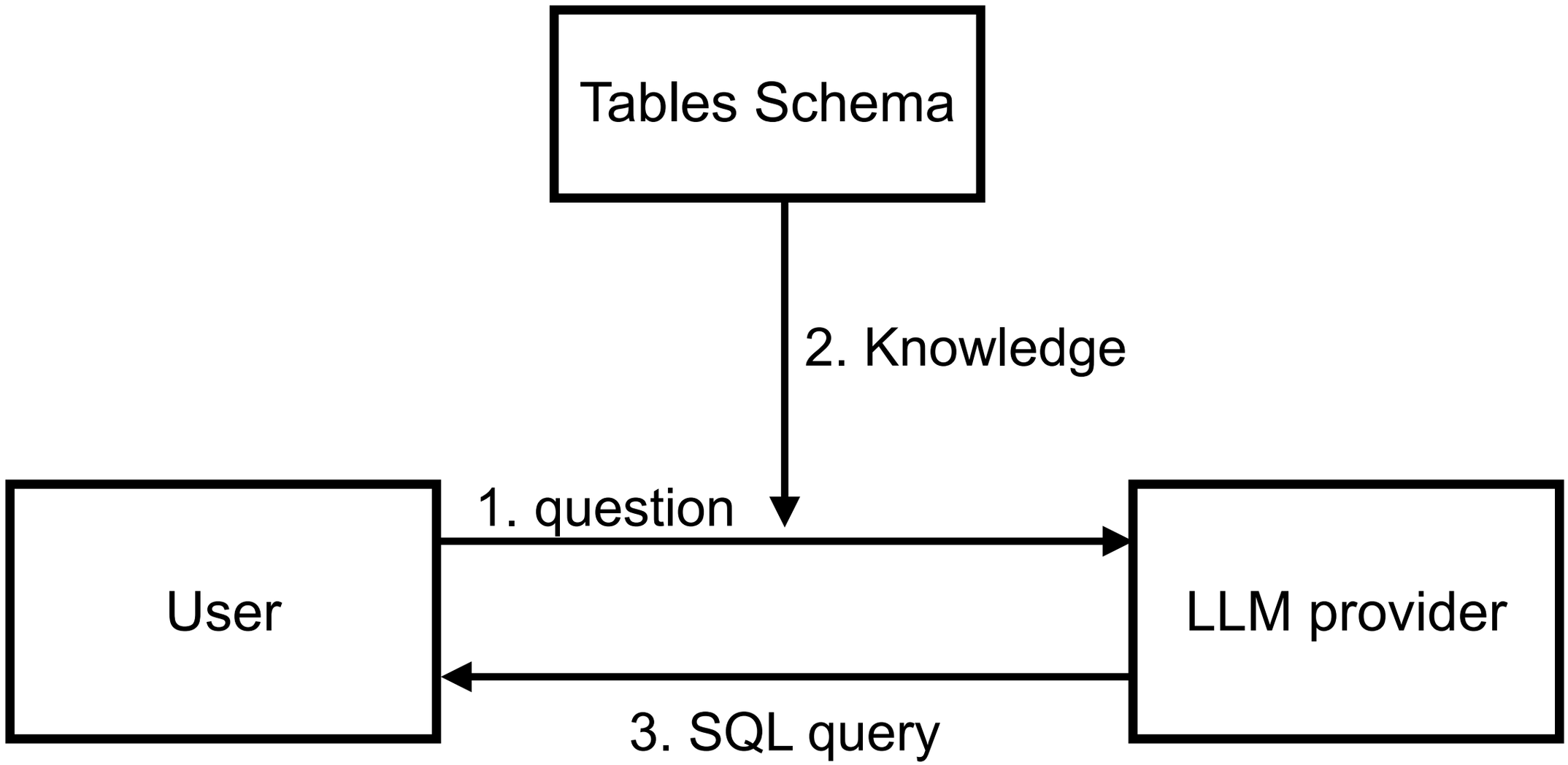
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Workshop



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Text-to-SQL



https://github.com/up1/workshop-ai-with-technical-team/tree/main/workshop/demo-sql

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Software Development Life Cycle

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SDLC



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Planning

- Written planning process
- Arch diagrams

Testing

- Automated testing
- TDD
- Testing environments
- Testing in prod
- Performance testing
- Load testing
- Generate test data

Development

- Automated dev env
- CI/CD
- Prototyping
- Code review
- Code generation
- Templates
- Cross-platform dev
- Preview env
- Post-commit code review
- Linting
- Static code analysis
- Project mgmt

Shipping

- FF & experimentation
- Logging
- Monitoring & alerting
- Staged rollouts

Maintenance

- Debug production
- Documentation
- Runbook examples
- Mitigation runbook

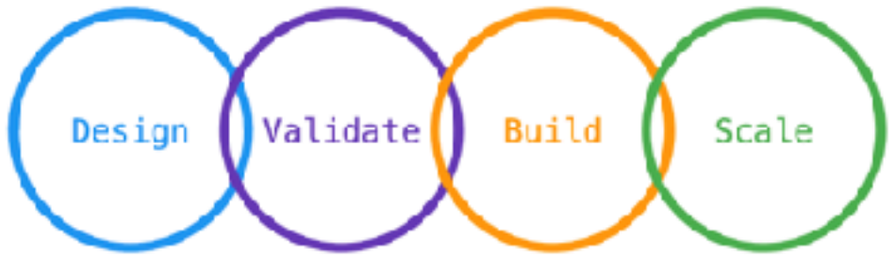
pragmaticengineer.com



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AI in Software Development

BEFORE

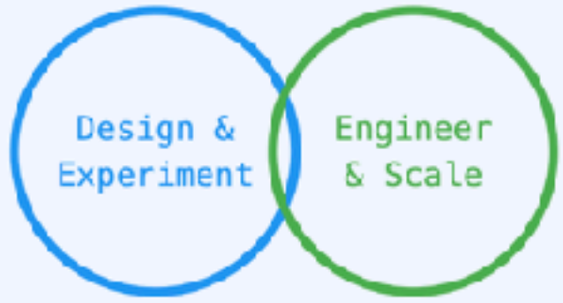


→ Fragmented SDLC Ownership;

→ Scaling Challenges;

→ Siloed Teams Increasing friction.

NOW



→ Democratized development;

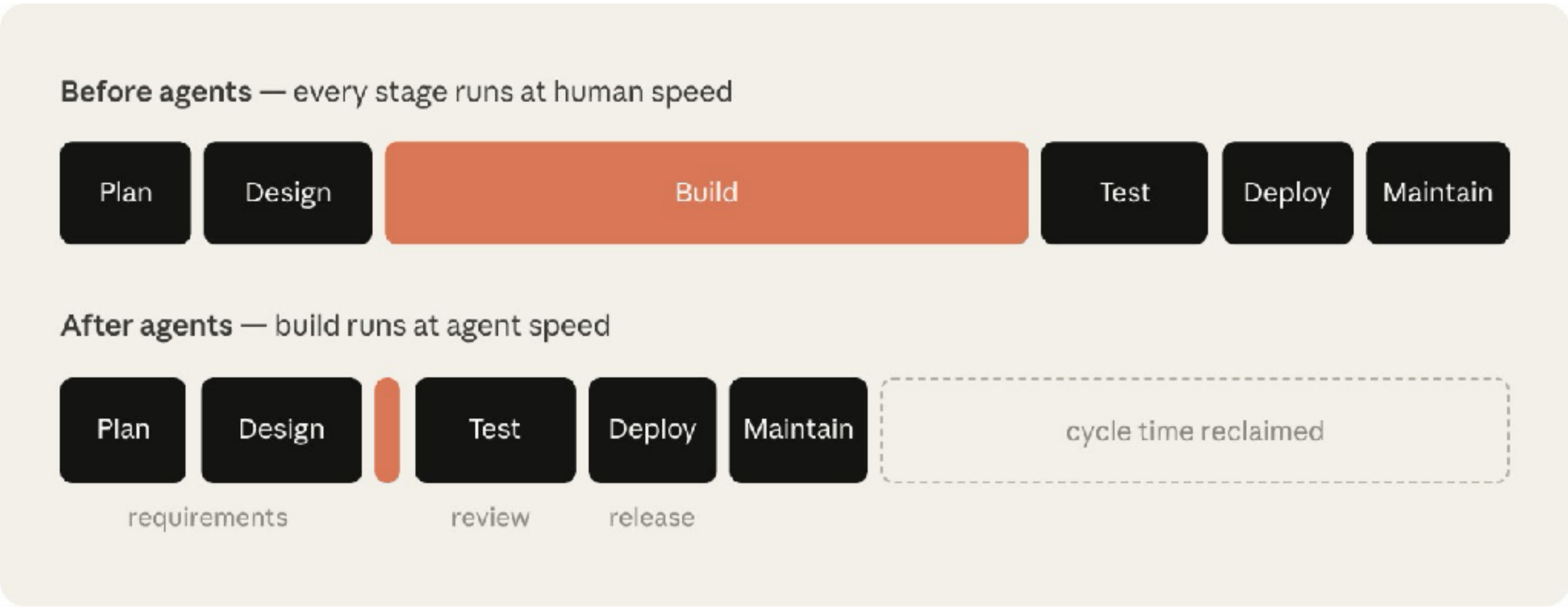
→ Bridge between business and IT;

→ Composability for business agility.

AI-native SDLC drives faster, customer-centric, and higher-quality innovation.

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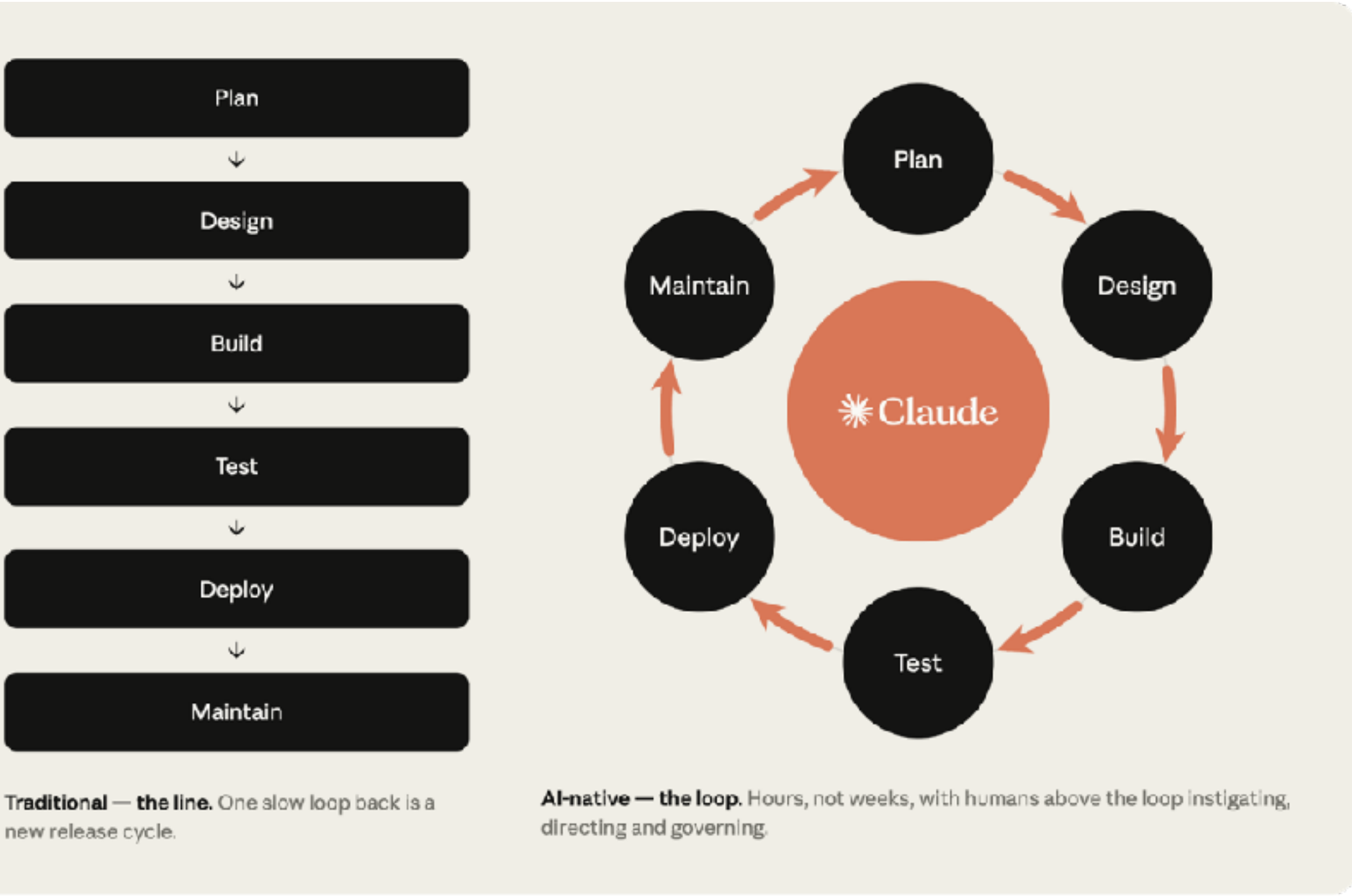
AI in SDLC



https://claude.com/resources/articles/the-ai-native-sdlc-playbook

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AI in SDLC



https://claude.com/resources/articles/the-ai-native-sdlc-playbook

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Impacts with productivity ?

Automated simple tasks

Improve quality and reliability

Improve communication

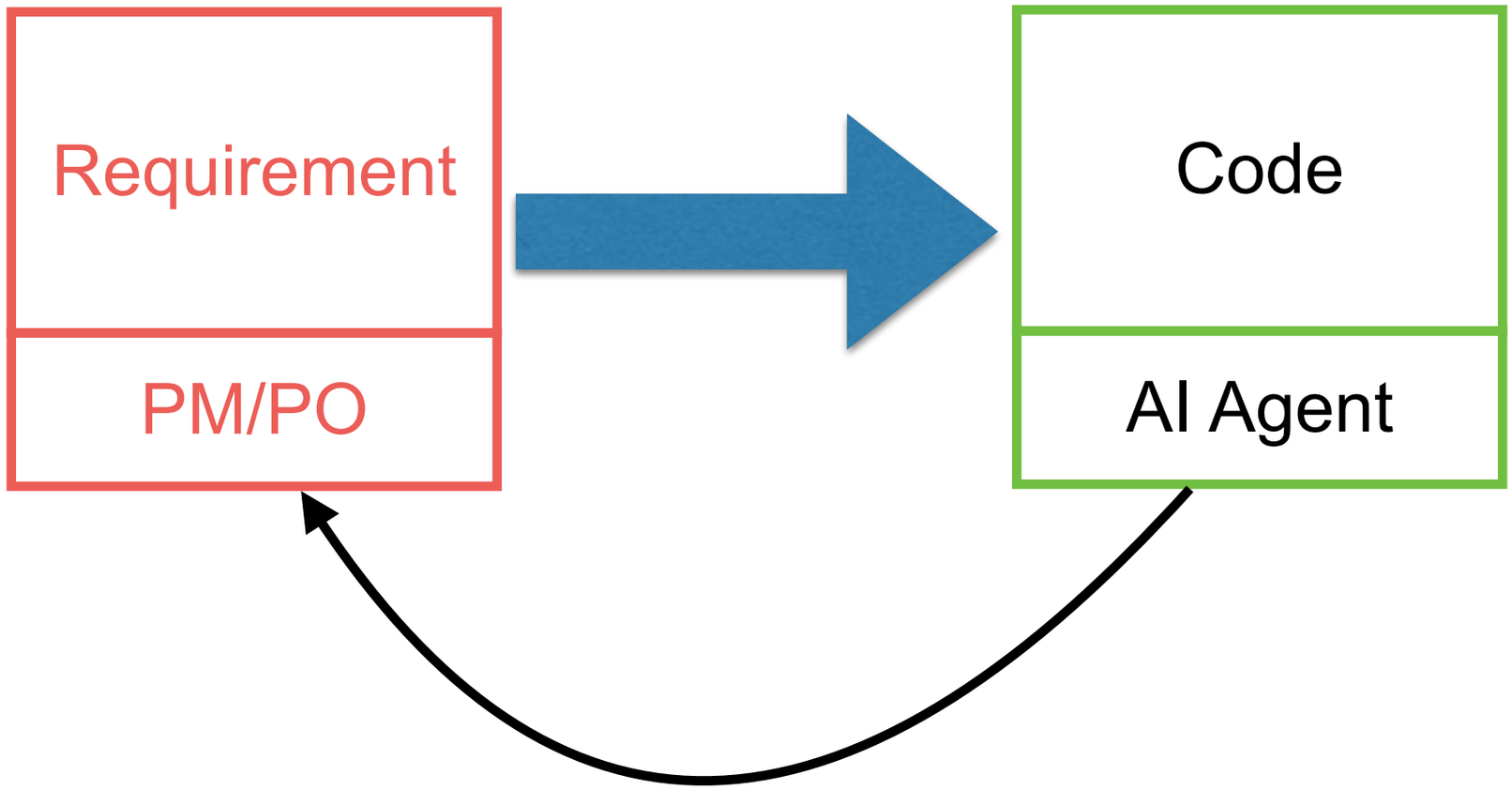
Faster prototype



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New Bottleneck ?

Decide what to **build**, not code



Get feedback faster

https://www.theaiopportunities.com/p/is-ai-slowing-down-the-new-bottleneck

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Generative AI isn't just a tool

it's your **team member**

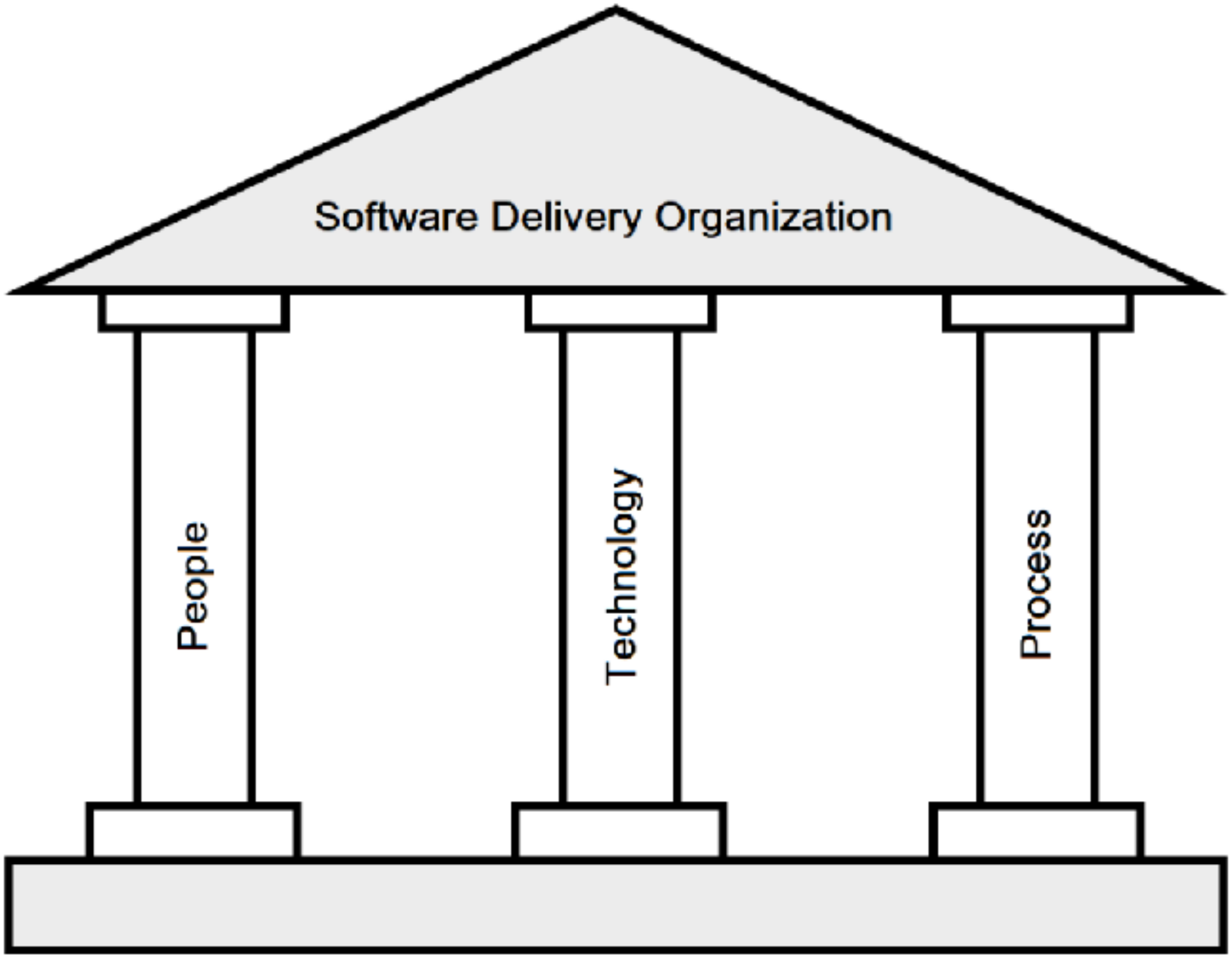
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Trust, but verify output !!



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3 Pillar of Software Development



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Requirement and Analysis



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Requirement and Analysis



Clarify requirement
Requirements writing and analysis
User story generation

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Requirement analysis

Clarify of User requirement ?

https://github.com/up1/workshop-ai-with-technical-team/wiki/Requirement-analysis



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Design Process

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Design



Architecture writing assistance
Sequence flow diagram generation
Data modeling
UX/UI design assistance

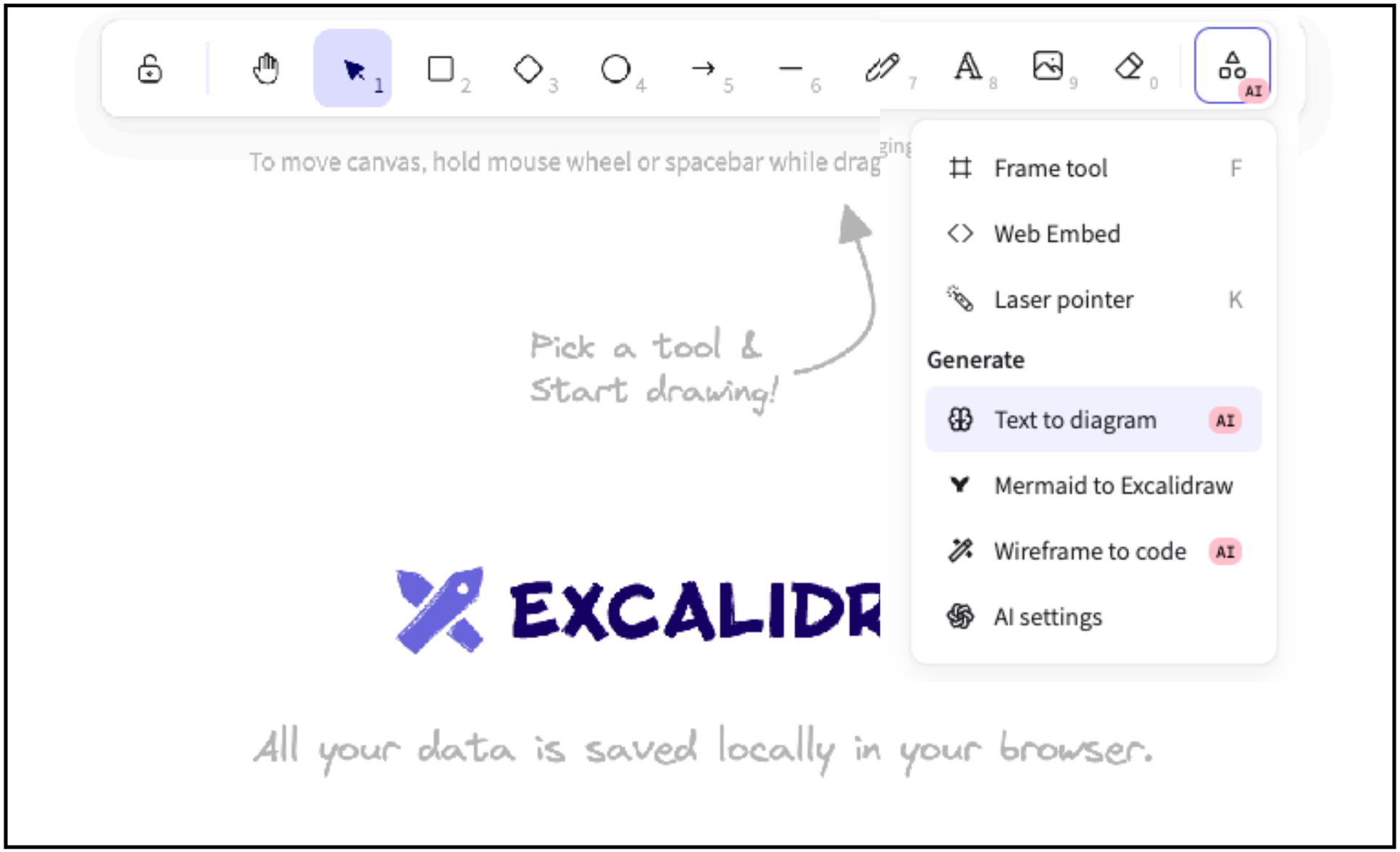
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Diagram



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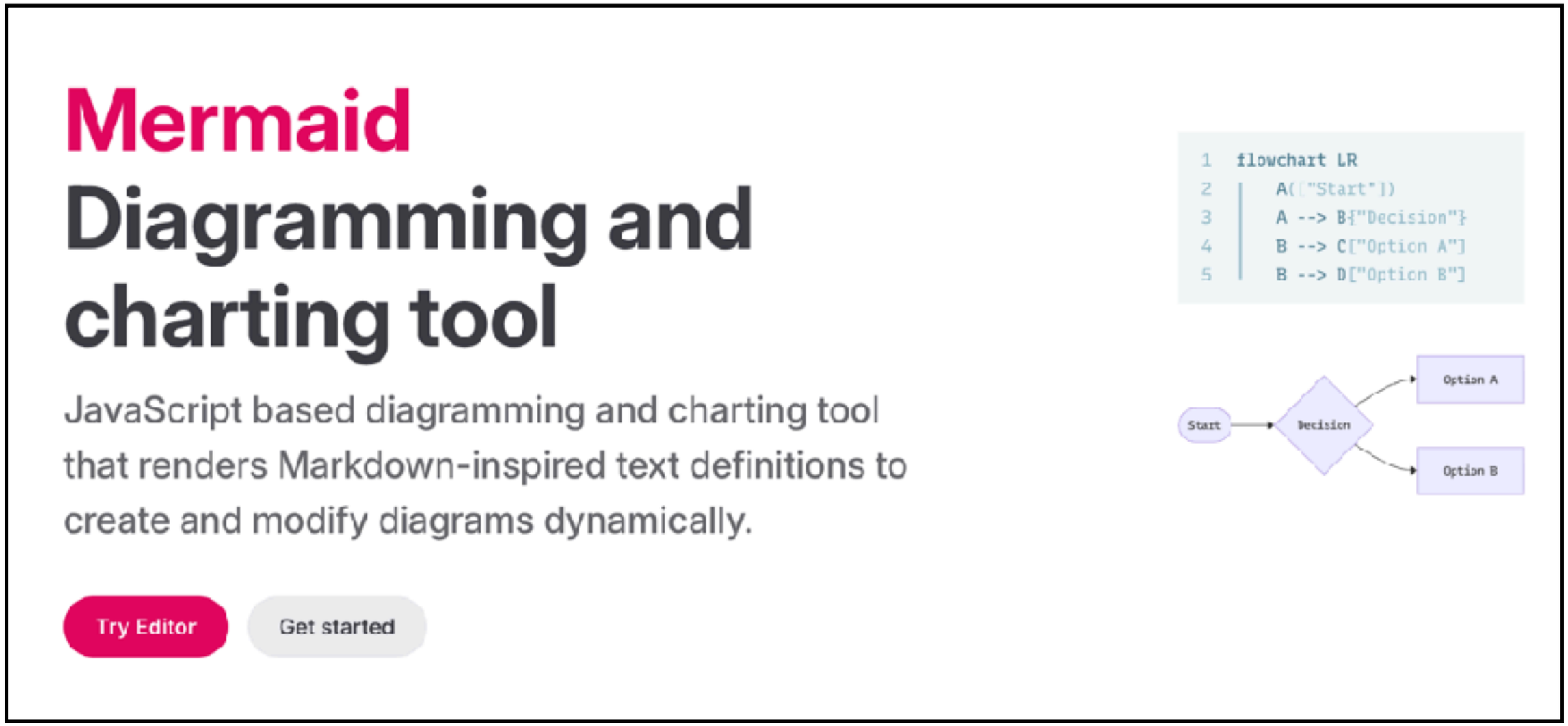
Excalidraw with AI



https://excalidraw.com/

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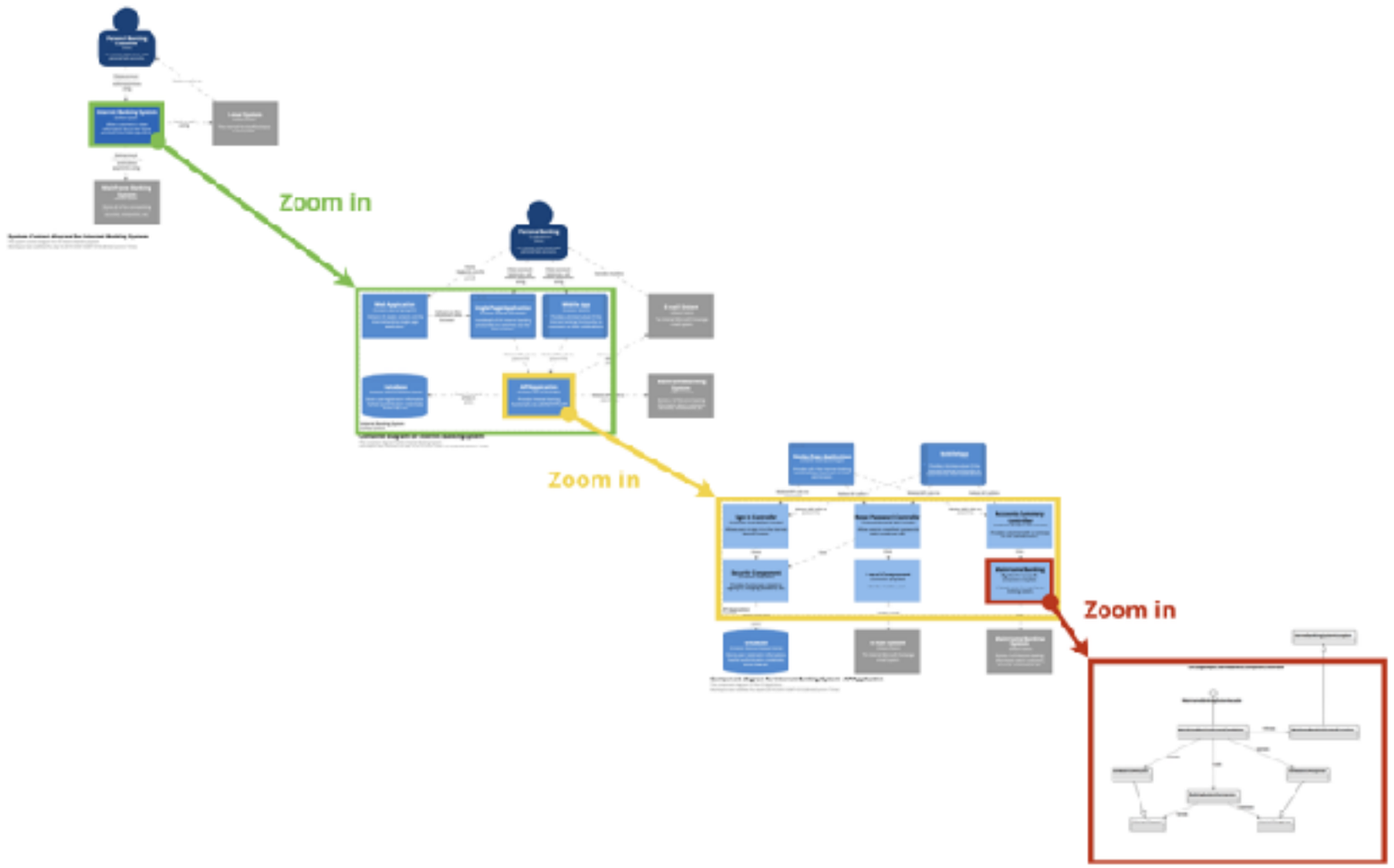
Mermaid Diagram



https://mermaid.js.org/

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Software Architecture



Level 1
Context

Level 2
Containers

Level 3
Components

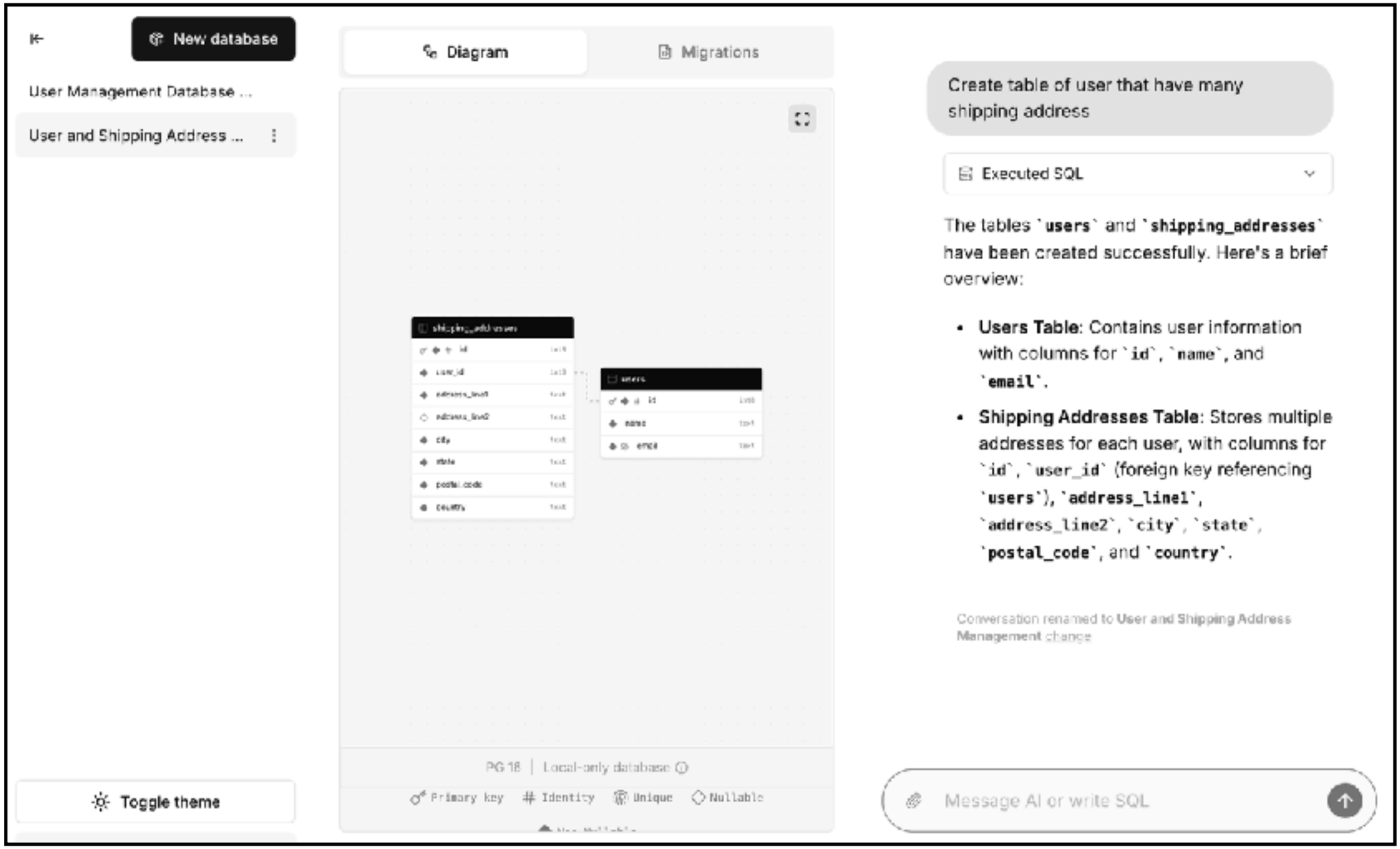
Level 4
Code

https://c4model.com/



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Database



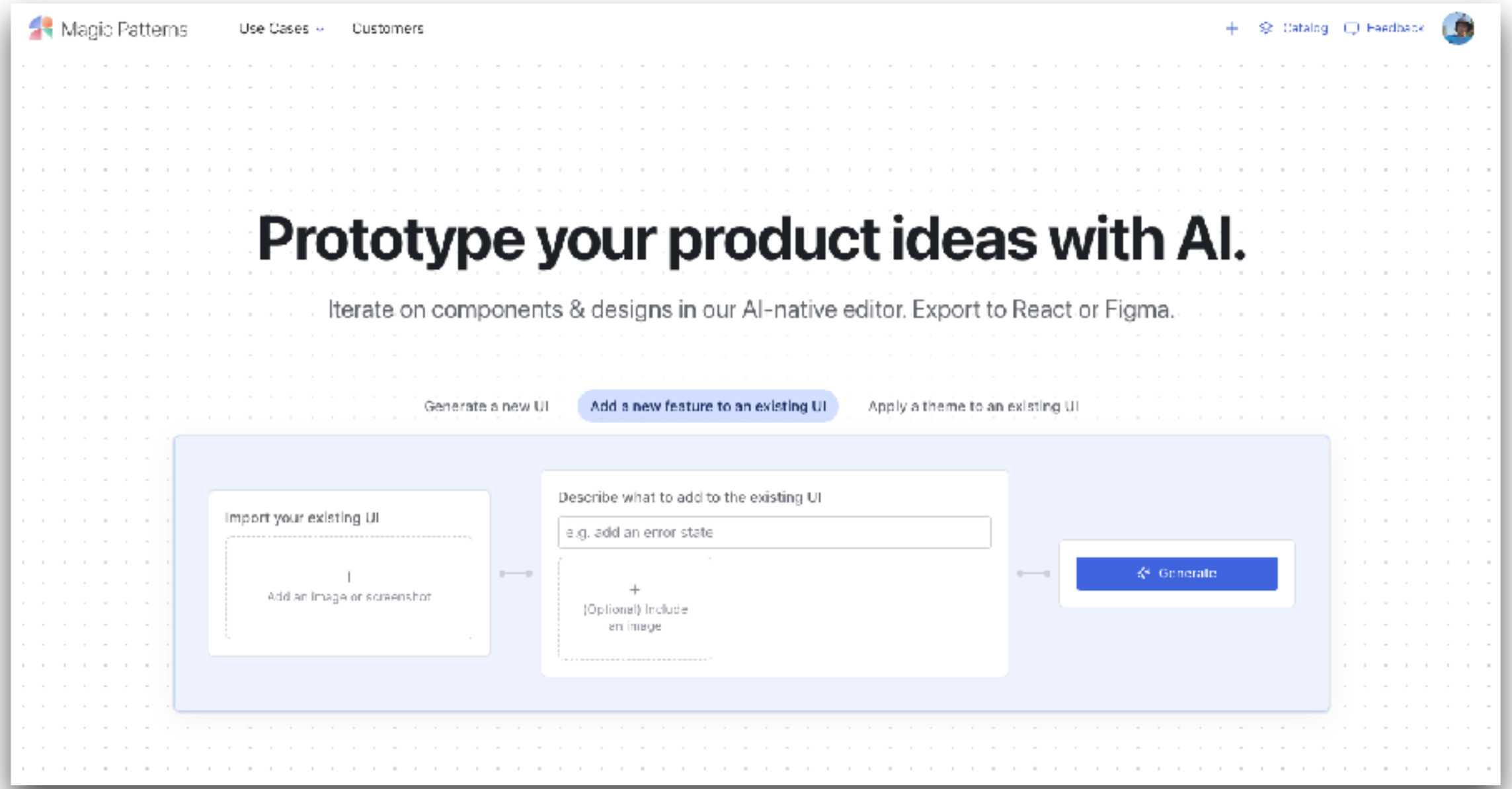
https://database.build/

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User Interface

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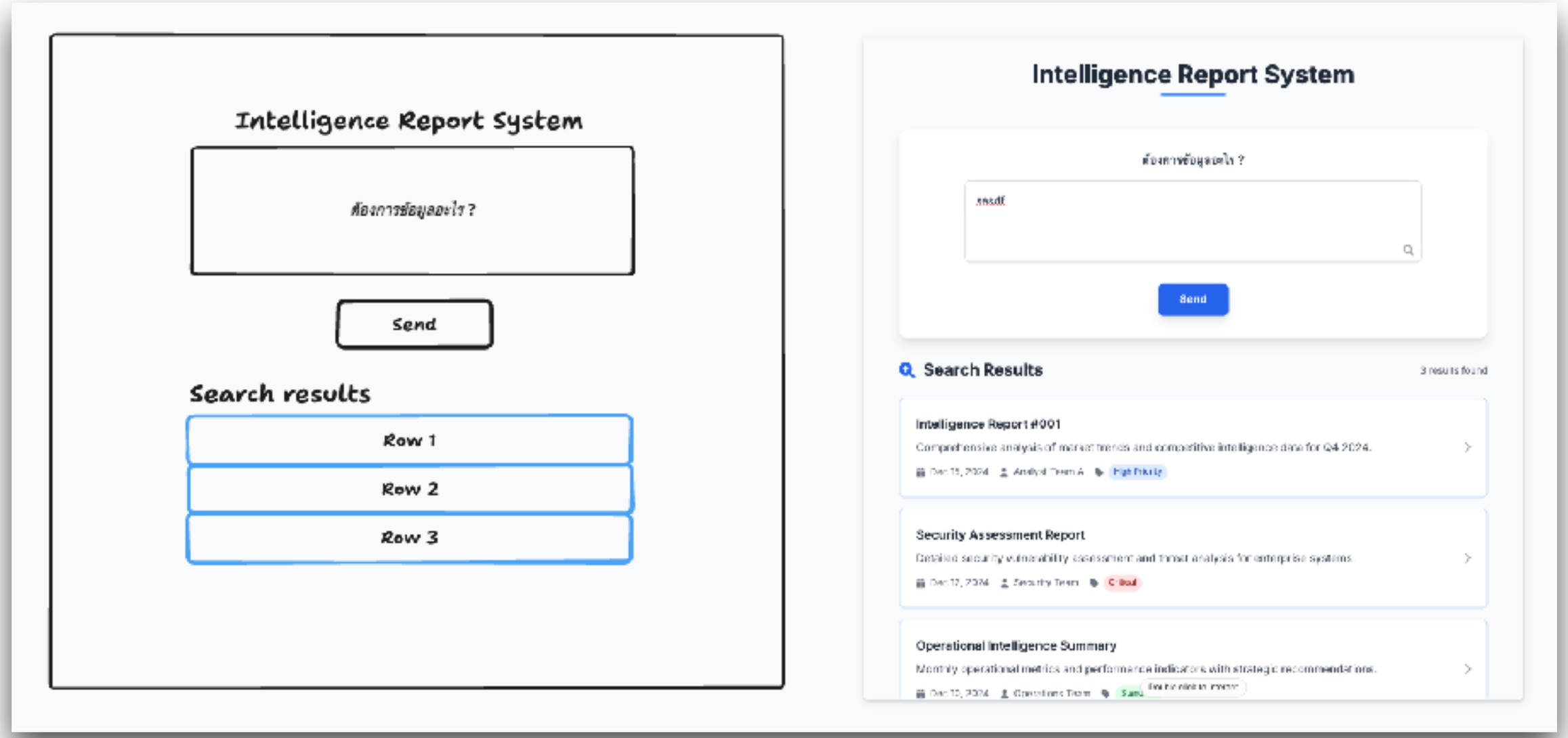
Magic Pattern



https://www.magicpatterns.com/

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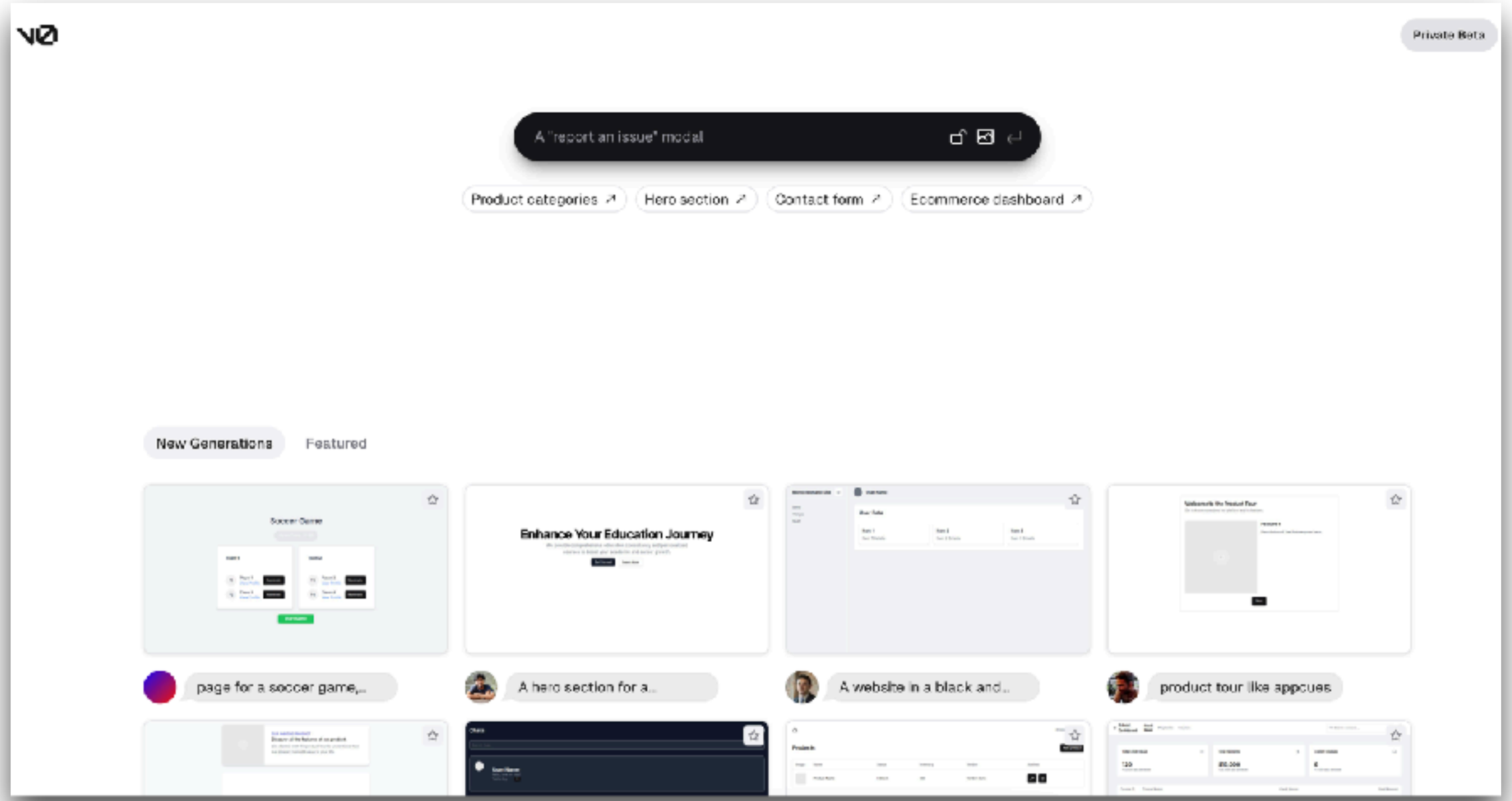
Make Real



https://github.com/tldraw/make-real

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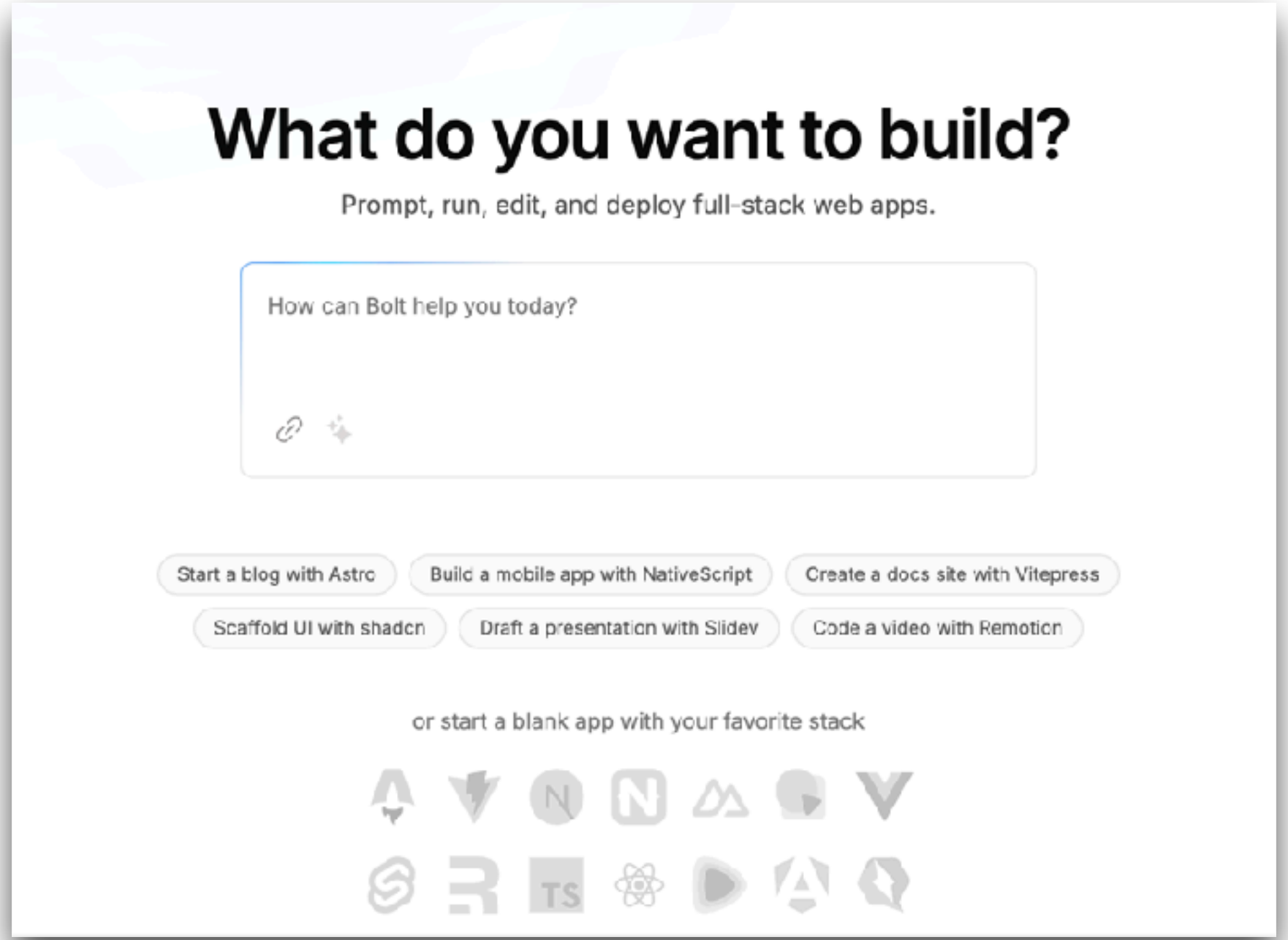
V0.app



https://v0.dev/

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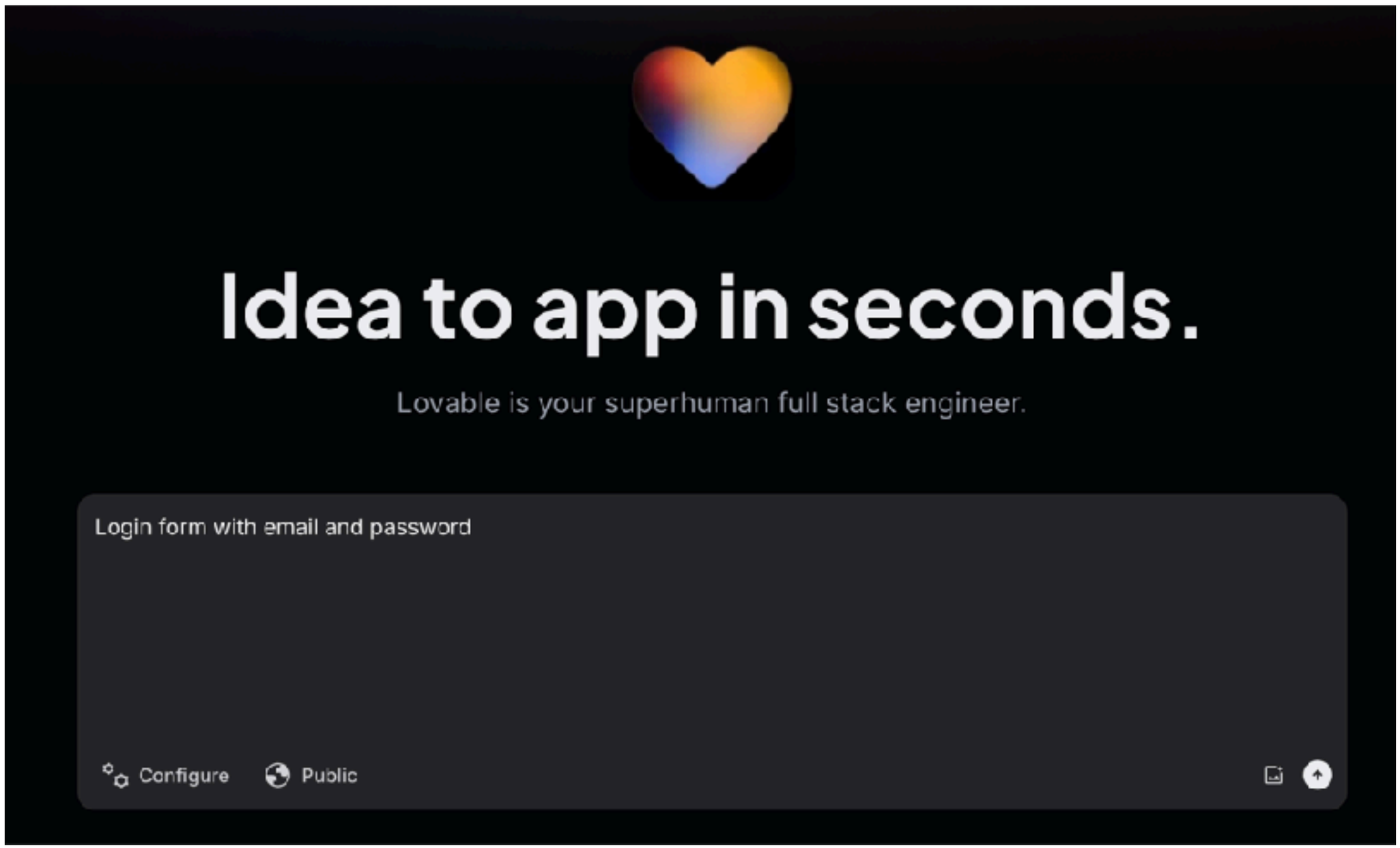
Bolt.new



https://bolt.new/

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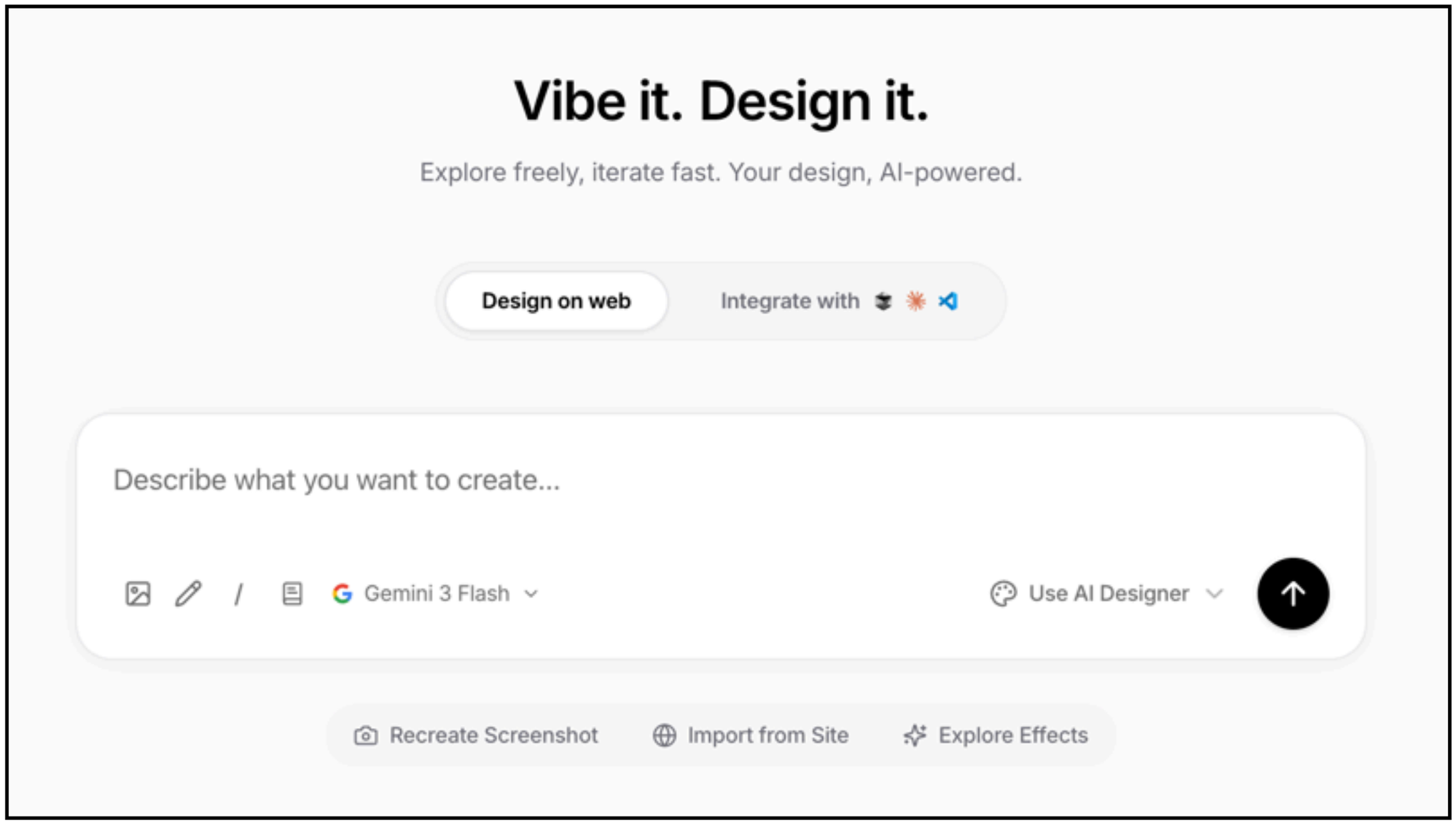
Lovable



https://lovable.dev/

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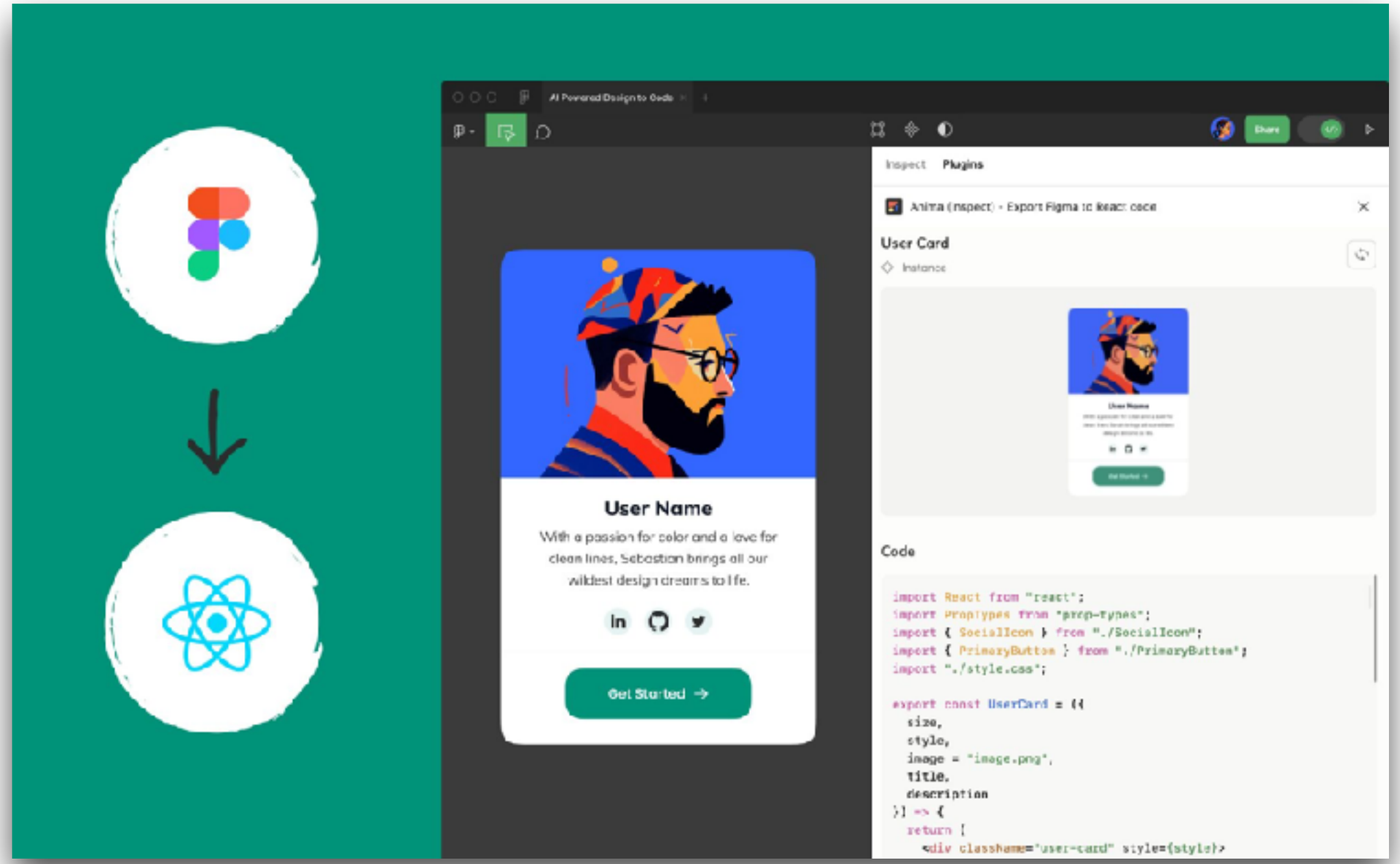
SuperDesign



https://app.superdesign.dev/

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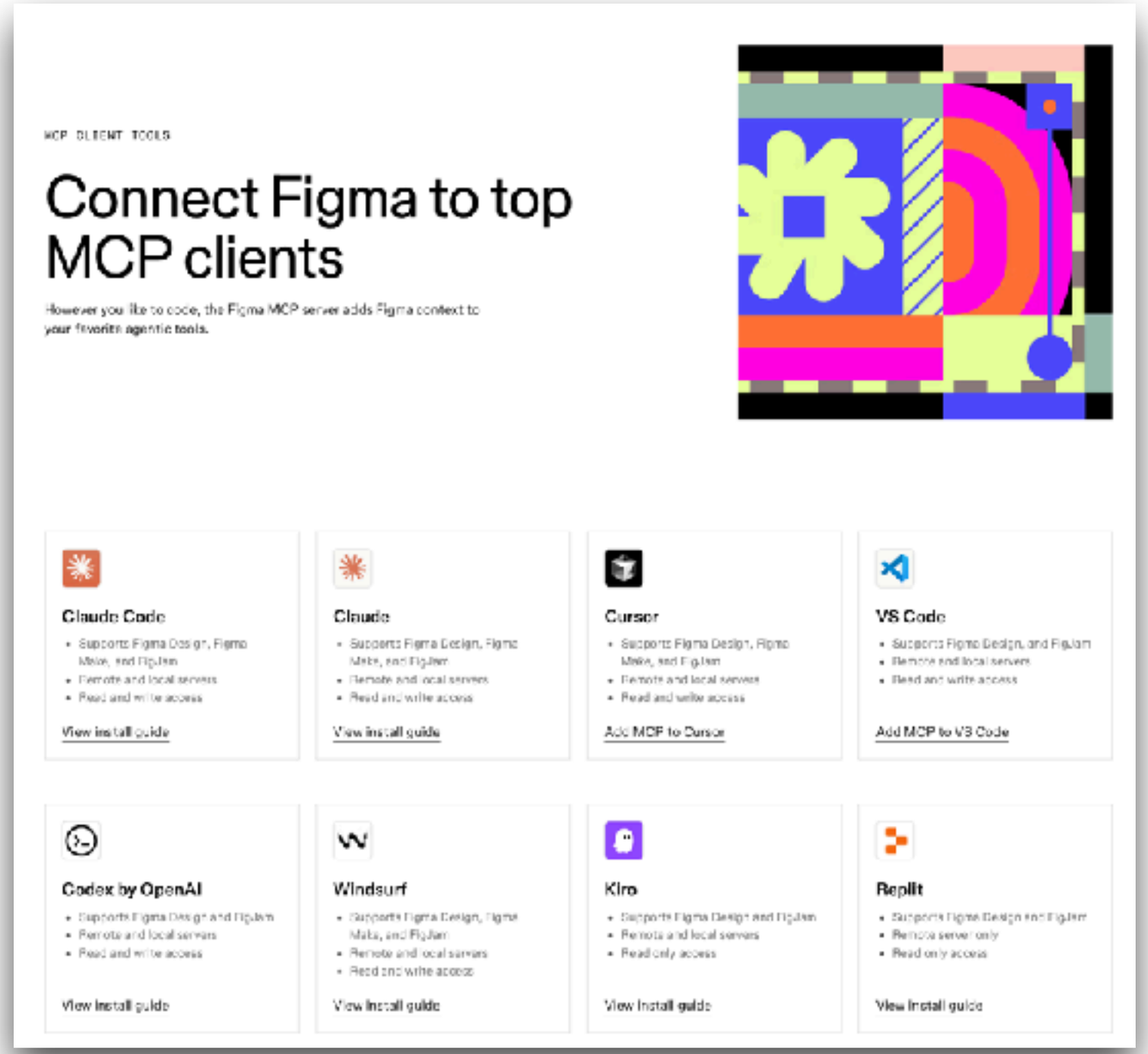
Figma



https://www.figma.com/mcp-catalog/

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Figma MCP

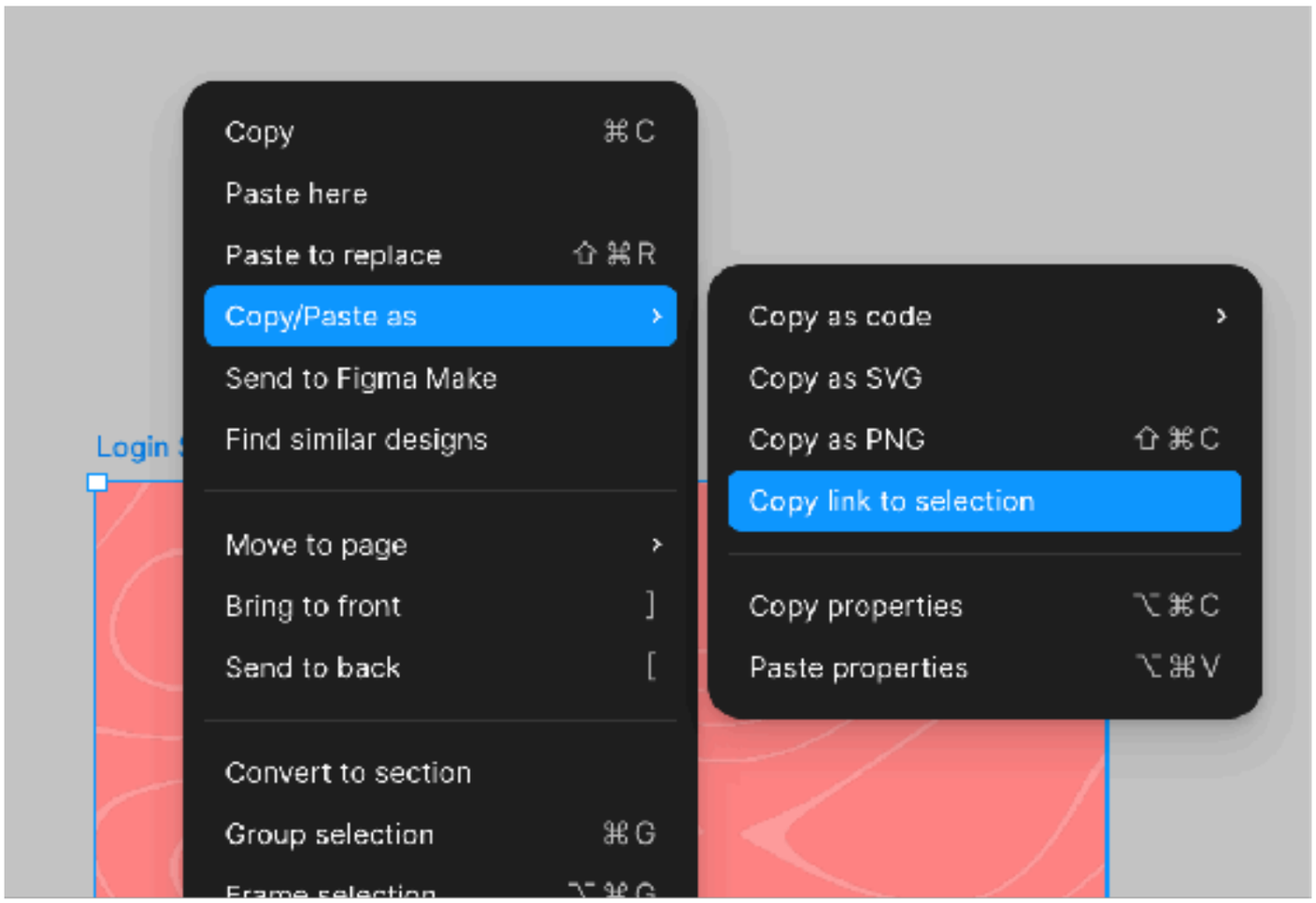


https://www.figma.com/mcp-catalog/

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Design to Code in Figma

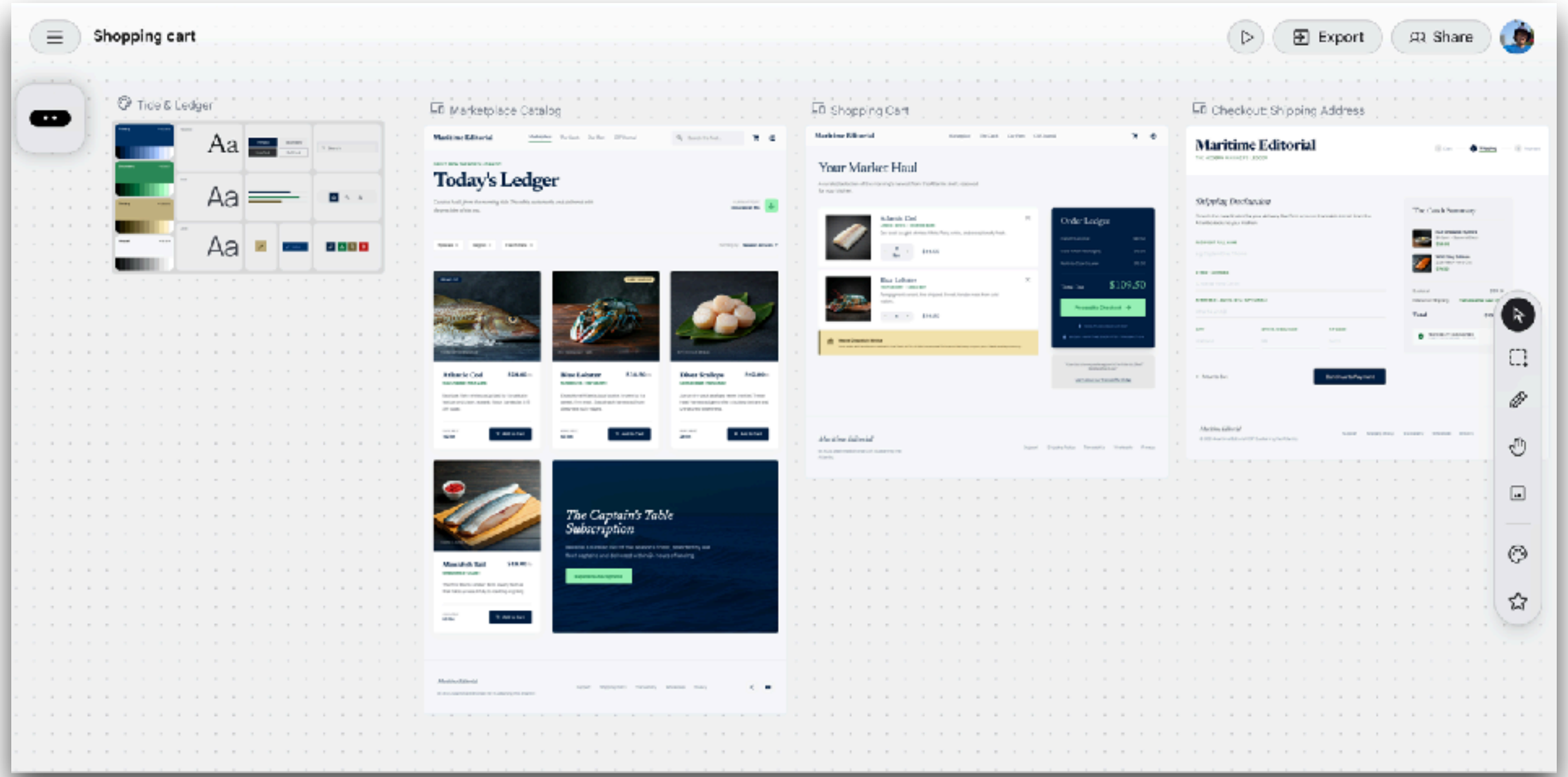
Select frame -> Copy/Paste as -> link to selection



https://www.figma.com/mcp-catalog/

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Google Stitch

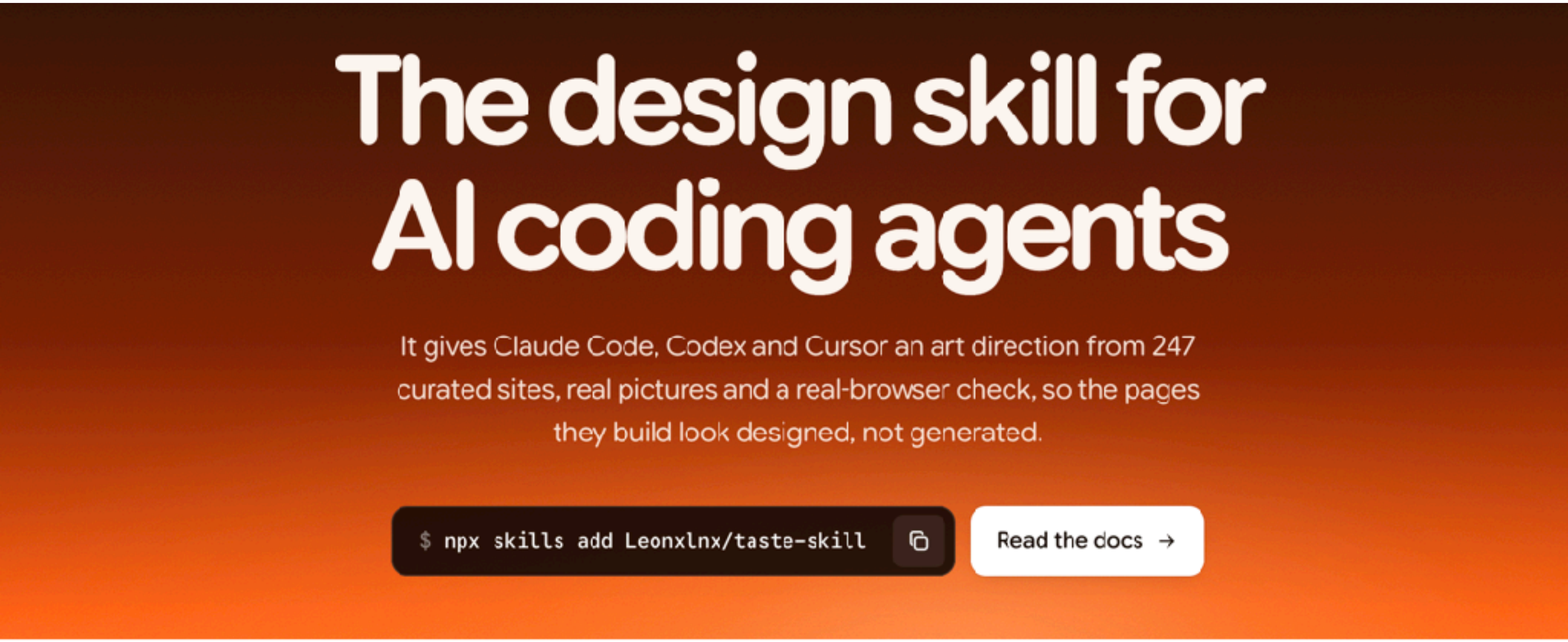


https://stitch.withgoogle.com/

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Taste Skill

The design skill for AI coding agent



https://www.tasteskill.dev/

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Development Process

Plan before Act

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Develop

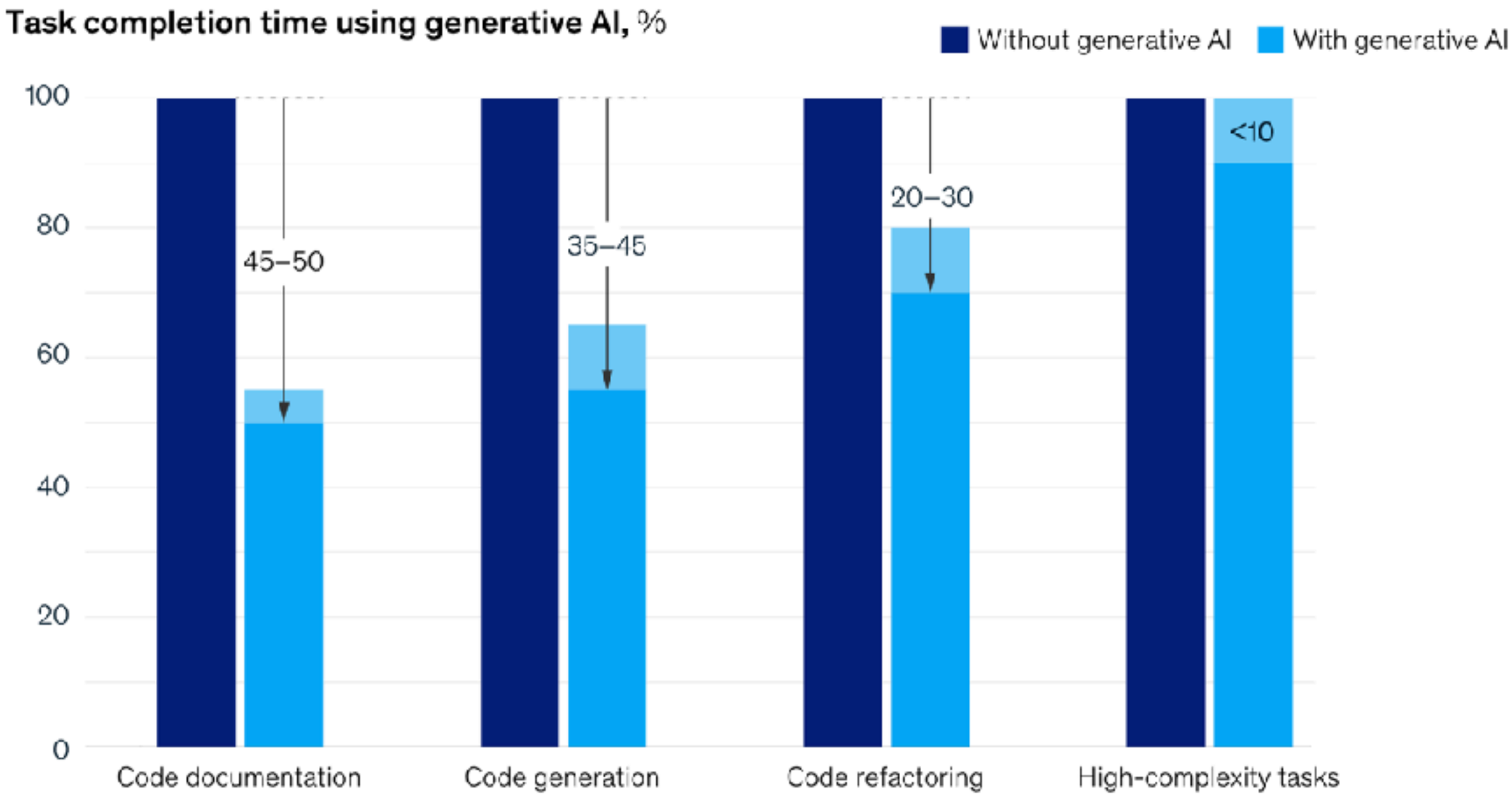


Code generation
Review and explain code
Debugging code
Improve consistency
Code translation

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Development

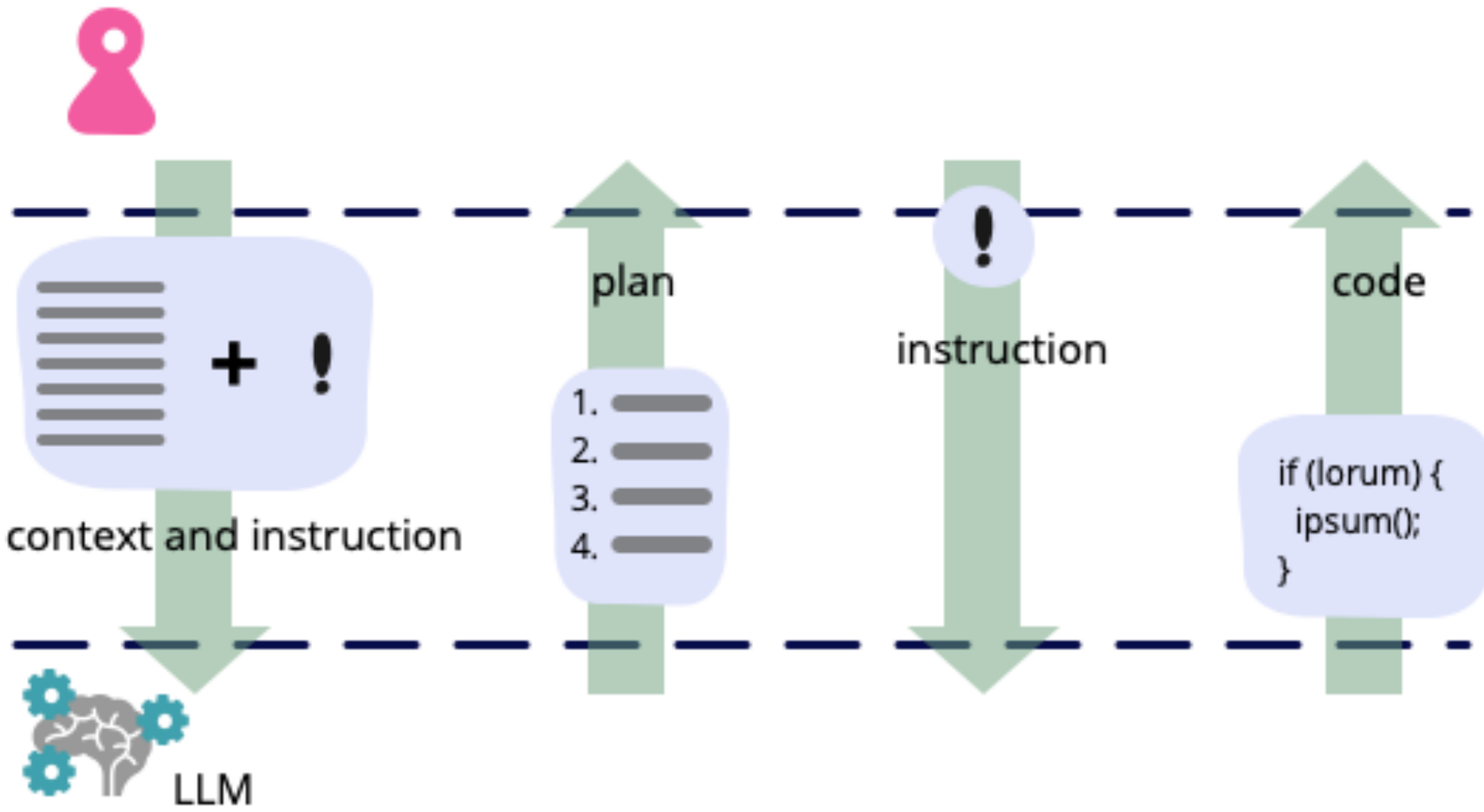
Generative AI can increase developer speed, but less so for complex tasks.



https://www.mckinsey.com/capabilities/mckinsey-digital/our-insights/unleashing-developer-productivity-with-generative-ai

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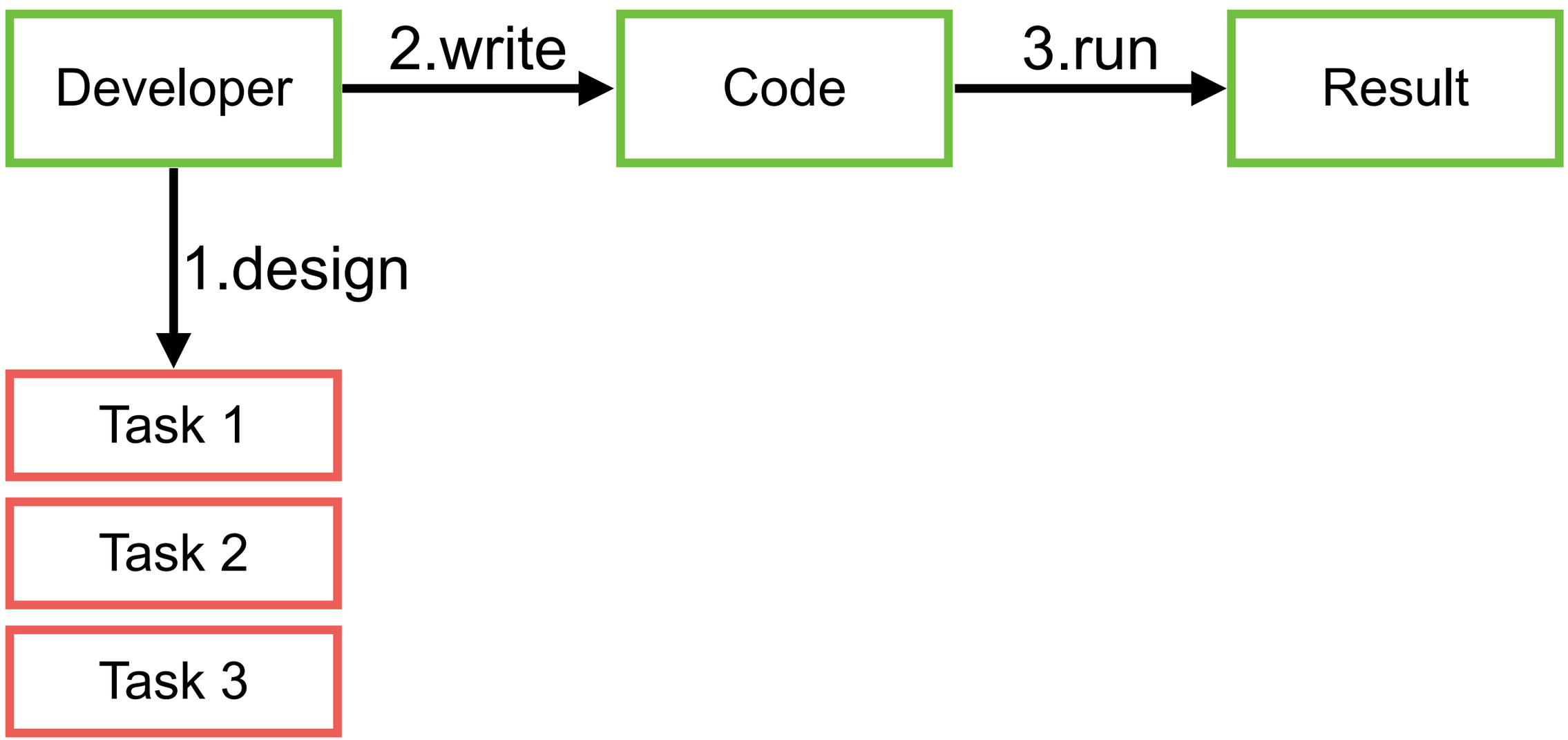
Development



https://martinfowler.com/articles/2023-chatgpt-xu-hao.html

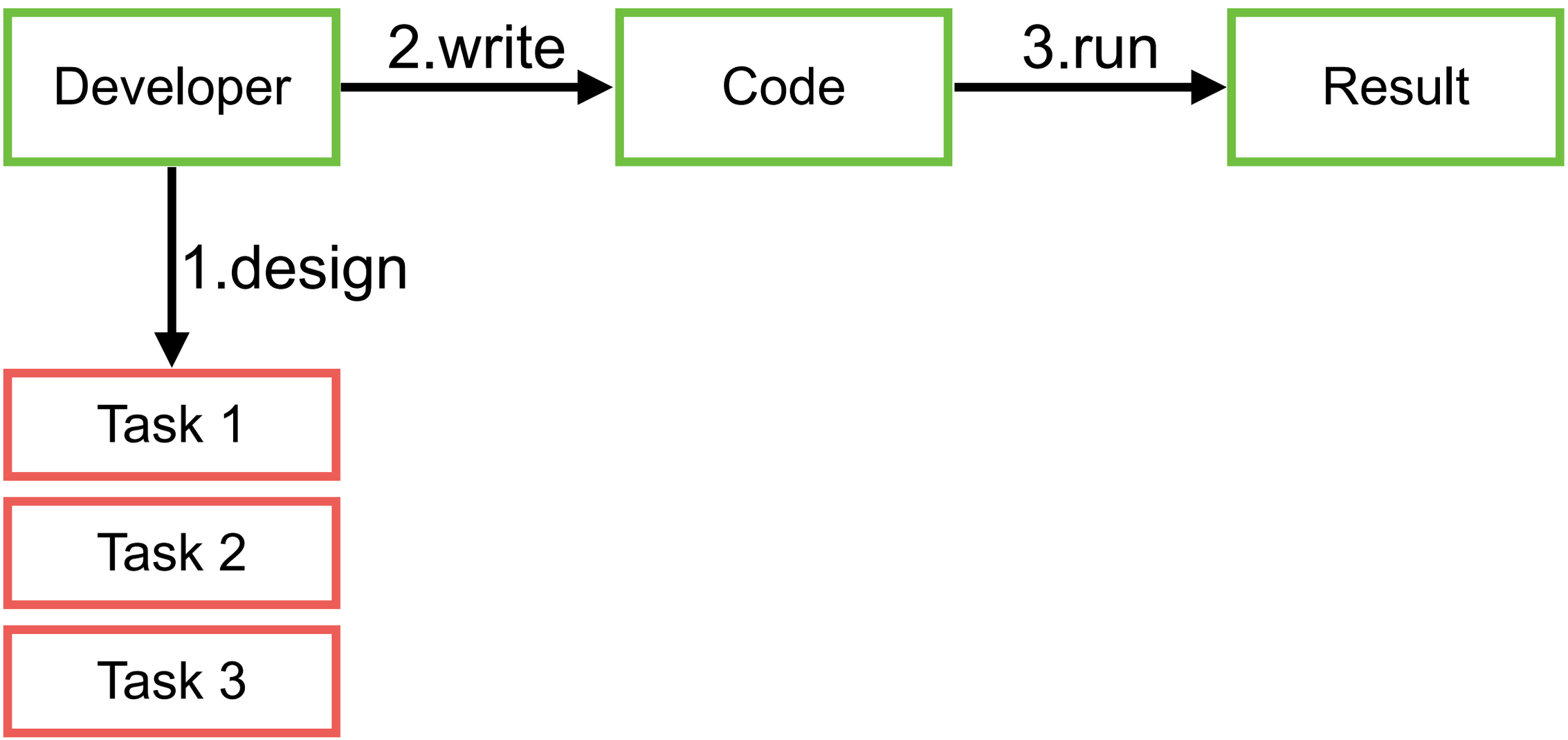
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Development



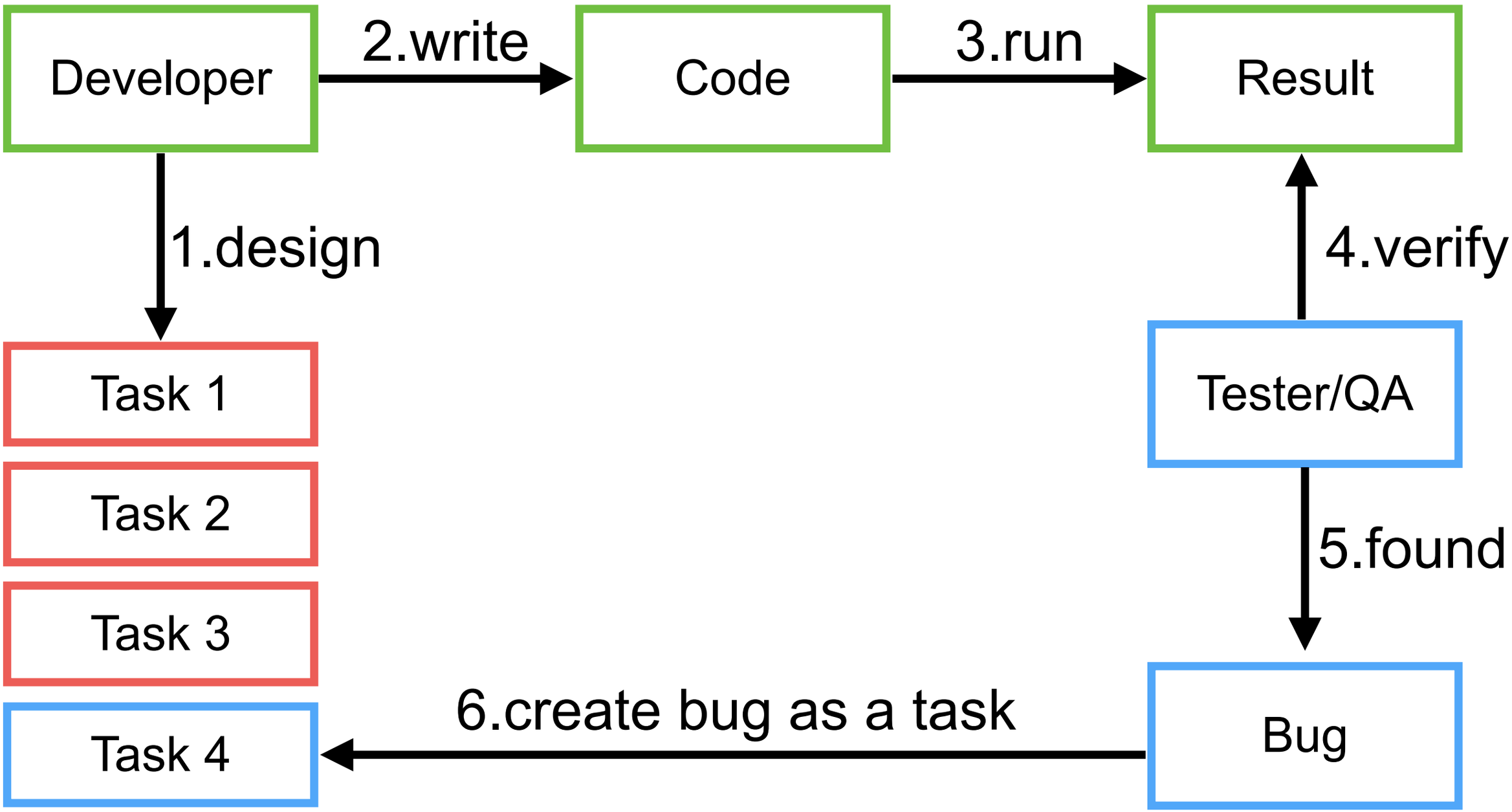
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Development



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Development + Testing



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Main Features

Ask question

Generate code

Refactor code

Document code

Find problems in your code

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Tips and Techniques with AI

Scope of prompt

Error in result

Setup and config project

Latest information

Use technical keywords

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Tips and Techniques with AI

Understand what you want

Modular approach

Clear and Precise inputs

Make sure you understand the code



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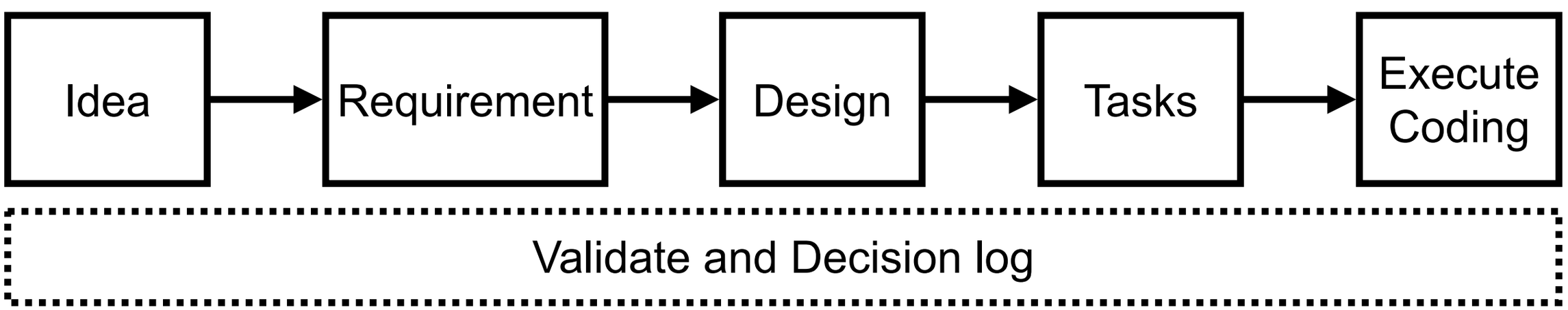
Specification-Driven Development (SDD)

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SDD ?

Specification or Documentation-First

Modern software development approach where detailed, structure specifications are created before writing code



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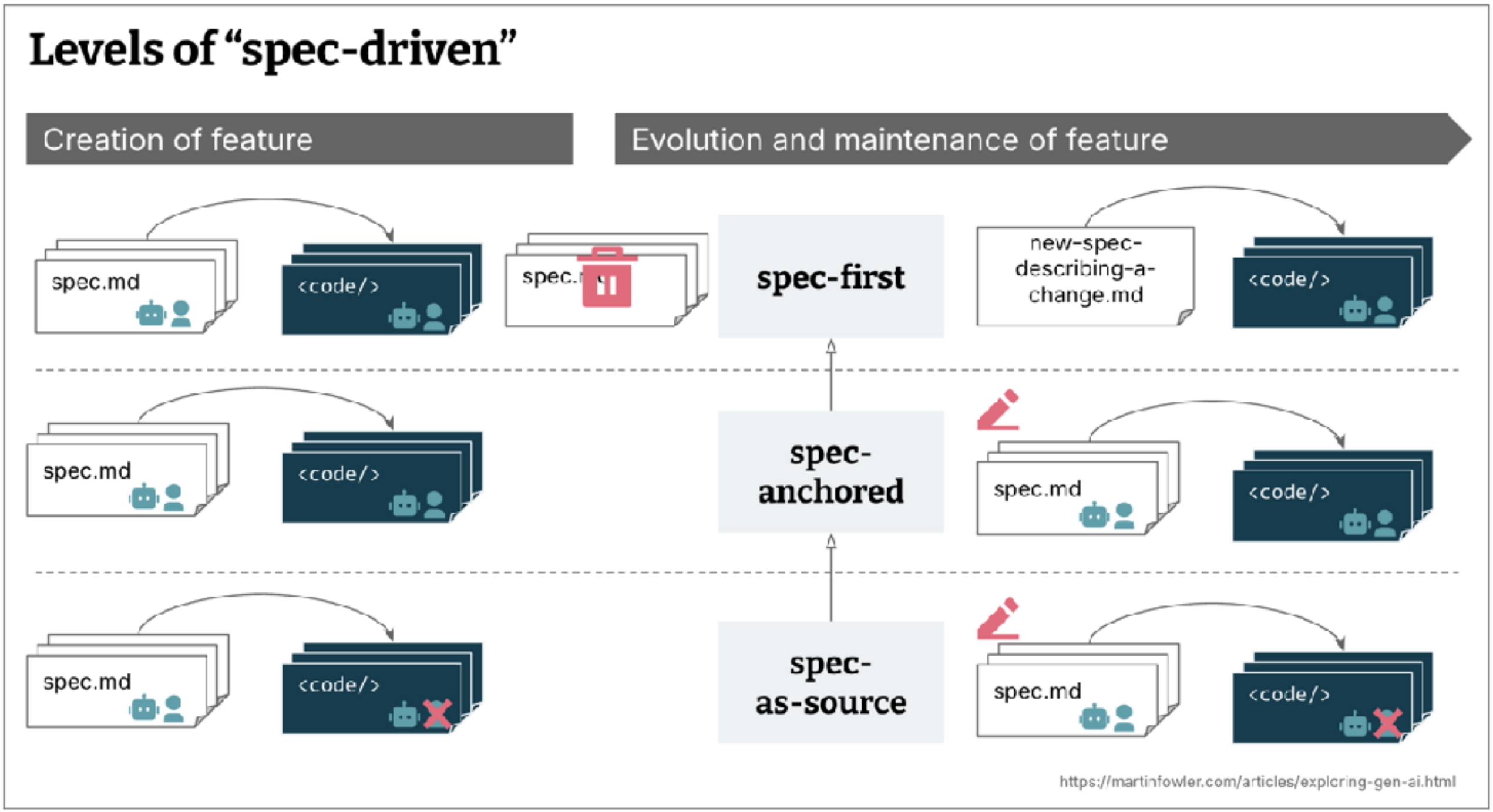
Types of Specification ?

User requirements
Design documents
Interface contracts (UI, API)
Data schemas
Validation rules
Data/business flows
Security policy and constraints
Resources and performance constraints



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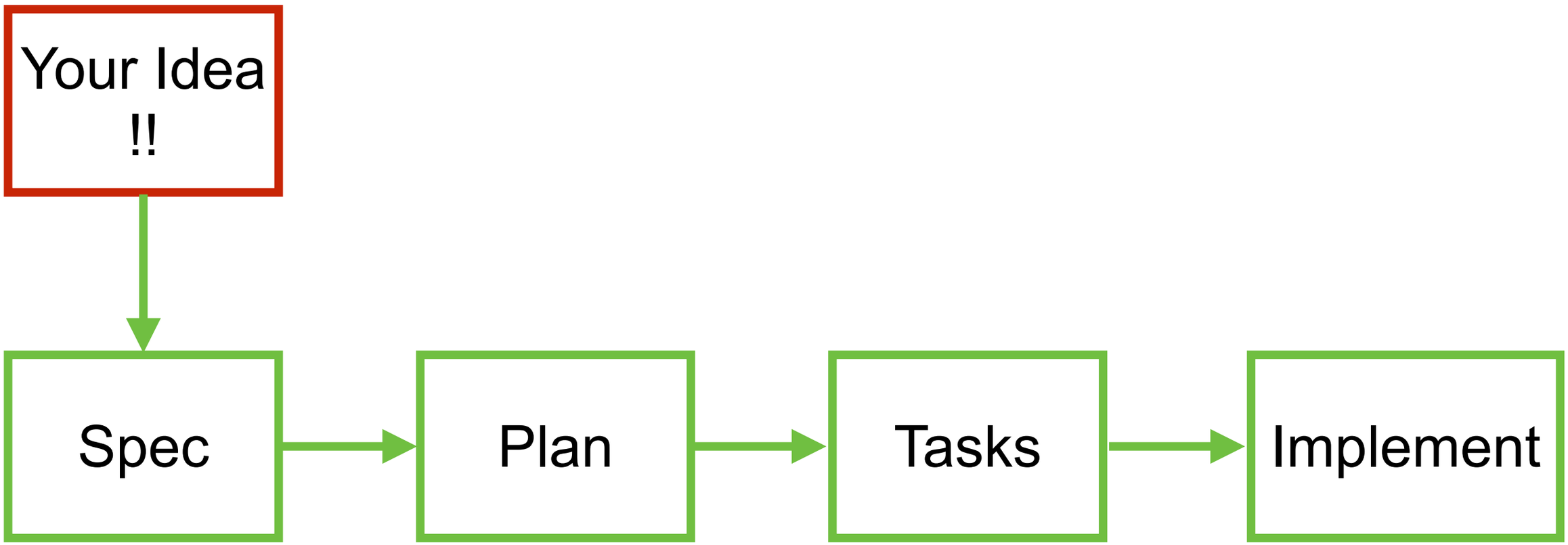
Workflow SDD



https://martinfowler.com/articles/exploring-gen-ai/sdd-3-tools.html

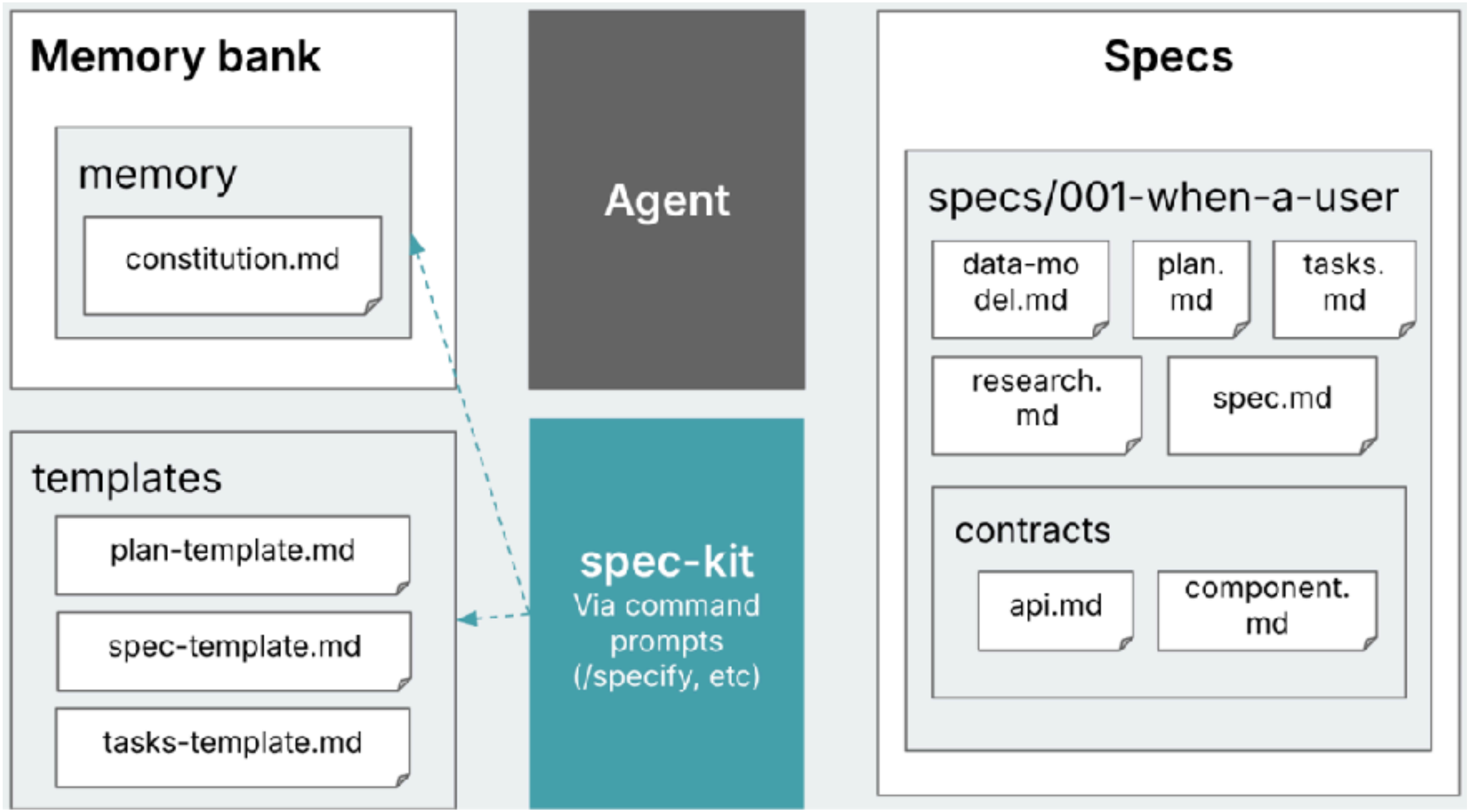
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Workflow SDD



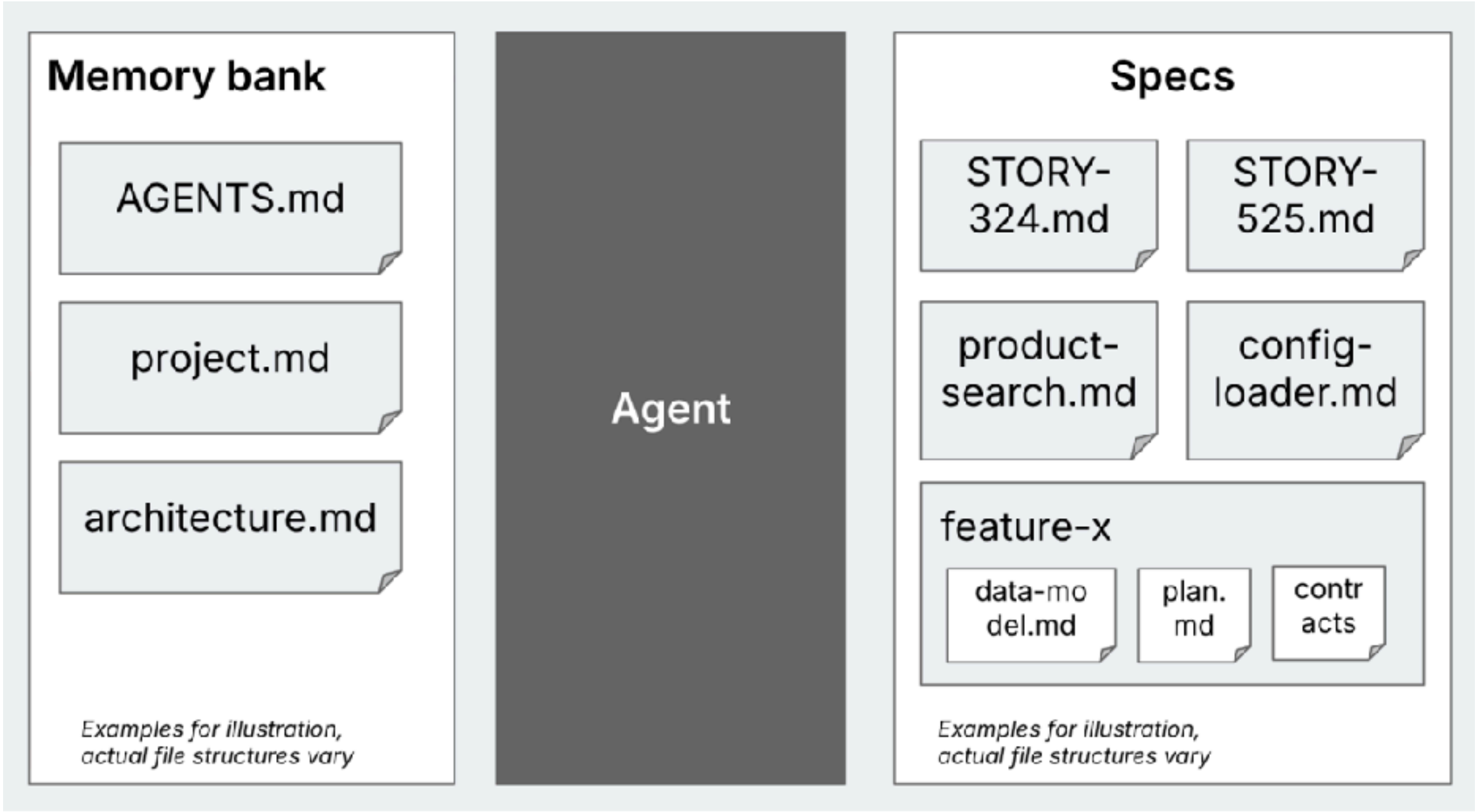
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Spec-Driven Development



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Memory Bank



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Testing Process
with High quality process

Functional

Non-Functional

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Testing



Test cases writing
Test code generation
Bug detection
Test planning
Data test generation

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6 Keys Software Quality

Defect density

Code duplication

Hardcode token/key

Security vulnerabilities

Outdated package

Non-permissive opensource libraries



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Risks when using Generative AI



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Risks

Quality of output generated
Explainability of decisions
Security policy !!
Sensitive data !!



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Start your journey



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